

Hacker Dropping **Mid-Heist Selfies**

LLM Identifies Information Stealer Infection Vectors and Extracts IoC

Olivier Bilodeau, Principal Cybersecurity Researcher
Estelle Ruellan, Threat Intelligence Researcher

Who Are We?



Olivier Bilodeau

- 15 years cybersecurity industry experience
- Principal Cybersecurity Researcher at Flare
- Former GoSecure, ESET. Founder MontréHack
- NorthSec's President
- Serial presenter: DEFCON, BlackHat, SecTor, Botconf, CERT-EU, AtISecCon



Estelle Ruellan

- Cyber Threat Intelligence Researcher
- Mathematics and Criminology Background
- Former student athlete
- Loves data science, shapes and colors
- Baby serial presenter: NorthSec, ShmooCon, Botconf, Hack.lu, eCrime APWG, EUROCRIME

Honorable mentions:



Agenda

1. The Information Stealer Malware Phenomenon
2. Mid-Heist Selfies
3. The LLM Pipeline
4. Prompt Engineering
5. LLM Assessment
6. Discriminating IoCs
7. Inside the Infostealer Playbook
8. Successful Campaigns: 2 Case Studies
9. Strength and Limits
10. Conclusion

Agenda

1. The Information Stealer Malware Phenomenon

2. Mid-Heist Selfies

3. The LLM Pipeline

4. Prompt Engineering

5. LLM Assessment

6. Discriminating IoCs

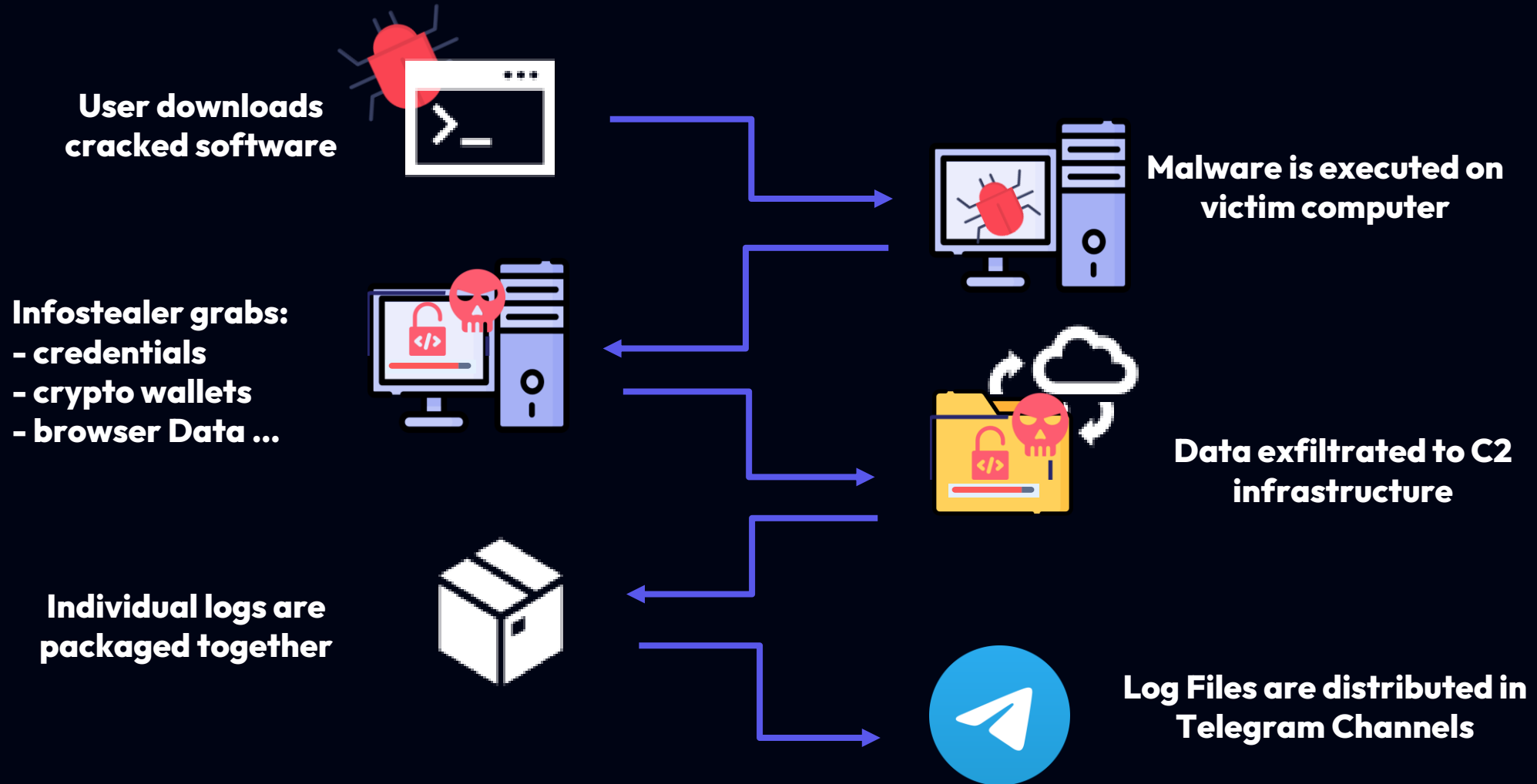
7. Inside the Infostealer Playbook

8. Successful Campaigns: 2 Case Studies

9. Strength and Limits

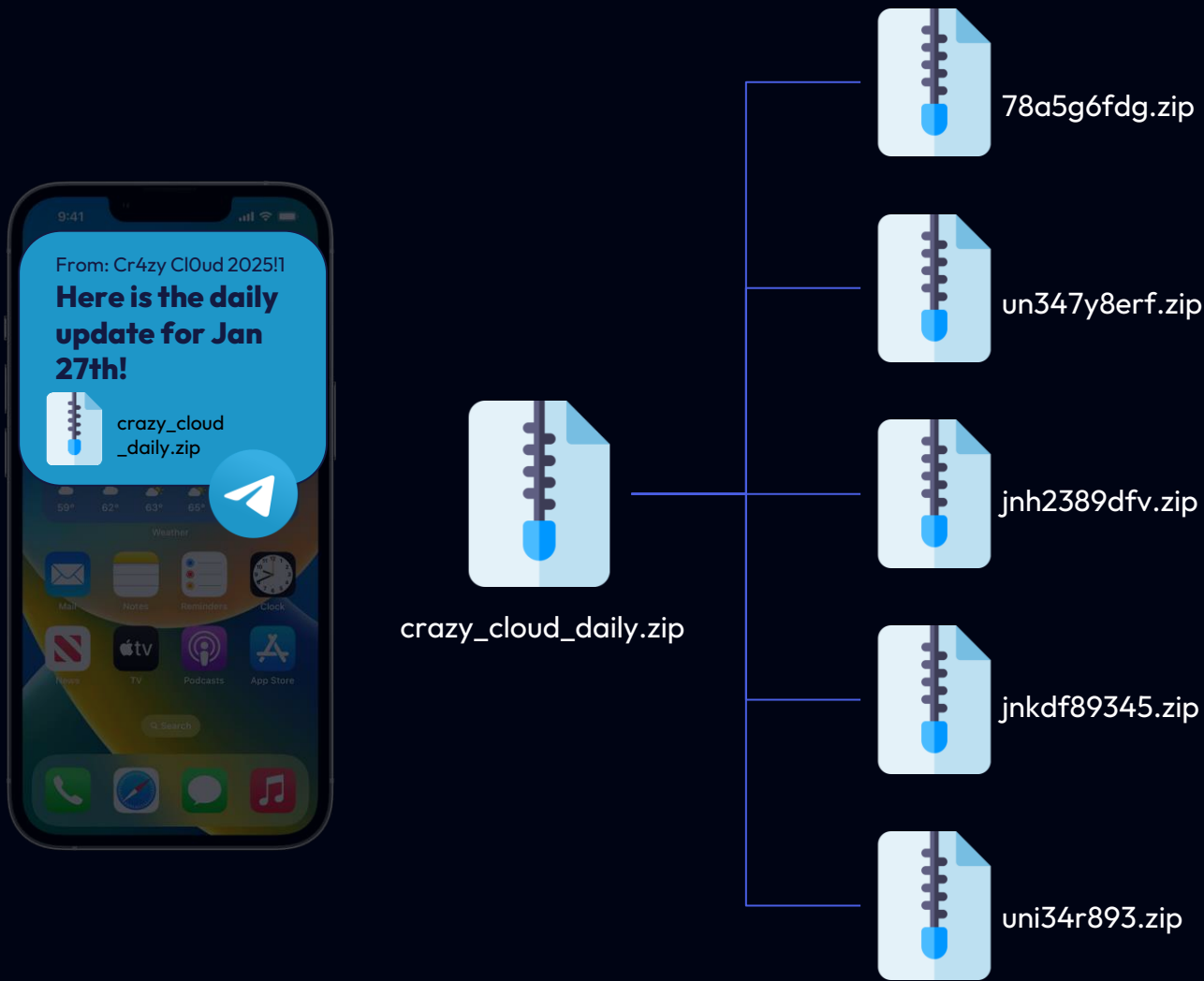
10. Conclusion

The Malware you (may) have never heard of:

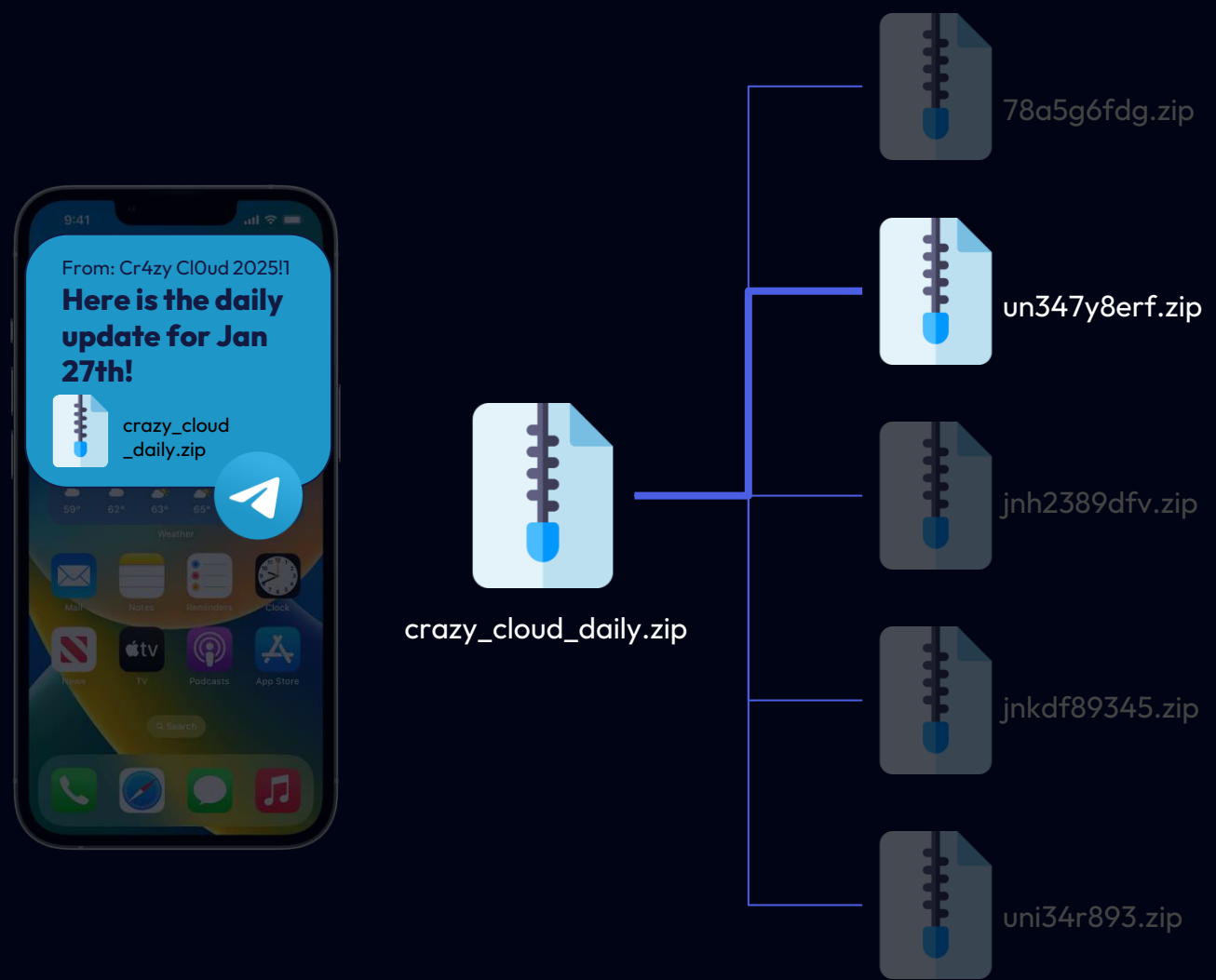


Administrative rights NOT required! && No Persistence!

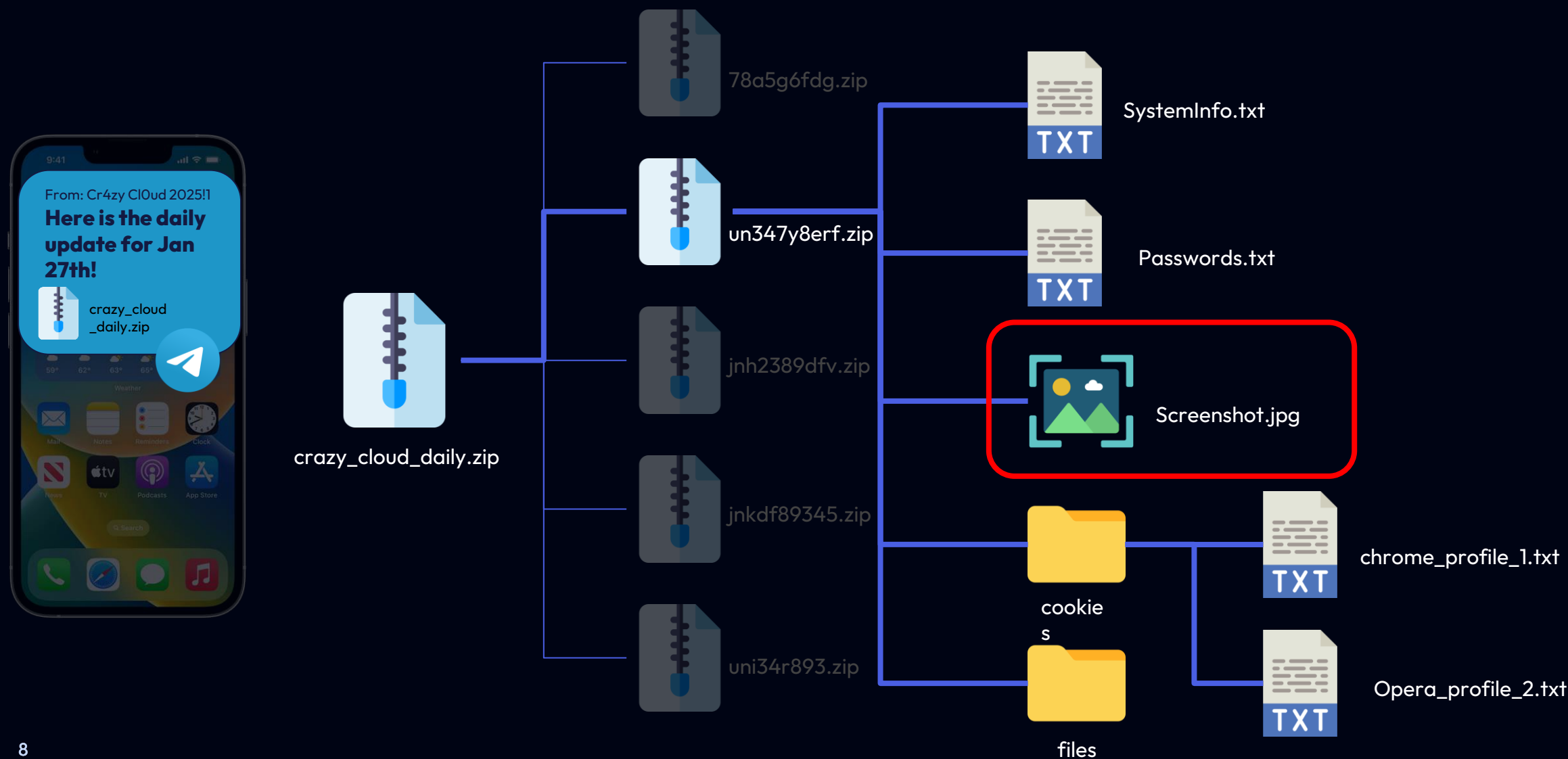
Stealer Log Structure



Stealer Log Structure



Stealer Log Structure



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Mid-Heist se

**REALISTIC PICTURE OF
BRO BEHIND HIS SCREEN**



Mid-Heist selfies



Mid-Heist selfies

Screen_011e67e8

how to install mod menu fortnite pc 2022

SHGZC

FORTNITE HACK
FREE DOWNLOAD

Download the archive
link in the descripton

DOWNLOAD - "CLICK"

password - shgzchee
*updated + undected

!! IF YOU HAVE TROUBLES
DOWNLOADING/LAUNCHING FILE JUST TURN OFF
YOUR ANTI-VIRUS ITS ABSOLUTELY SAFE !

0:02 / 1:04

FORTNITE HACK | UNDETECTED | FORTNITE MOD MENU | DOWNLOAD FREE

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FORTNITE HACK | UNDETECTED | FORTNITE MOD MENU | DOWNLOAD FREE

DOWNLOAD LINK: <https://telegra.ph/FORTNITE-HACK-FREE-DOWNLOAD-09-14>

ARCHIVE PASSWORD: shgzchee

Working

Instructions:
1) Download the archive.
2) Unzip the file to any folder/desktop.

Copy
Copy link to highlight
Go to <https://telegra.ph/FORTNITE-HACK-FREE-DOWNLOAD-09-14>
Print...
Inspect

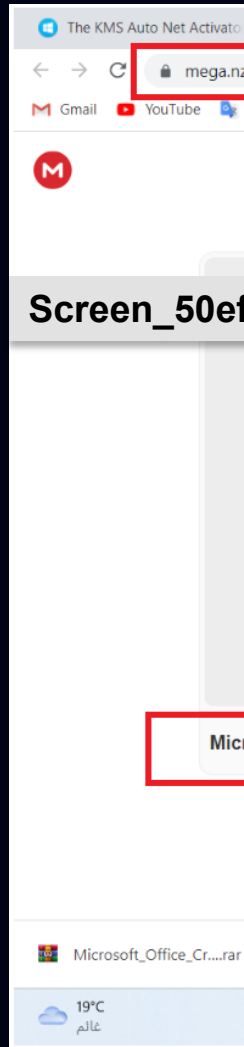
Ctrl+C
Ctrl+P

Fortnite Hack.rar

Fortnite_Hack_2022.zip

Some examples

Mid-Heist s



Mid-Heist selfies

Infection screenshots contain all the clues
and hints needed to solve the mystery of
infection

Mid-Heist selfies

11

Malware Families

+15M

Screenshots

*

+25%

of all logs

*Including duplicates

Mid-Heist

TOO MUCH
INFORMATION

COULD USE
AN LLM

1
Malware

%
gs

*Including duplicates

Agenda

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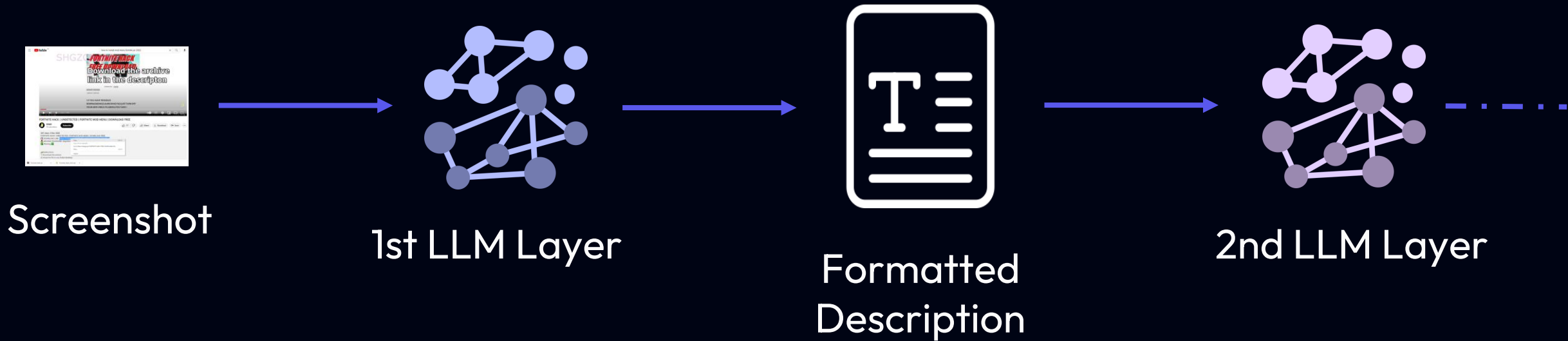
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Overview of the Pipeline



Overview of the Pipeline



Why 2 Layers ?

Overview of the Pipeline



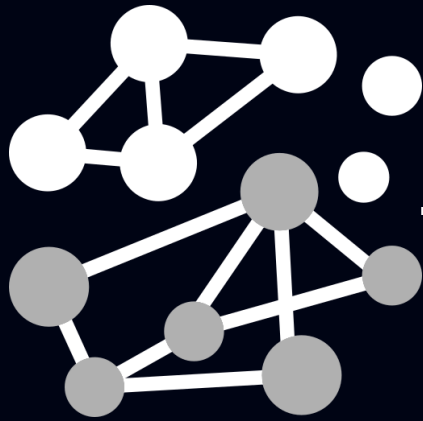
Screenshot



ayer

Why 2 Layers ?

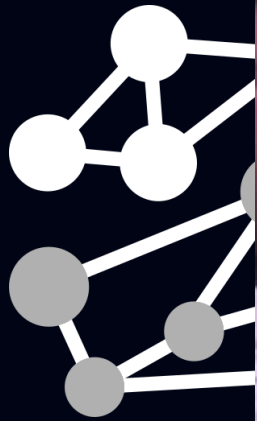
Overview of the Pipeline



“Identify the infection vector”

Why 2 Layers ?

Overview of the Pipeline



vector”

Prompt Engineering



WHY 2 LAYERS?

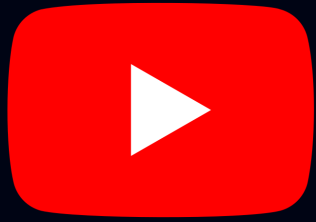
Overview of the Pipeline



1. Visually assess the screenshot

Why 2 Layers ?

Overview of the Pipeline



FORTNITE

Screen_011e67e8

how to install mod menu fortnite pc 2022

SHGZC

FORTNITE HACK
FREE DOWNLOAD
Download the archive
link in the descripton

DOWNLOAD - "CLICK"

password - shgzhcqs
*updated + undected

! IF YOU HAVE TROUBLES
DOWNLOADING/LAUNCHING FILE JUST TURN OFF
YOUR ANTI-VIRUS ITS ABSOLUTELY SAFE !

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DOWNLOAD LINK: <https://telegra.ph/FORTNITE-H4CK-FREE-DOWNLOAD-09...>

ARCHIVE PASSWORD: shgzhcqs

Working

Instructions:
1) Download the archive.
2) Move the file to any folder on desktop.

Fortnite Hack.rar

Fortnite_Hack_2022.zip



DOWNLOAD LINK:

<https://telegra.ph/FORTNITE-H4CK-FREE-DOWNLOAD-09...>

Why 2 Layers?

Overview of the Pipeline



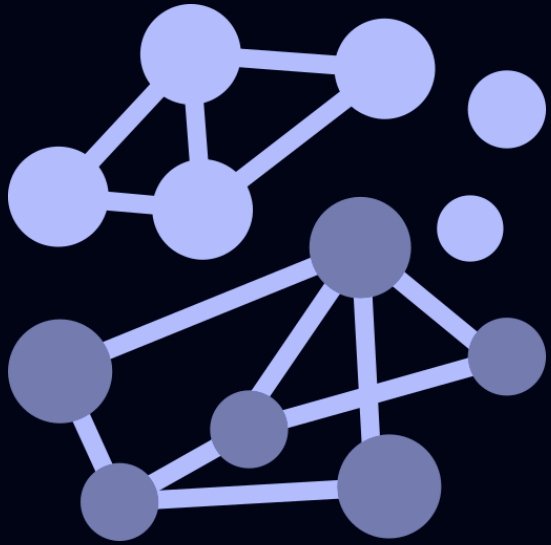
1. Visually assess the screenshot
2. Point out potential infection vectors based on field knowledge

Overview of the Pipeline

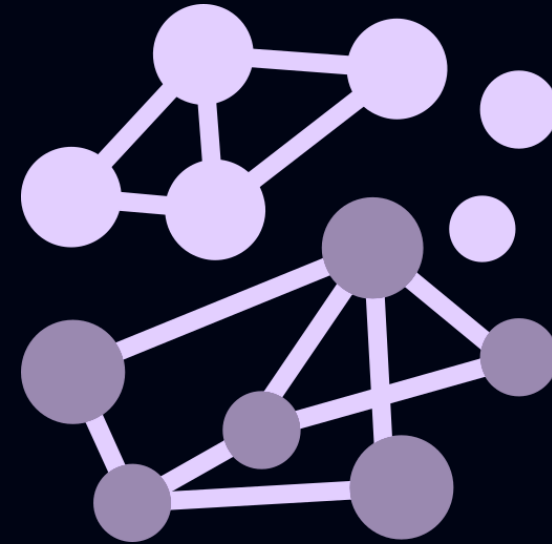
An LLM can't just 'figure it out':
we must **translate analyst intuition** into
instructions.

Why 2 Layers ?

Overview of the Pipeline



1. Visually assess the screenshot

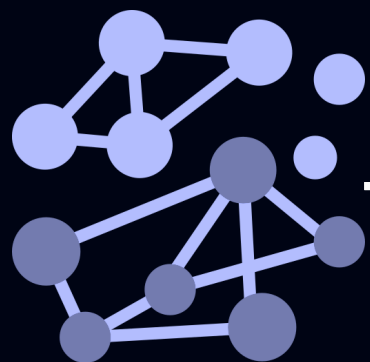


2. Point out infection vectors

Agenda

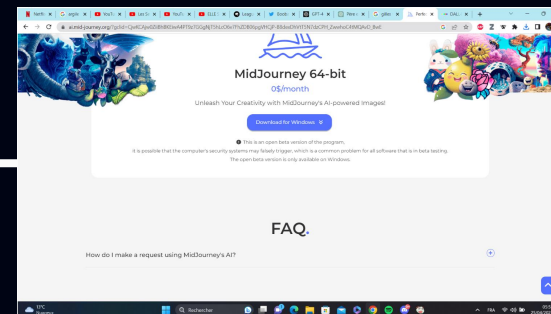
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The first layer
Prompt Engineering

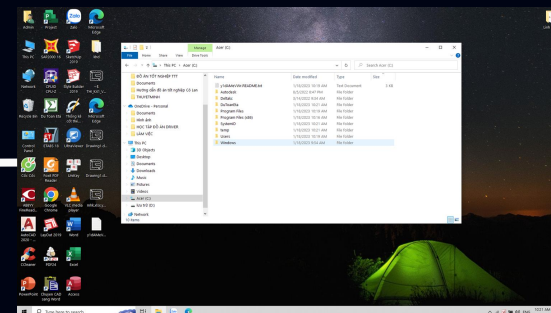


Visual Assessment

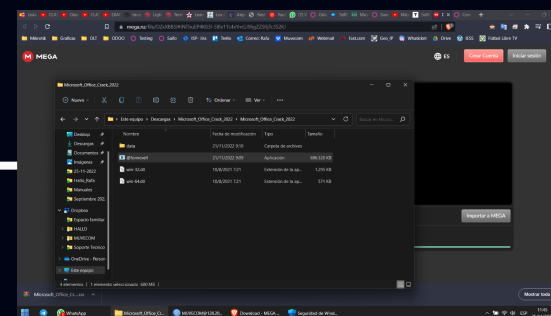
Web Content



File System



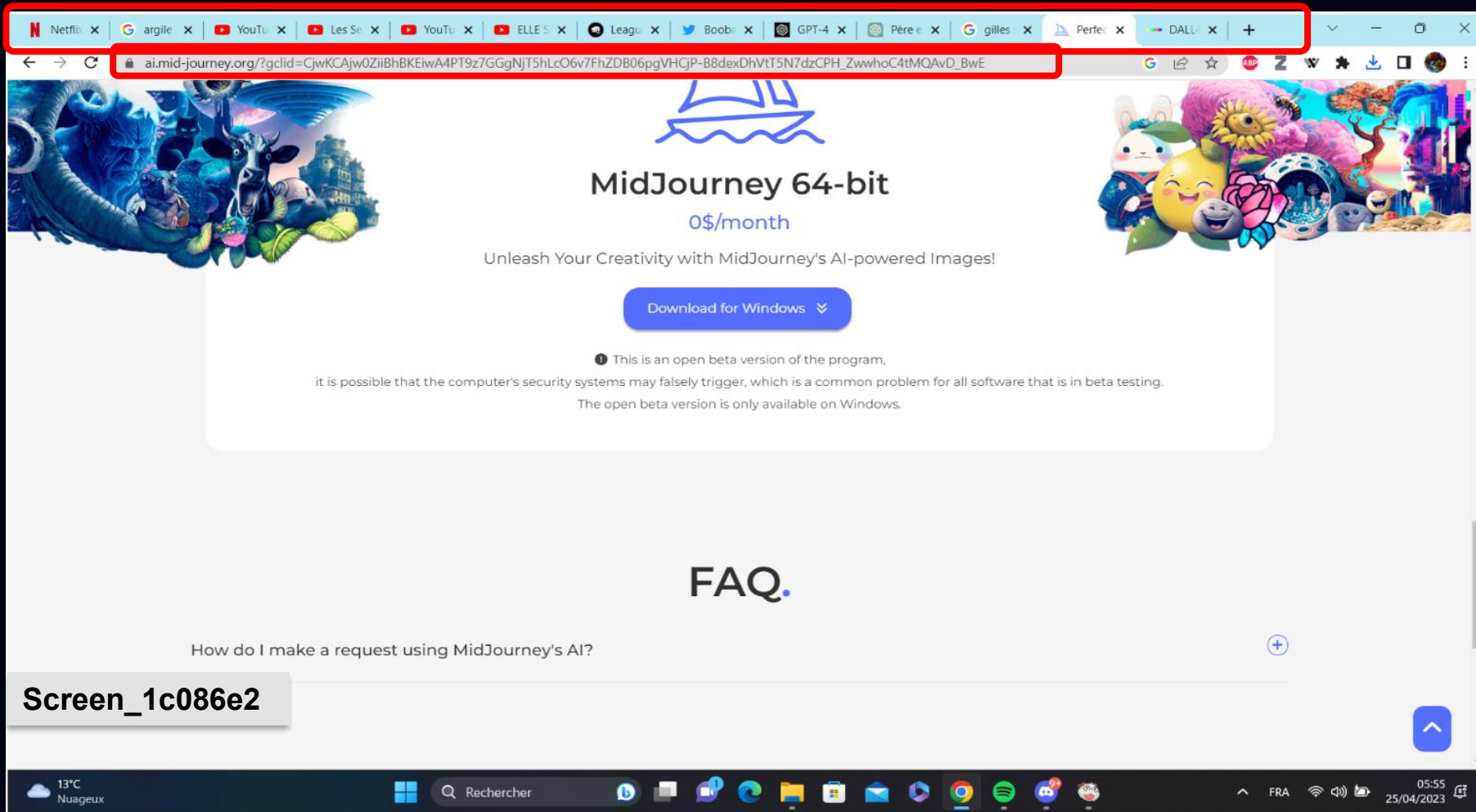
Hybrid



The first layer

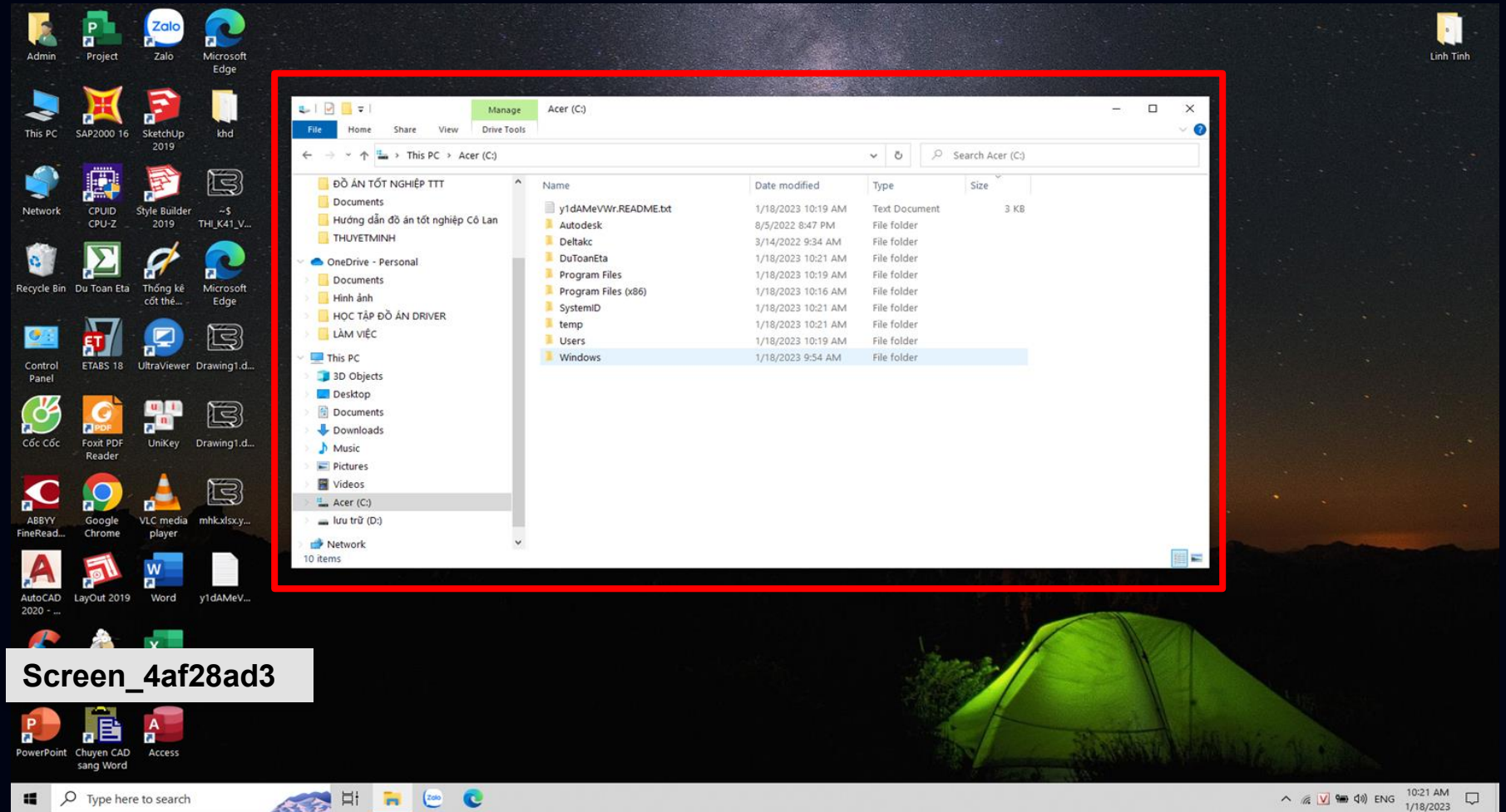
Prompt Engineering

Web Content



The first layer Prompt Engineering

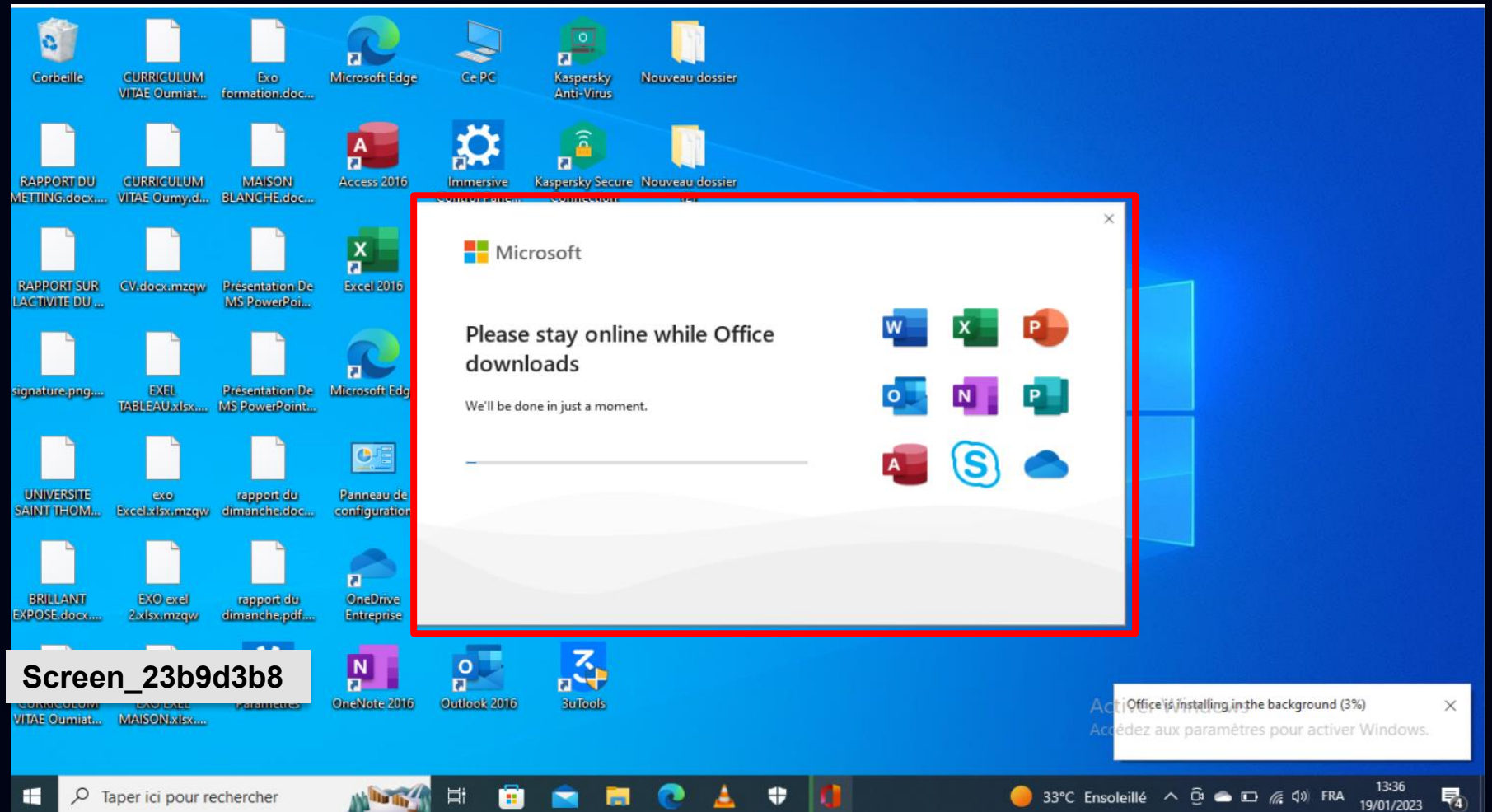
File System



The first layer

Prompt Engineering

File System



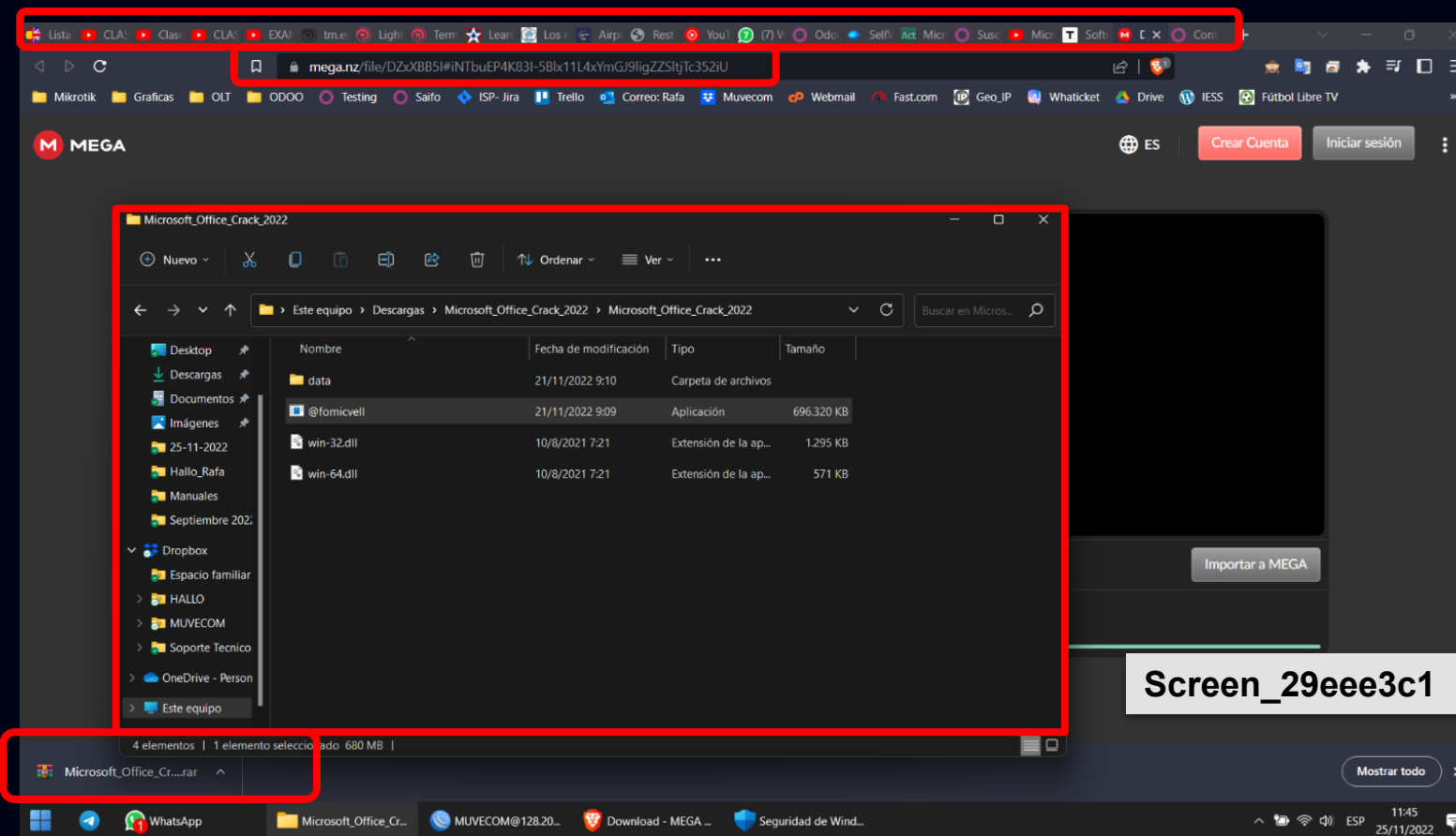
Screen_23b9d3b8

The first layer

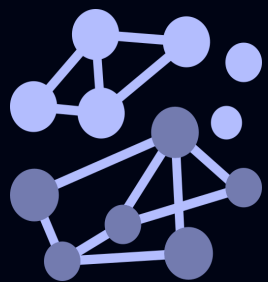
Prompt Engineering



Hybrid



The first layer
Prompt Engineering



Visual
Assessment



Scene Description



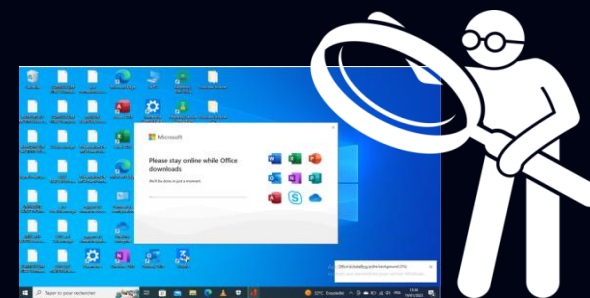
File Explorer & Installer



URL



Browser Tab Identification



Suspicious Elements

Main Content:

Describe the main content visible on the screen, include as much detail as possible.



1ST LAYER INPUT PROMPT

Files/Programs:

Installer: Focus on installers or install window, put the name of the file being installed. When there is a name for the installer window, get the name of file/folder or the path.

File explorer: Focus on file explorer if there is one. Put the names of files and their extensions in this section. If the path of the file explorer reveals the name of a folder/file, get it. Ignore all desktop programs and icons. Seperate filenames by a ",". If there aren't any file, executable or program put "X".

URL

Put all URLs you see. If there aren't any URLs, put "X".

Browser Tabs Analysis:

Ignore bookmarks. For each active browser tab in the top row, list in this format:
- [logo: {logo name}] [text: {visible text}] (meaning/context if apparent). If there aren't any webpage, put "X".

Suspicious Elements:

Highlight any file, executable, program, URL or download link that could contain malware. These could be youtube videos, blogs, google drive, etc.

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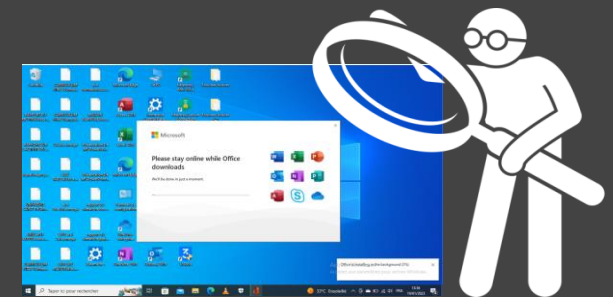
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URL

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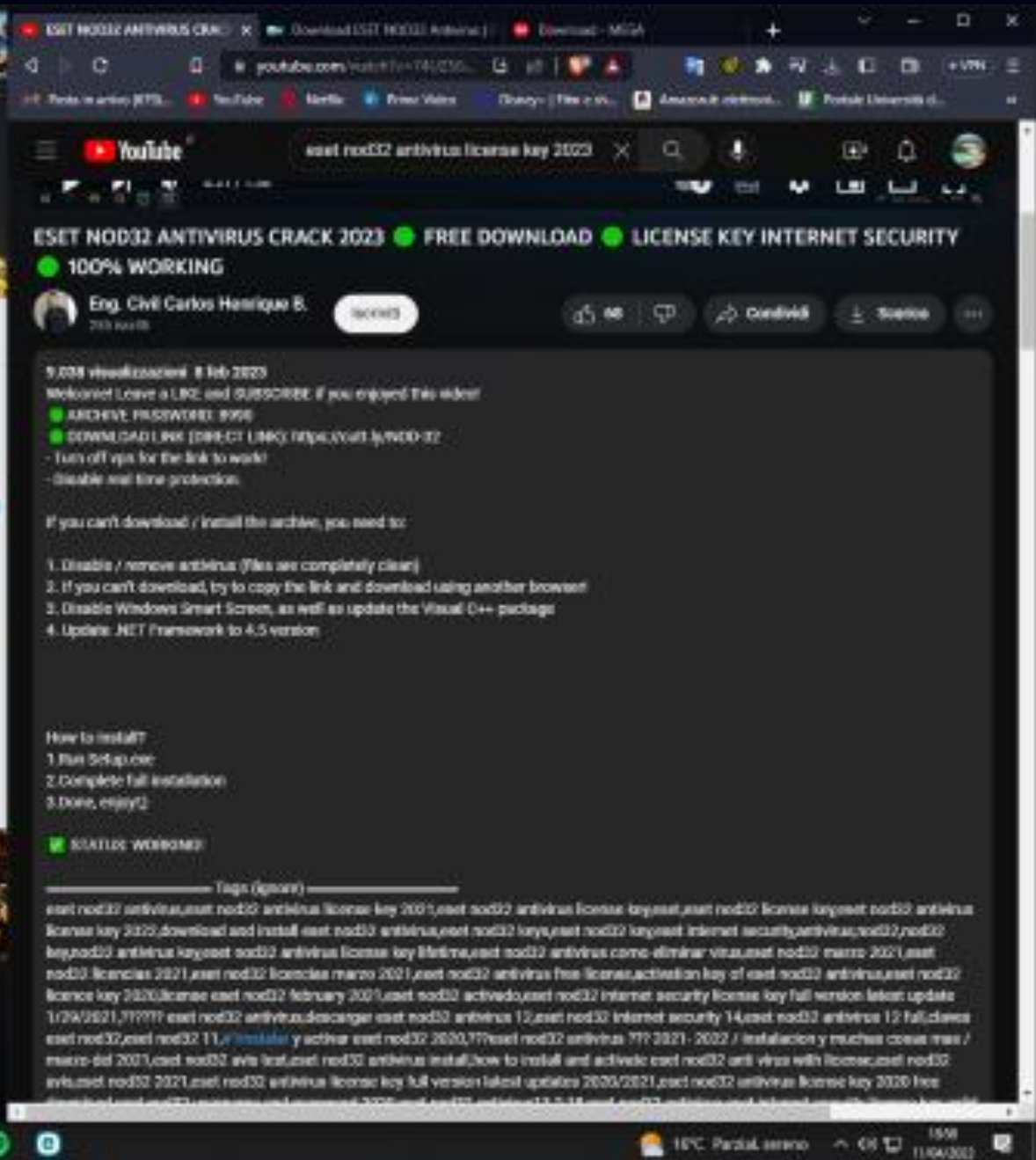
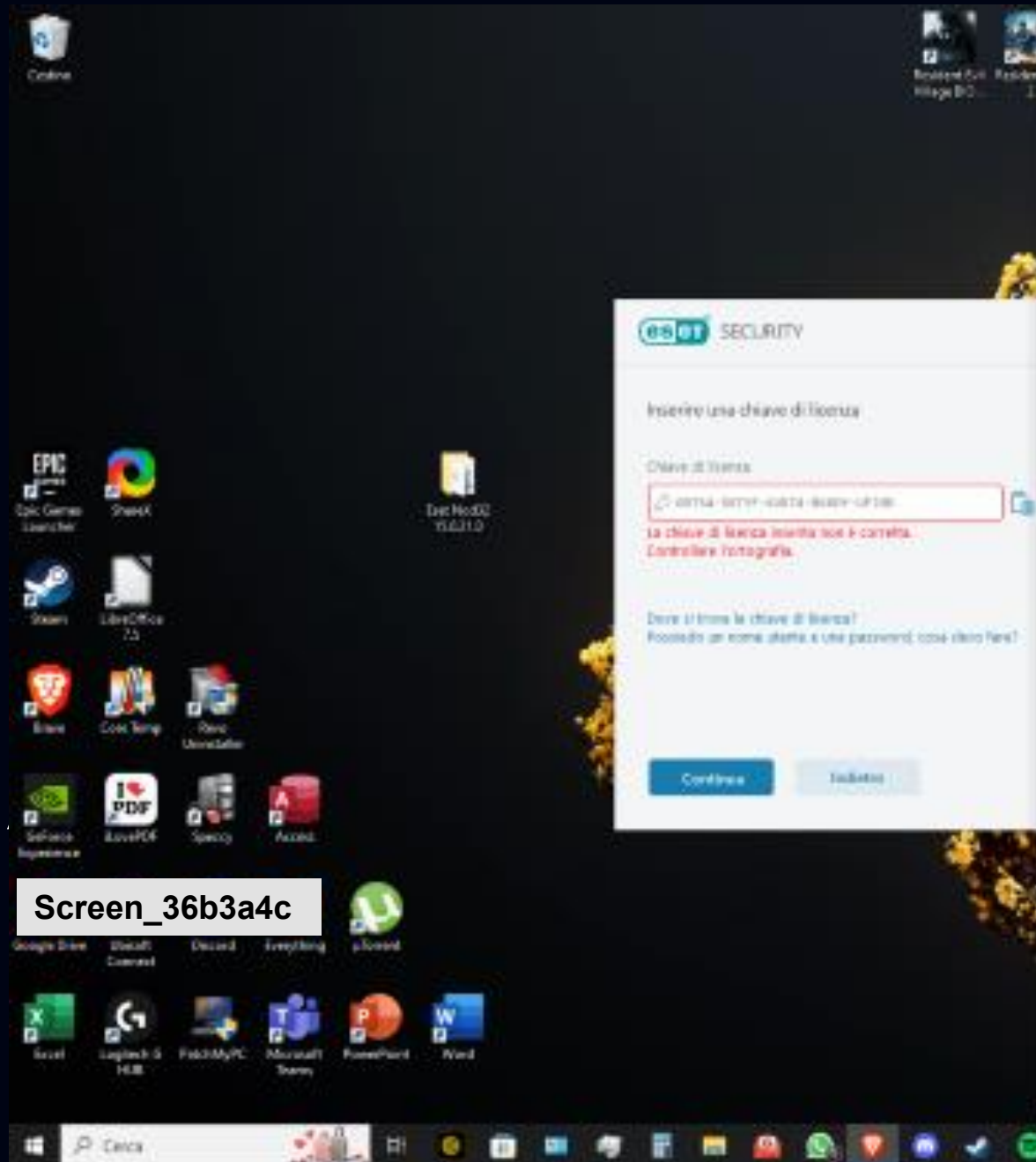
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Suspicious Elements:

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Main Content:

The screenshot displays a computer desktop environment with an ESET Security window on the left side, prompting the user to input a license key. There is a red error message indicating that the entered license key is not correct. On the right side, there is a YouTube video titled "ESET NOD32 ANTIVIRUS CRACK 2023" which claims to offer a free download of a license key for ESET antivirus software. It includes a description with instructions for installation and cautionary steps such as disabling antivirus and Windows Smart Screen.

Files/Programs:

Installer: ESET NOD32 ANTIVIRUS CRACK 2023

File explorer: X

URL

1. <https://www.youtube.com/watch?v=HBG5nZQ7ThA>

Tabs

1. {Youtube} : [Download ESET NOD32 ANTIVIRUS CRACK 2023]
2. {ESET} [License key internet security 100% working]

Suspicious Elements:

- The YouTube video titled "ESET NOD32 ANTIVIRUS CRACK 2023" and the associated download link (<https://cutt.ly/NOD-32>) are highly suspicious, as they suggest accessing cracked software, which typically contains malware.
- The license key entry prompt in the ESET window may indicate that the user is following instructions from the video to illegally activate software.

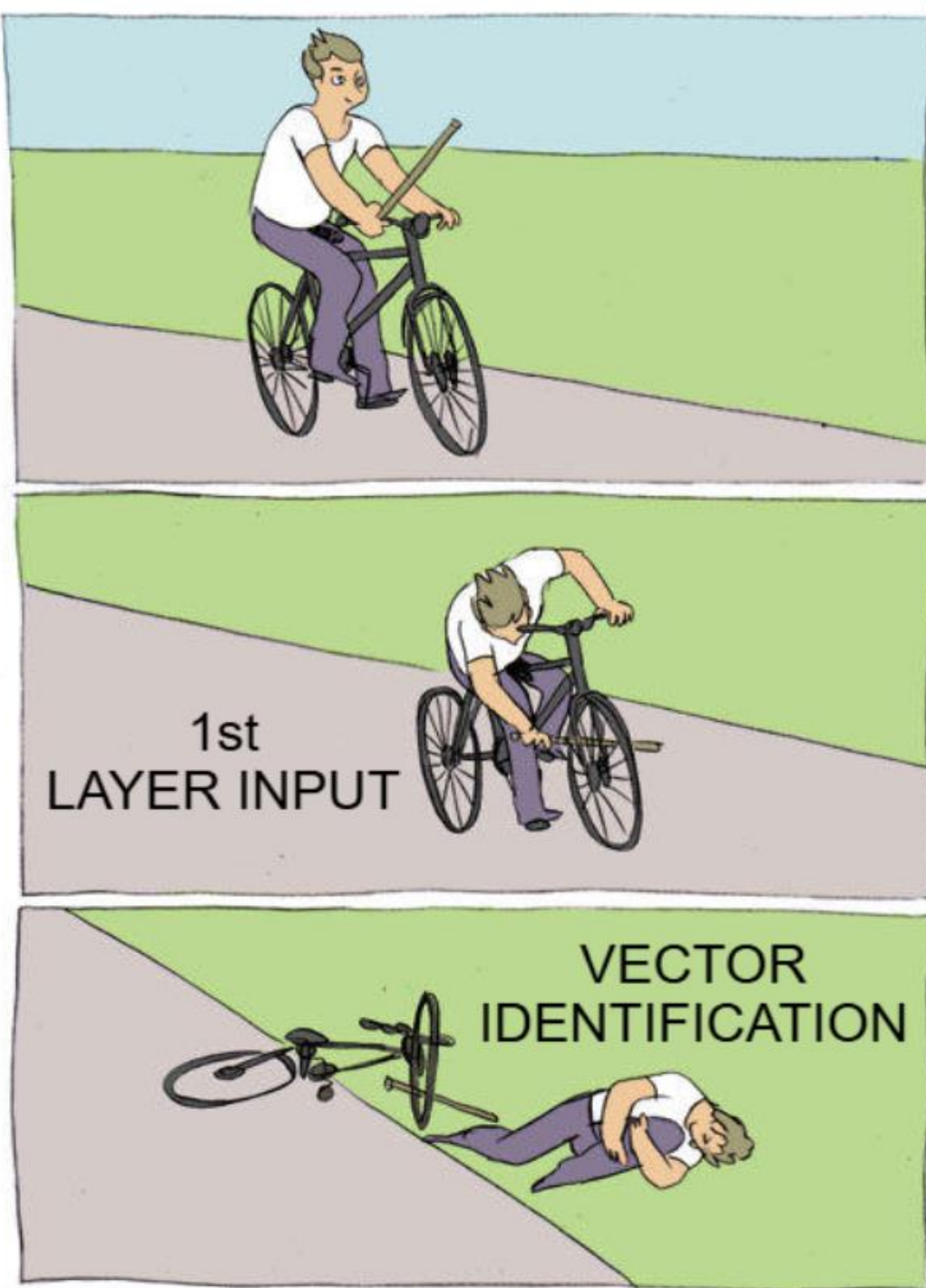
1ST LAYER OUTPUT

The first layer
Prompt

Web
Content

File
System

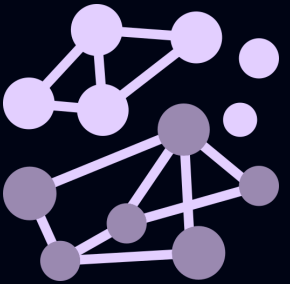
Hybrid



INPUT



Formatted
Description



Vector
Identification

The first layer
Prompt

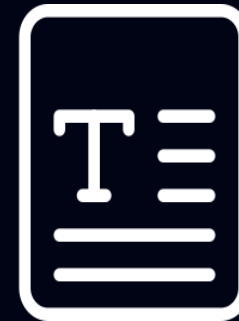
Web
Content

File
System

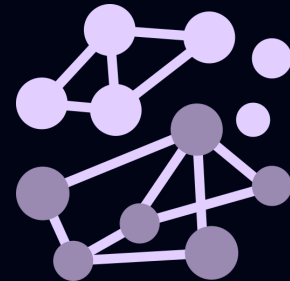
Hybrid



INPUT



Formatted
Description

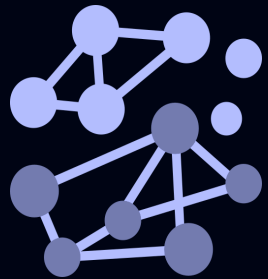


Vector
Identification

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The first layer
Evaluation



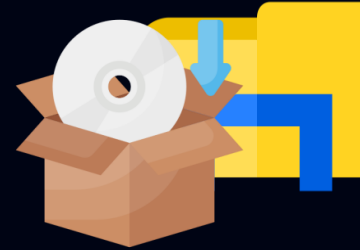
Visual
Assessment

96%



Scene Description

100%



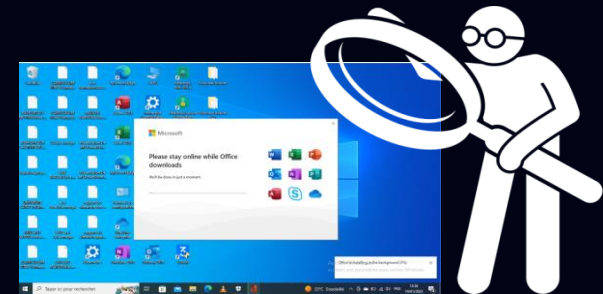
File Explorer & Installer

100%



URL

85%



Suspicious Elements



Browser Tab Identification

The first layer

Evaluation - Browser Tab Identification



30%



32%



36%



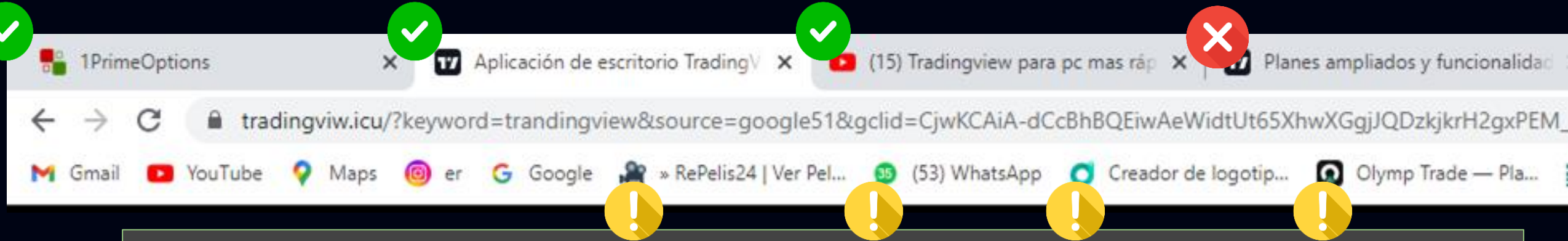
The first layer

Evaluation – Browser Tab Identification – Case Study



The first layer

Evaluation - Browser Tab Identification - Case Study

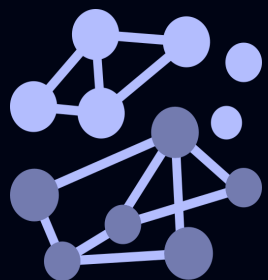


Browser Tabs Analysis:

- [logo: TradingView] [text: Aplicación de escritorio TradingView] (A webpage for the desktop application of TradingView)
- [logo: YouTube] [text: (15) Tradingview para pc mas rápido] (A video related to using TradingView on PC)
- [logo: Google] [text: 1PrimeOptions] (A tab likely related to trading or financial information)
- [logo: WhatsApp] [text: (53)] (Notifications for WhatsApp)
- [logo: Google] [text: RePelis24 | Ver Pel...] (A site for watching movies or series)
- [logo: Google] [text: Creador de logotipo...] (Logo creator link)
- [unknown logo] [text: Olymp Trade - Pla...] (A financial trading platform)

The first layer

Evaluation - Solution



Visual
Assessment



Scene Description



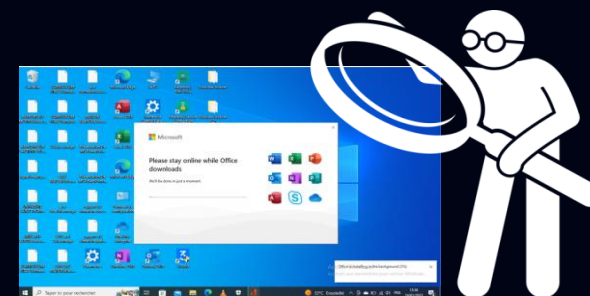
File Explorer & Installer



URL



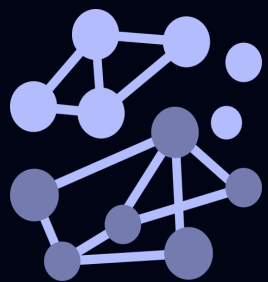
Browser Identification



Suspicious Elements

The first layer

Evaluation - Solution



Visual
Assessment

Sc



http://

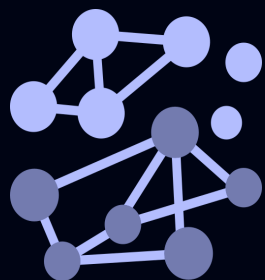
URL



ents

The first layer

Evaluation - Solution



Visual
Assessment



Scene Description



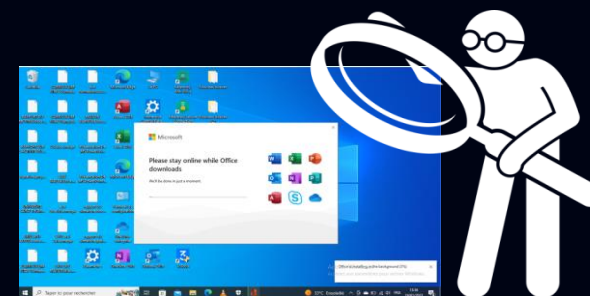
File Explorer & Installer



URL



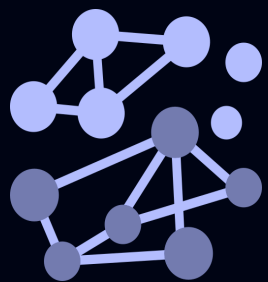
Browser Identification



Suspicious Elements

The first layer

Evaluation - Solution



Visual
Assessment

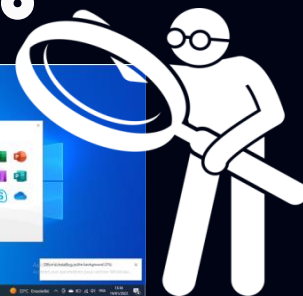


& Installer



URL

85%



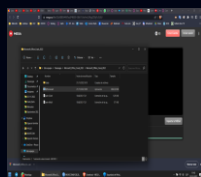
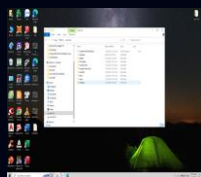
Suspicious Elements

The first layer
Evaluation

Web
Content

File
System

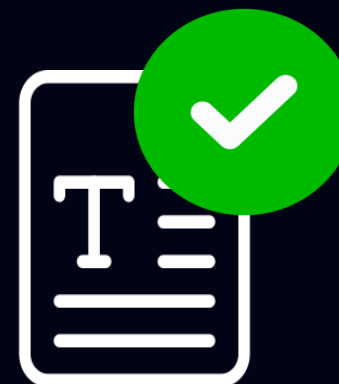
Hybrid



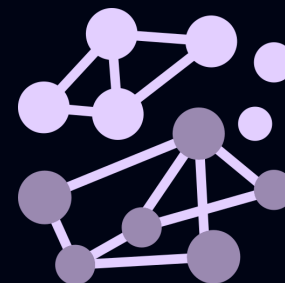
Visual
Assessment



Formatted
Description

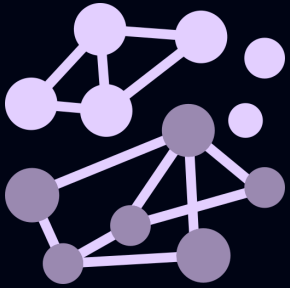


2nd LLM
Layer

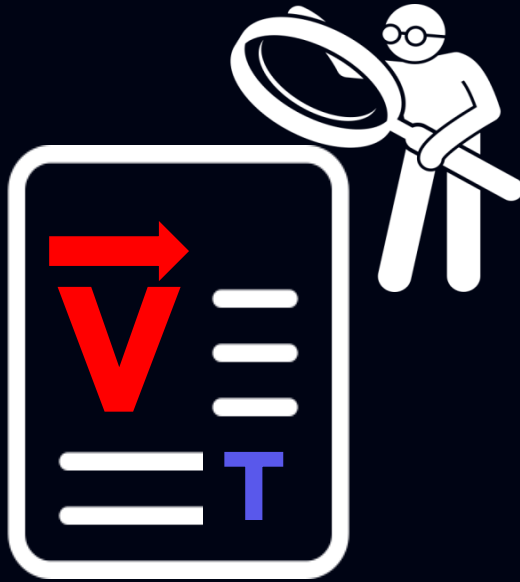


THE 2ND LAYER

THE LLM LAYERS



Vector identification
+ formatting



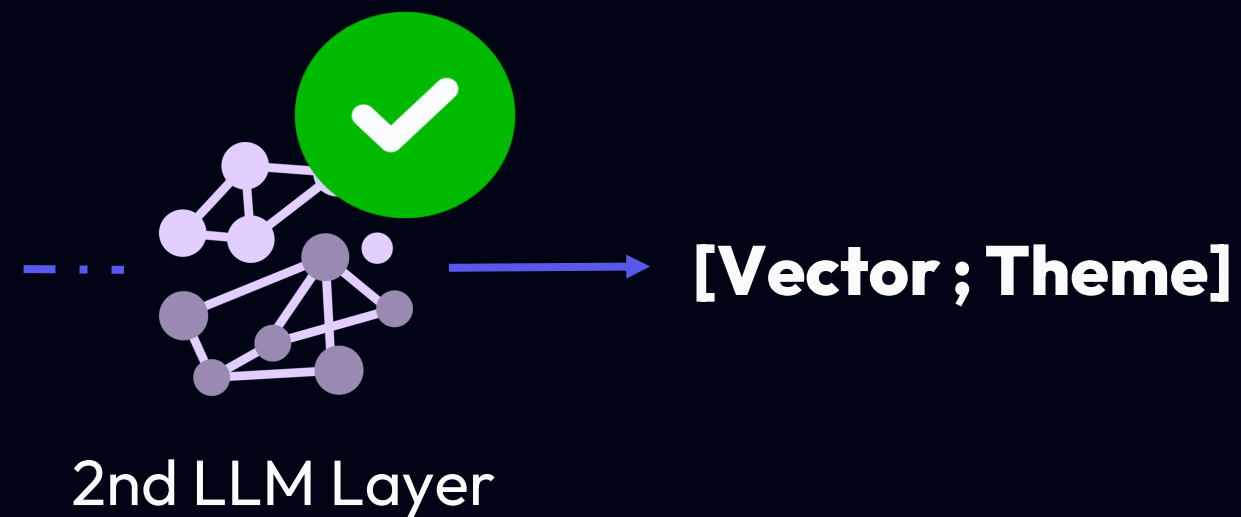
[**V**ector; **T**heme]



<https://mega.nz/folder/GEkRCKaT#f93dJ6myfe3fENhDS4wqxQ;>
[KMSAuto++ v1.8.7 for Microsoft product activation](#)

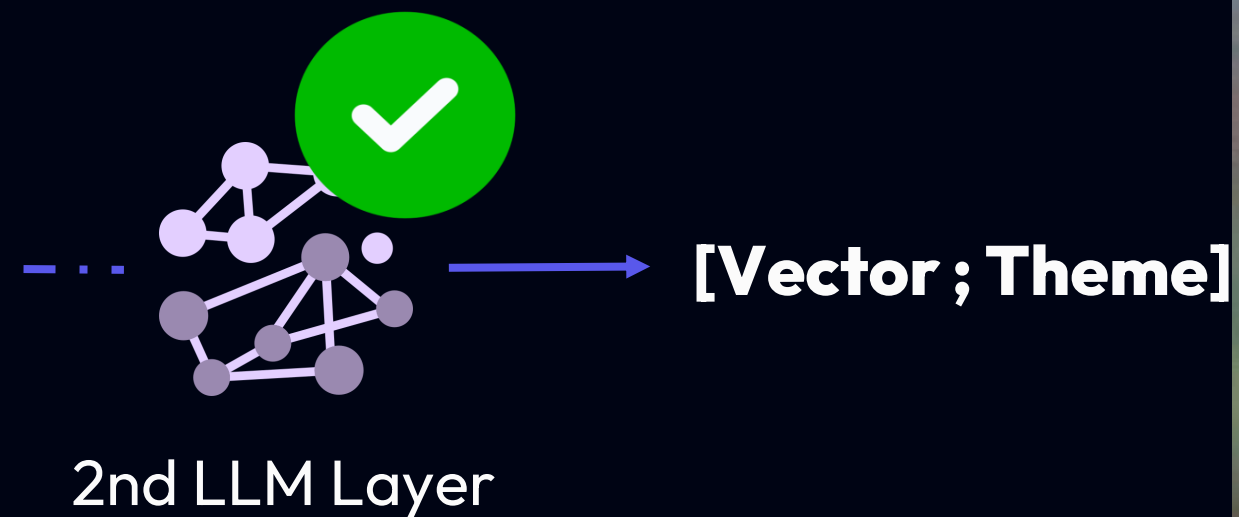
Getting out the trash

Discriminating IoCs



Getting out the trash

Discriminating IoCs



Getting out the trash

Discriminating IoCs

<https://gameshpc.com/2021-teu-aio-full-espanol/>

IoC - <https://mediafire.com/file/hD54wxq...>

<https://mediafire.com/file/hD54wxq...>

2nd L

lor

<https://zdescarga.com/>

(me)

DEAD

RIES

<https://elamigosgames.com/>

[LicenseKey/](#)

windows/descargar

Getting out the trash

Discriminating loC

<https://github.com/loC>

loC - <https://github.com/loC>



<https://mediafire.com/>

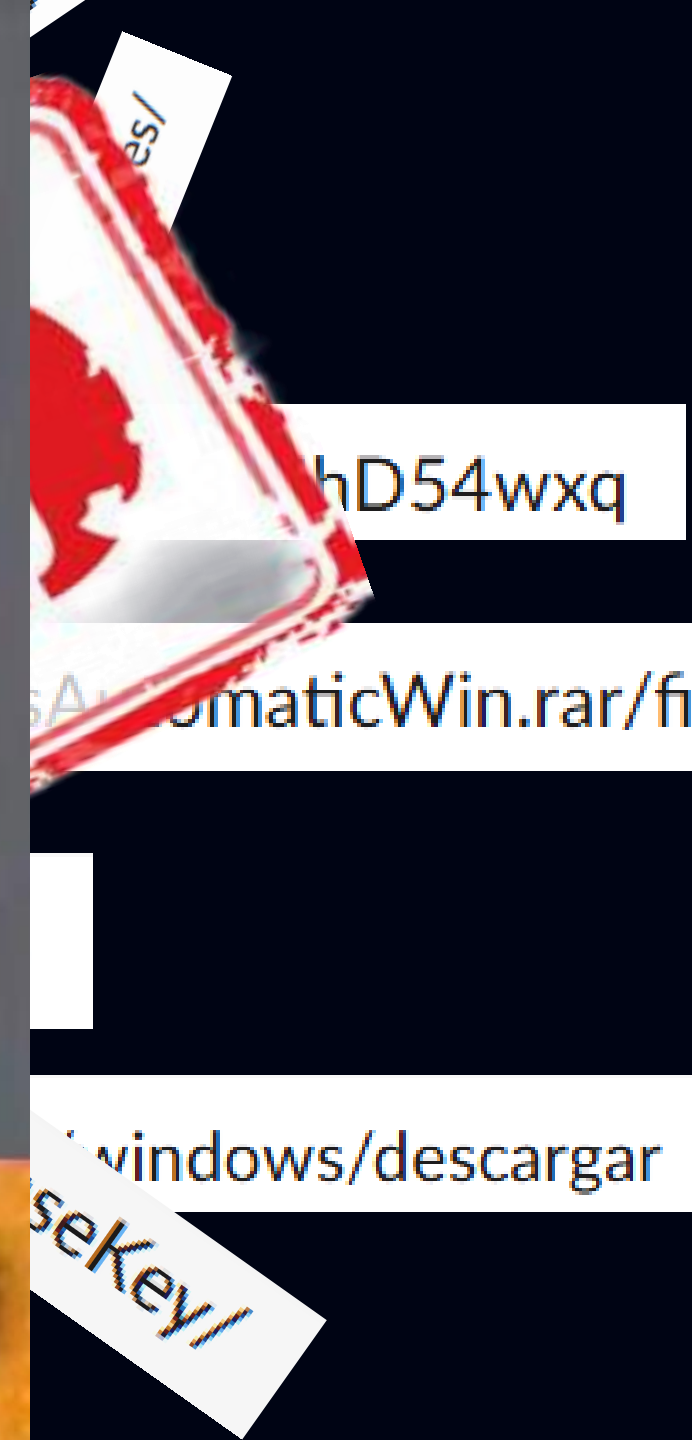
2nd L

lor

<https://zdescarga.com/>

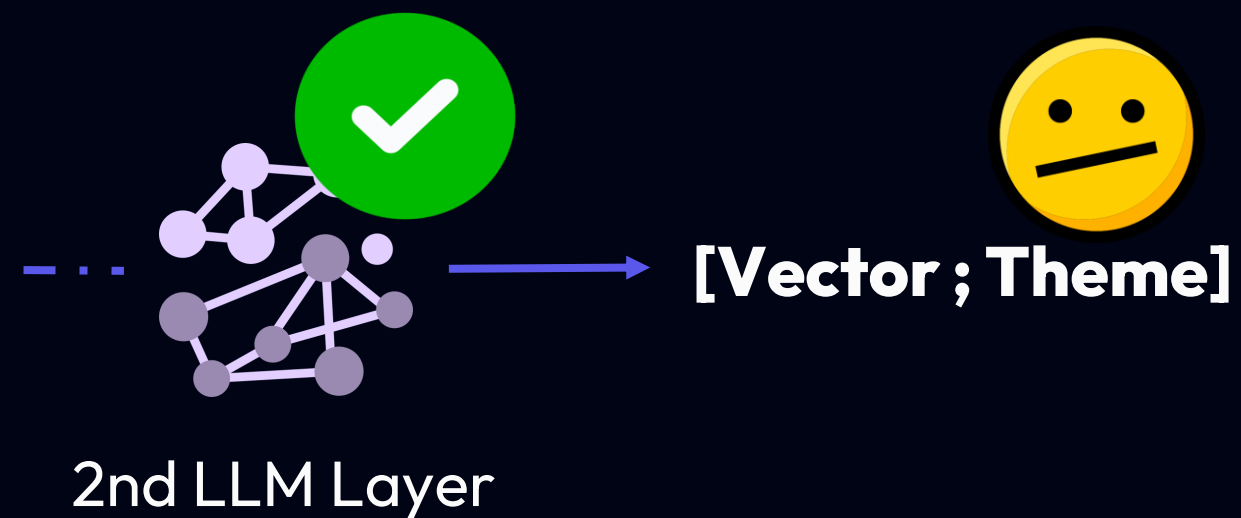


TRIGGERED



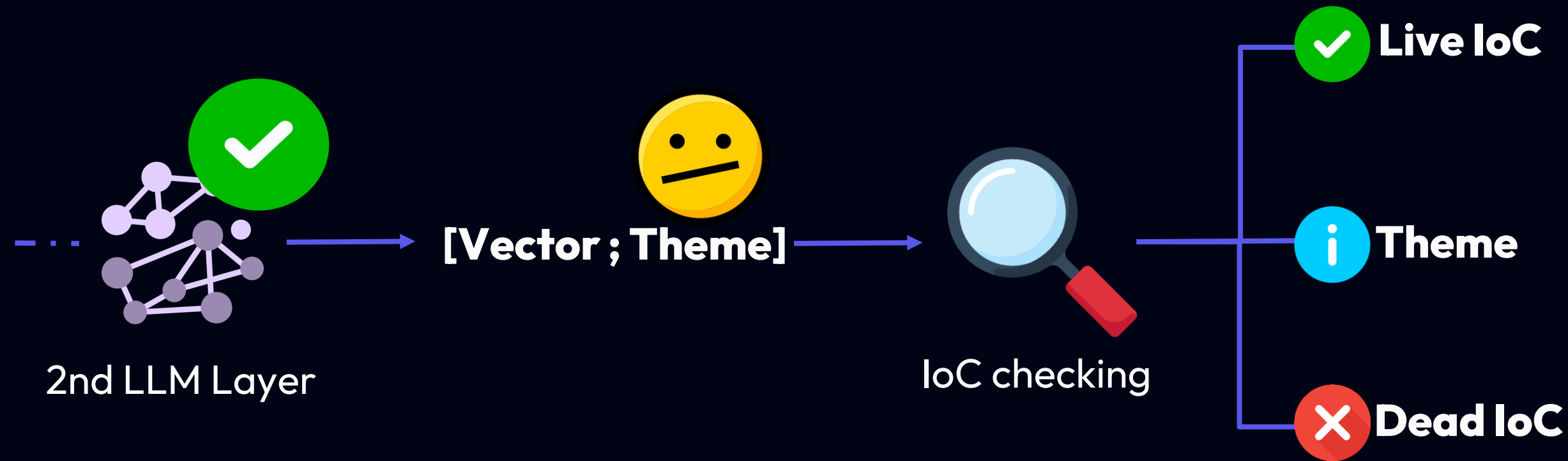
Getting out the trash

Discriminating IoCs



Getting out the trash

Discriminating IoCs



Agenda

1. The Information Stealer Malware Phenomenon

2. Mid-Heist Selfies

3. The LLM Pipeline

4. Prompt Engineering

5. LLM Assessment

6. Discriminating IoCs

7. Inside the Infostealer Playbook

8. Successful Campaigns: 2 Case Studies

9. Strength and Limits

10. Conclusion

Getting out the trash

Discriminating IoCs



IoC checking



File sharing platforms



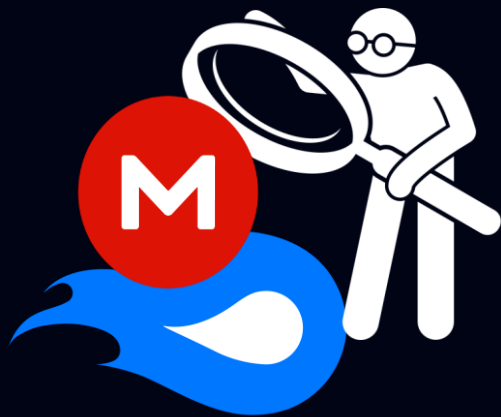
YouTube Videos



Others

Getting out the trash

Discriminating IoCs



IoC checking



The file you are trying to download is no longer available

- This link is unavailable as the user's account has been closed for gross violation of MEGA's [Terms of Service](#).



The file you are trying to download is no longer available

This could be due to the following reasons:

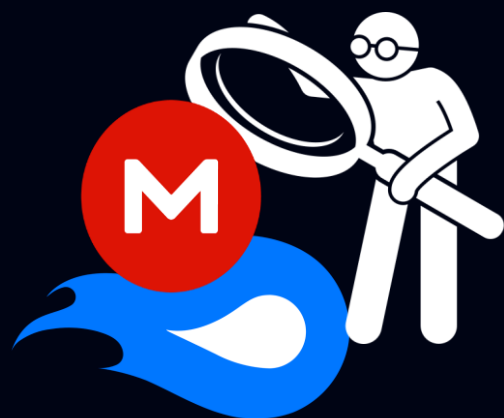
- The file has been removed as it violated our Terms of Service.
- Invalid URL - the link you are trying to access does not exist.
- The file has been deleted by the user.



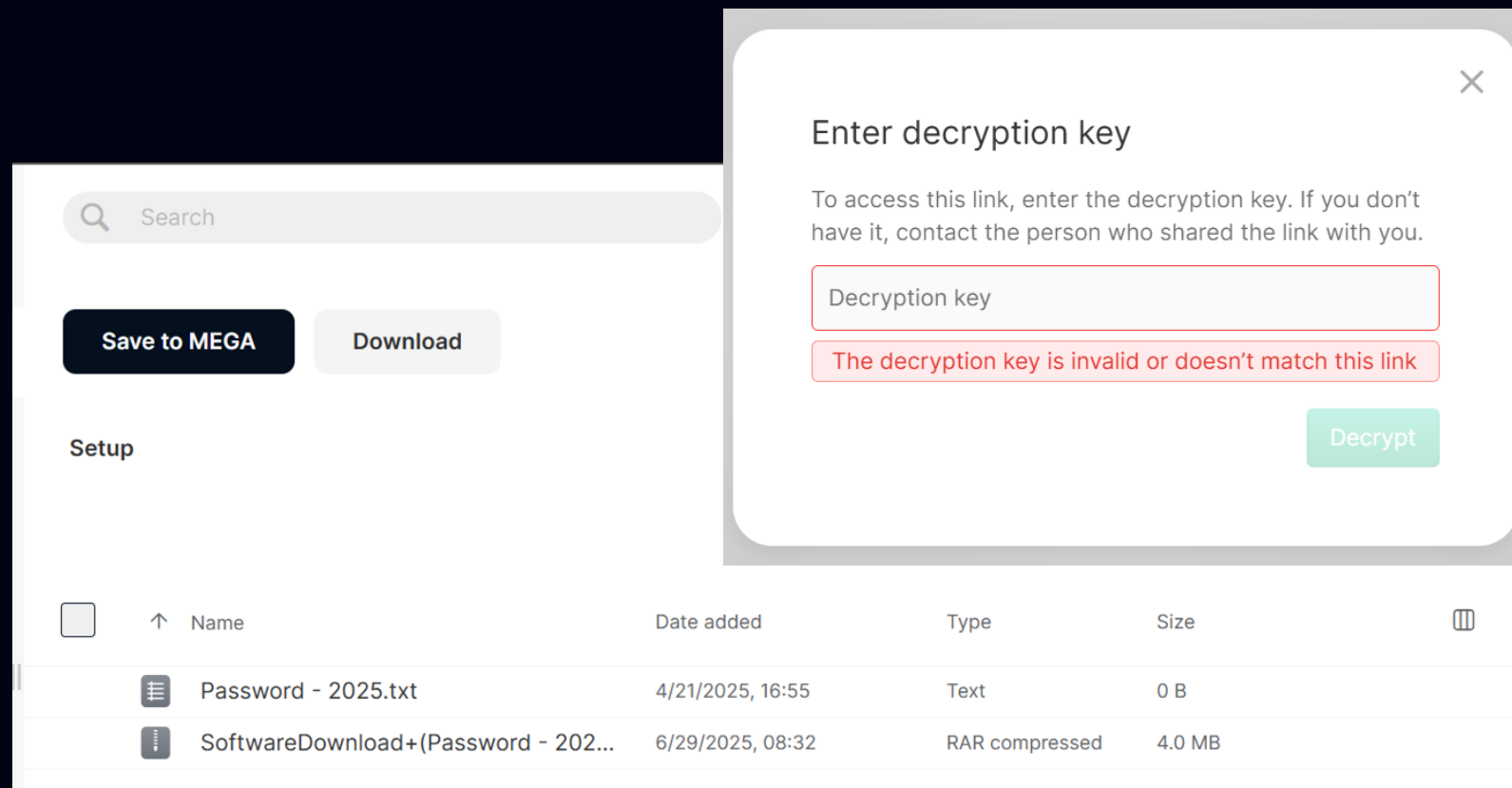
Dead IoC

Getting out the trash

Discriminating IoCs



IoC checking



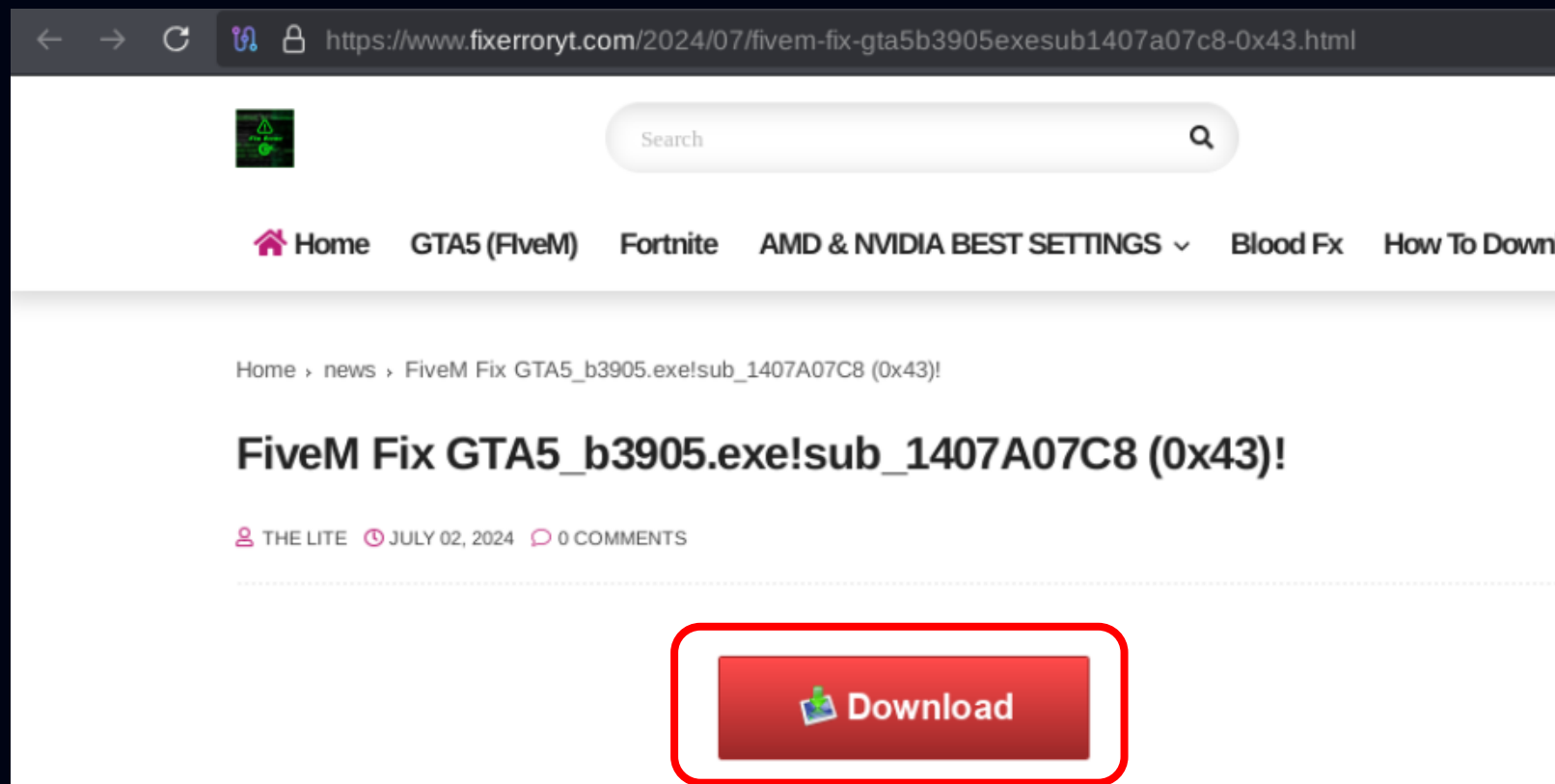
Live IoC

Getting out the trash

Discriminating IoCs



IoC checking



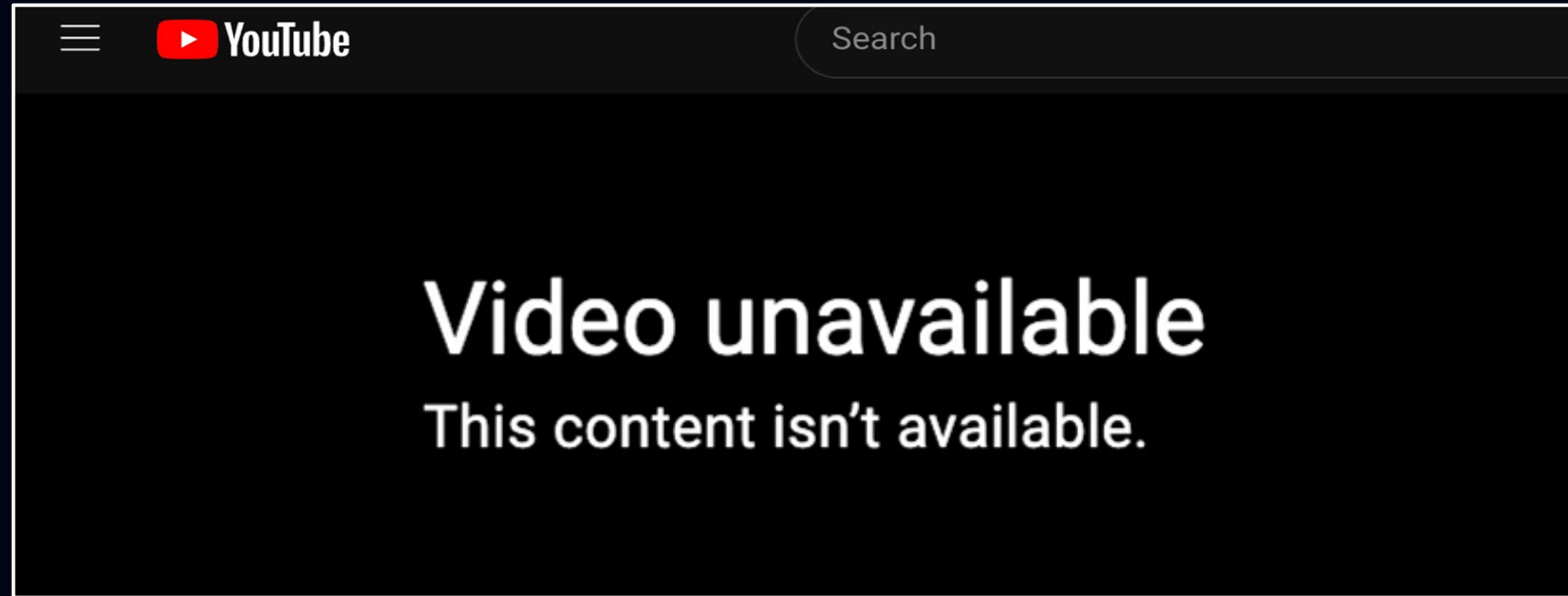
Live IoC

Getting out the trash

Discriminating IoCs



IoC checking



Dead IoC

Getting out the trash

Discriminating IoCs



IoC checking

Microsoft Office Crack Free Download Full Version 2022



V4ctor

S'abonner

33 vues 31 oct. 2022

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ARCHIVE PASSWORD: 7521

DOWNLOAD LINK (DIRECT LINK): <https://bit.ly/3N8nHDp>



Live IoC

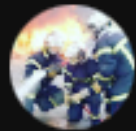
Getting out the trash

Discriminating IoCs



IoC checking

Microsoft Office Crack Free Download Full Version 2022



V4ctor

S'abonner

33 vues 31 oct. 2022

Welcome! Leave a LIKE and SUBSCRIBE if you enjoyed this video!

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Theme

DEMO

python -m fetch_scre ~/f/r/s/stealerlogs_ioc_feed

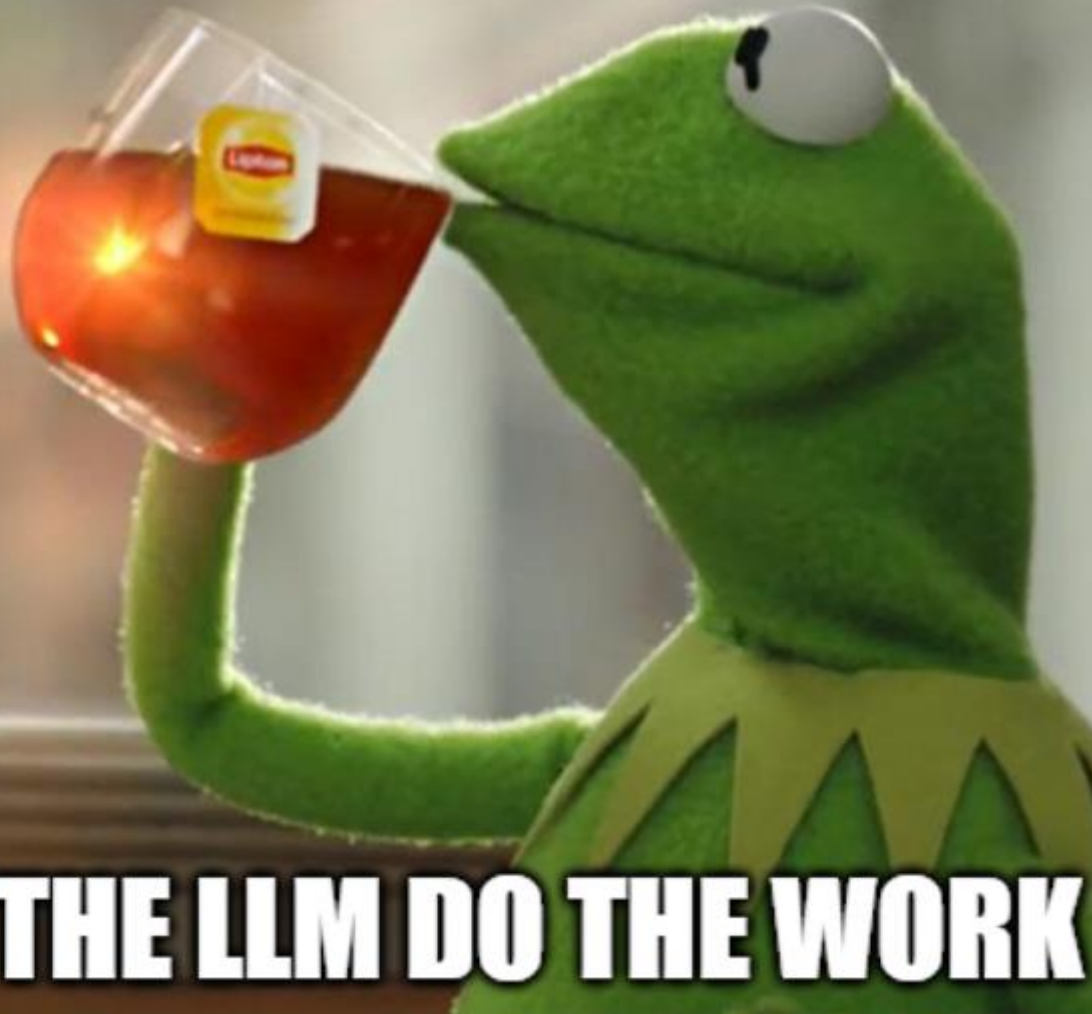
python -m fetch_scre ~/f/r/s/stealerlogs_ioc_feed

~/f/r/s/stealerlogs_ioc_feed

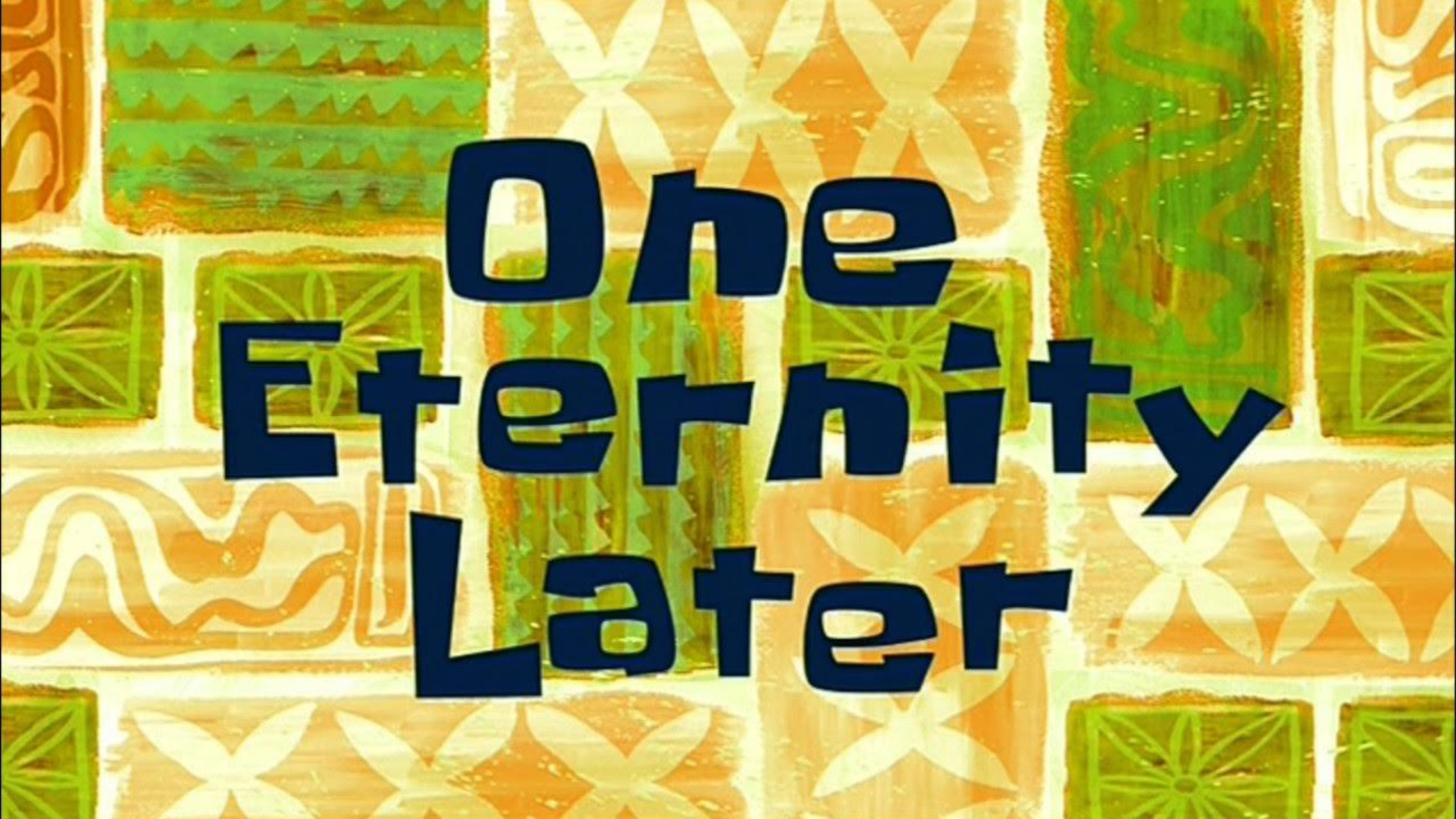
python -m fetch_scre ~/f/r/s/stealerlogs_ioc_feed 118x29

```
(stealerlogs-ioc-feed) obilodeau@sparkle ~/f/r/s/stealerlogs_ioc_feed (main)> python -m fetch_screens.refactored_pipeline.fetch_analyze --download-screens --open-screens --openai-response --delay 10
```

NOW



LET THE LLM DO THE WORK



**One
Eternity
Later**

Agenda

1. The Information Stealer Malware Phenomenon
2. Mid-Heist Selfies
3. The LLM Pipeline
4. Prompt Engineering
5. LLM Assessment
6. Discriminating IoCs
7. Inside the Infostealer Playbook
8. Successful Campaigns: 2 Case Studies
9. Strength and Limits
10. Conclusion

Lure Themes - Cracked Software

Lure Themes - Cracked Software



\$\$\$



Office
Cracked

0\$

Lure Themes



Paying \$150
for lifetime access
to a secure,
legitimate license
from a globally
recognized corporation.



Free_Microsoft_Office_Crack.rar



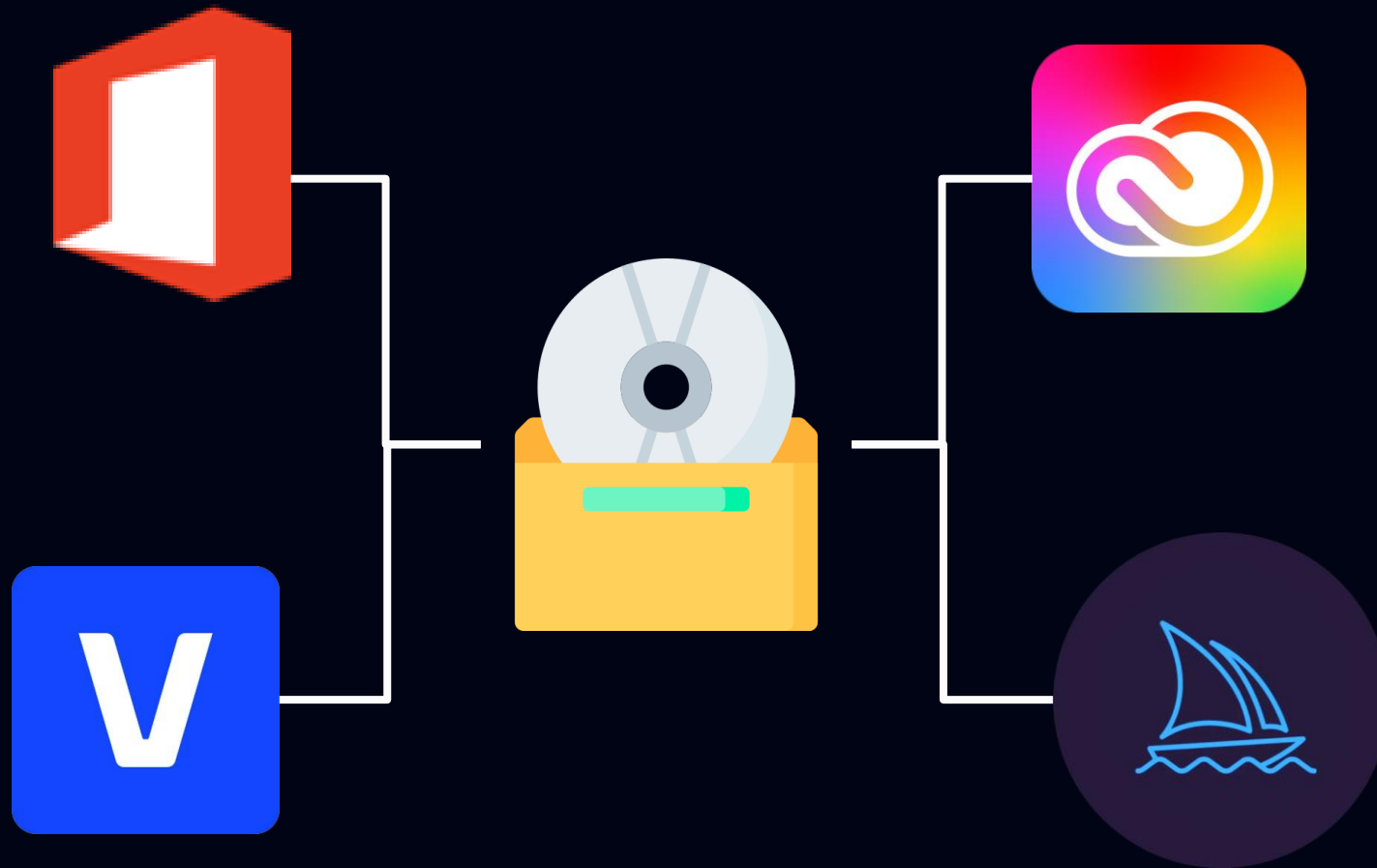
ce
ked

Lure Themes - Cracked Software

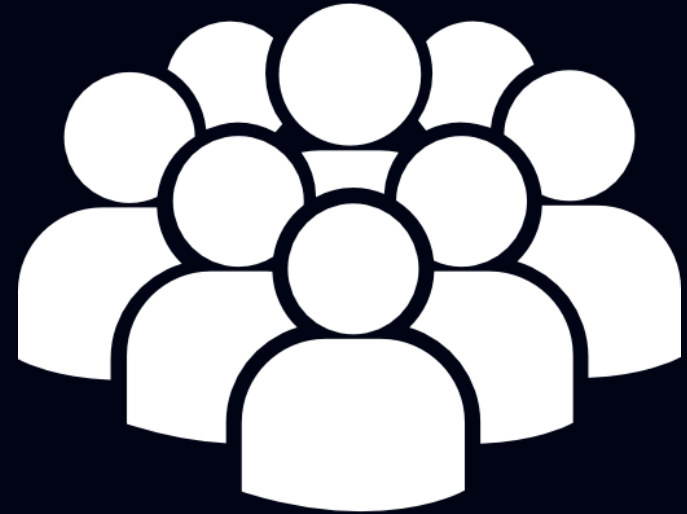


Threat actors **prey** on users' **willingness** to **bypass** legitimate licensing **fees** at the **cost** of their own **security**

Lure Themes - Cracked Software



Lure Themes - Cracked Software



MAINSTREAM

Infostealer Playbook - Lure Themes

Targeting mainstream products ensures
large pool of potential victims

Lure Themes - Gaming Cheats & Mods

Lure Themes - Gaming Cheats & Mods



FORTNITE



Lure Themes - Gaming Cheats & Mods



VALORANT

**VALORANT
TRIGGERBOT
HACK**



FORTNITE



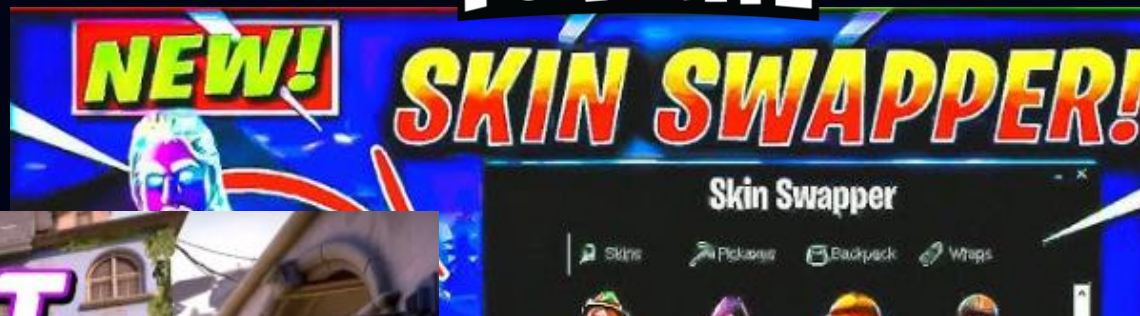
Lure Themes - Gaming Cheats & Mods



VALORANT

**VALORANT
TRIGGERBOT
HACK**

FORTNITE



MINECRAFT

Lure Themes - Gaming Cheats & Mods



Skins



Weapons



Mods

Lure Themes

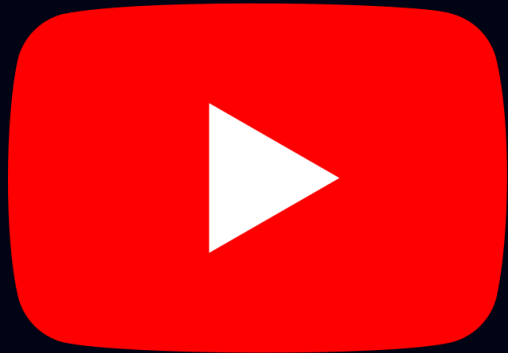
If it's free and shady, you're likely the
victim

Distribution Strategies – **YouTube as a Distribution System**

The background of the slide features a complex, abstract pattern. It consists of numerous small, light blue dots arranged in a grid-like fashion, which is slightly distorted to create a sense of depth and movement. Overlaid on this dot pattern are several wavy, translucent lines in shades of blue and purple. These lines flow from the bottom left towards the top right, creating a dynamic, organic feel. The overall color palette is cool, dominated by blues and purples, set against a dark, almost black background.

Distribution Strategies – YouTube as a Distribution System

👑 BEST FORTNITE HACKS 2025 | AIM LIKE A PRO! FREE & EASY SETUP | 100% SAFE ✅



🔥 BEST FREE FORTNITE CHEAT 2025 |
SILENT AIM + ESP | UNDETECTED 🛡️

🔥 MICROSOFT 2022 CRACK (GENUINE + 100% FREE) 🔥

Distribution Strategies – YouTube as a Distribution System

👑 BEST FORTNITE HACKS 2025 | AIM LIKE A PRO! FREE & EASY SETUP | 100% SAFE ✓

SEEMS LEGIT



100% SAFE



22 CRACK (GENUINE + 100% FREE) 🔥

SHGZC



Download the archive
link in the c



DOWNLOAD LINK:

<https://telegra.ph/FORTNITE-H4CK-FREE-DOWNLOAD-09...>



ARCHIVE PASSWORD: shgzchea



FORTNITE HACK | UNDETECTED | FORTNITE MOD MENU | DOWNLOAD FREE



SS&B

78 subscribers

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17



Share



Download



Save

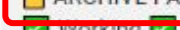


267 views 8 Dec 2022

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ARCHIVE PASSWORD: shgzchea



working

Instructions:

- 1) Download the archive.
- 2) Unzip the file to any folder/desktop.

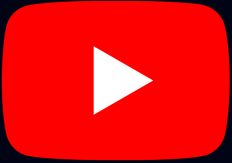


Instructions:

- 1) Download the archive.
- 2) Unzip the file to any folder/desktop.

Screen_011e67e8

Distribution Strategies – YouTube as a Distribution System



It is free

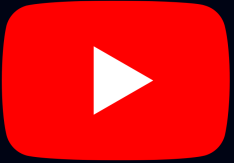


It works



**Disabling antivirus is
needed & safe**

Distribution Strategies – YouTube as a Distribution System



It is free



It works



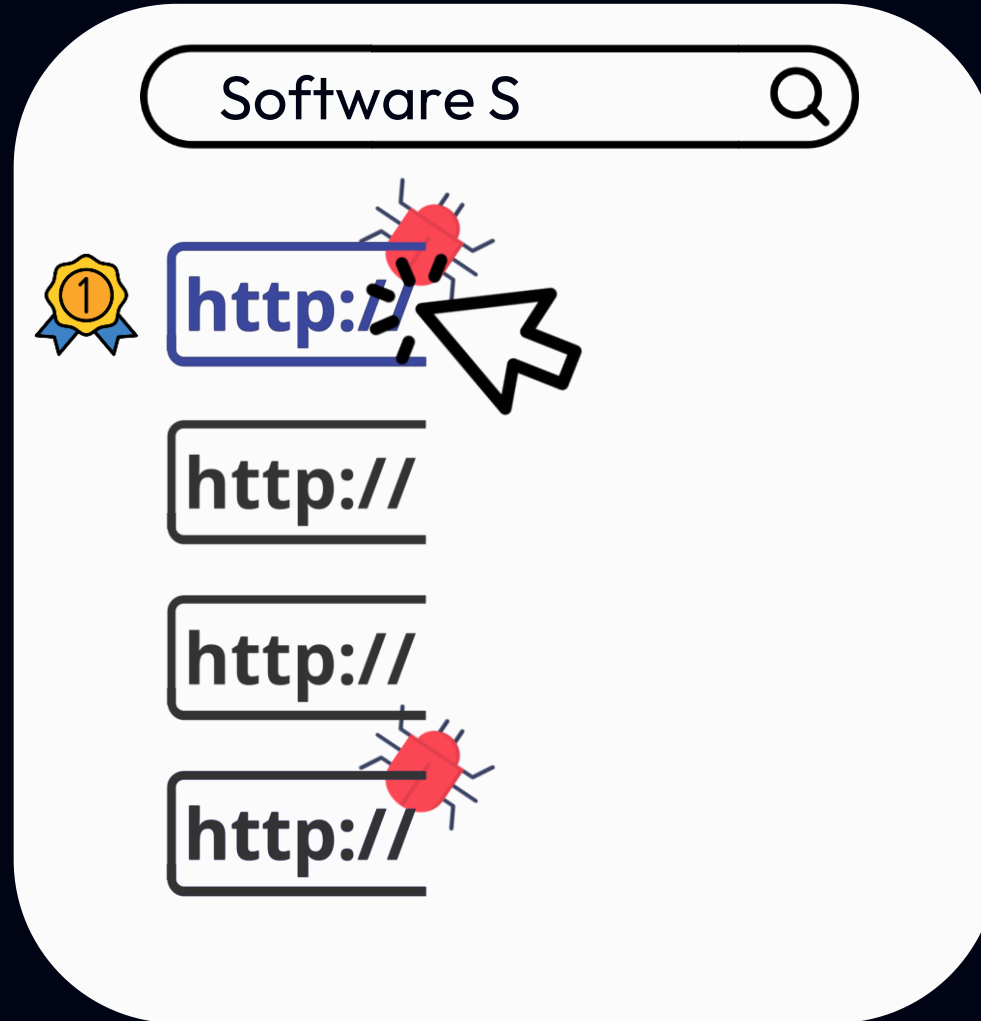
**Disabling antivirus is
needed & safe**

Distribution Strategies - YouTube as a Distribution System

YouTube's reach and tutorial-driven-content make it the perfect launchpad for infostealer malware

Distribution Strategies - Google Ads

Distribution Strategies - Leveraging Google Ads



Distribution Strategies – Leveraging Google Ads

Google Ads give threat actors a fast lane to users' trust — by placing malicious content where users expect safety: at the top.

Agenda

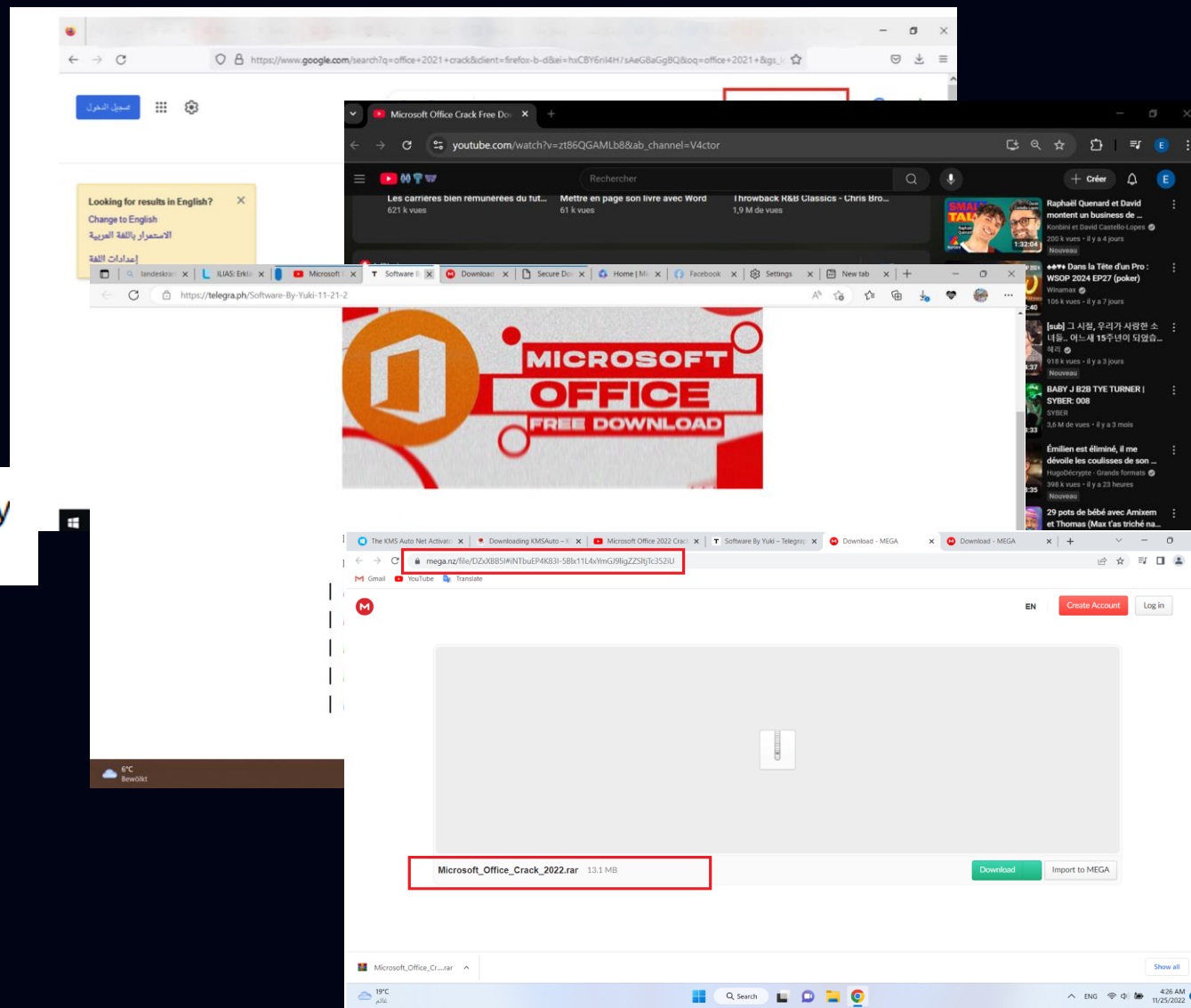
1. The Information Stealer Malware Phenomenon
2. Mid-Heist Selfies
3. The LLM Pipeline
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5. LLM Assessment
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8. Successful Campaigns: 2 Case Studies
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10. Conclusion

Successful Campaigns: 3 2 Case Studies

IoC - <https://mega.nz/folder/GErKCTaT#9dGmye>
Theme - Microsoft 2022

IoC - <https://mega.nz/folder/GErKCTaT#9dGmye>
Theme - Microsoft 2022 Cracked Free

IoC - <https://mega.nz/folder/GErKCTaT#9dGmye>
Theme - Yuki Microsoft 2022 DOWNLOAD



Successful Campaigns: 32 Case Studies



MidJourney

6.3%



Blitz Java

5.3%



MidJ0urney



Blitz Java



midjourney prompts with results

midjourney when blending with two text prompts, what do you put between them.

midjourney what are some of the best user prefer option set examples

midjourney ai

midjourney bot

midjourney discord

midjourney v5

midjourney api

midjourney free

midjourney v4

Google Search

I'm Feeling Lucky

Report inappropriate predictions

Screen_0b3c24da

Environ 39 800 000 résultats (0)



Sponsorisé

ai.mid-journey.org
https://ai.mid-journey.org

Get The Latest Updates - MidJourney

Comprehensive Tutorials on Working with **MidJourney**. Exclusive Updates and Features. Subscribe To our **MidJourney** Course.

Recherches associées

- midjourney image
- midjourney bot
- midjourney ai
- midjourney #macron
- midjourney gratuit
- midjourney how to use
- midjourney discord
- midjourney prix

Midjourney
https://midjourney.com · Traduire cette page

Midjourney

Midjourney is an independent research lab exploring new mediums of thought and expanding the imaginative powers of the human species.



Midjourney

Midjourney est un laboratoire de recherche indépendant qui produit un programme d'intelligence artificielle sous le même nom et qui permet de créer des images à partir de descriptions textuelles, suivant un fonctionnement similaire à celui de DALL-E d'OpenAI. [Wikipédia](#)

Créateur : Midjourney

Première version : 2022

Screen_0a4ffca2



This is an open beta version of the program,

it is possible that the computer's security systems may falsely trigger, which is a common problem for all software that is in beta testing. The open beta version is only available on Windows.

It is possible that the computer's security systems may FALSELY trigger

FAQ.

- How do I make a request using MidJourney's AI? +
- What stock images can I find on MidJourney +
- How can I use MidJourney's images? +

Screen_1c086e2



SEEMS LEGIT



MidJourney 64-bit

/month

MidJourney's AI-powered Images!

for Windows

beta version of the program
gger, which is a common problem for all software that is in beta testing.
is only available on Windows.



It is possible that the computer's security systems may FALSELY trigger

AQ.

How can I use MidJourney's images?

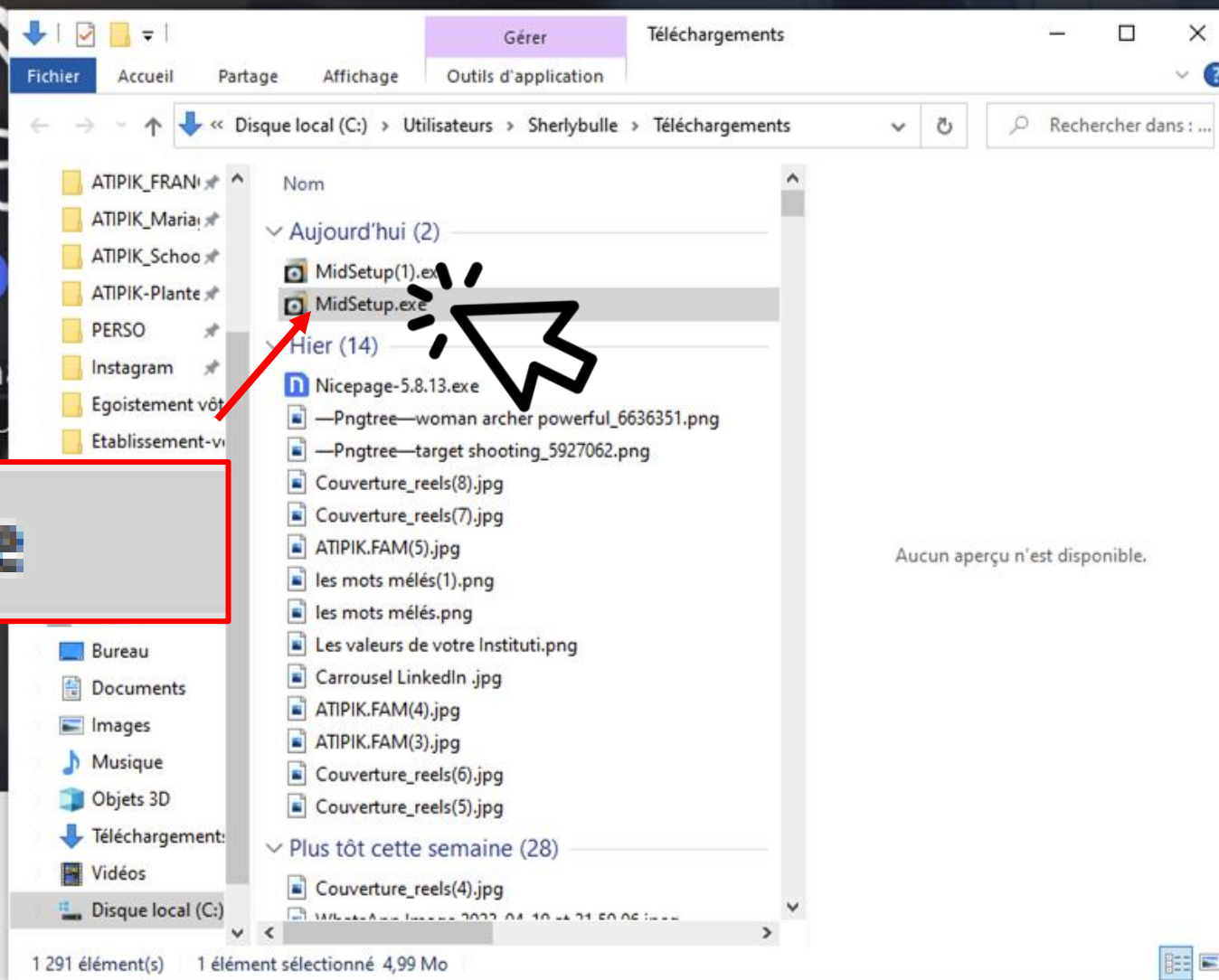
Screen_1c086e2

MidJourney

Innovative AI-powered program that
to generate stu

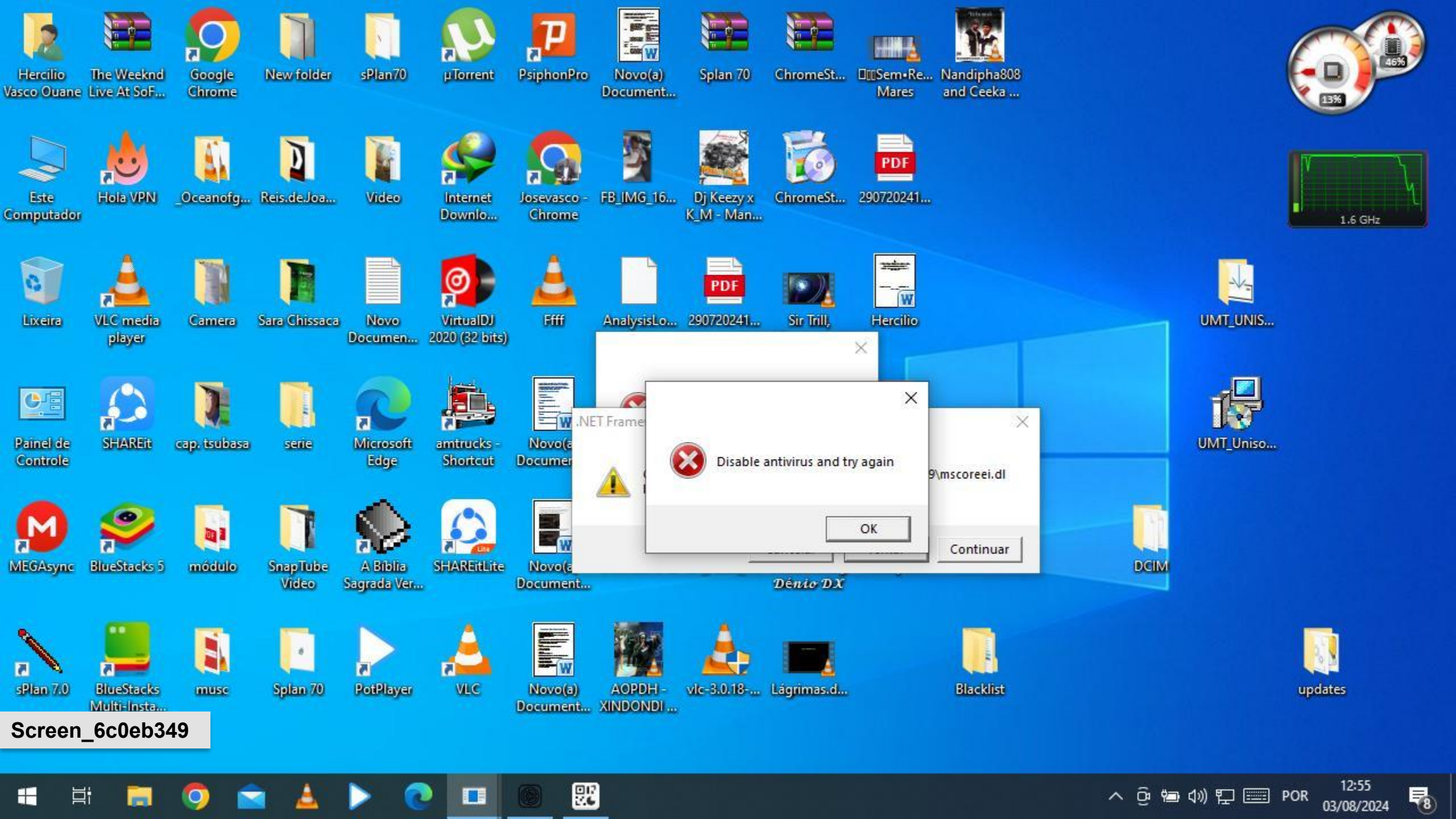


MidSetup.exe

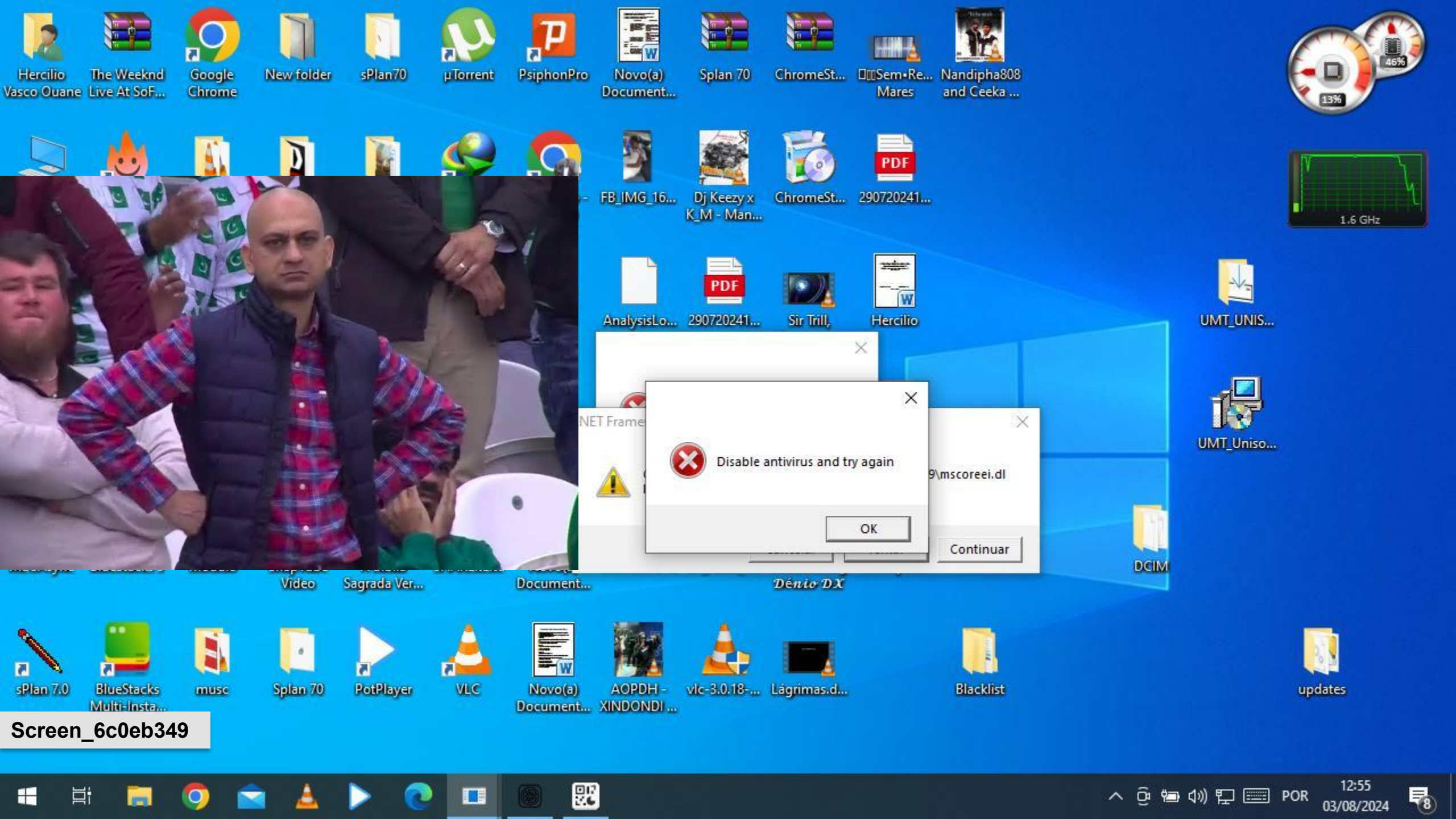


Screen_2d6e7a1f

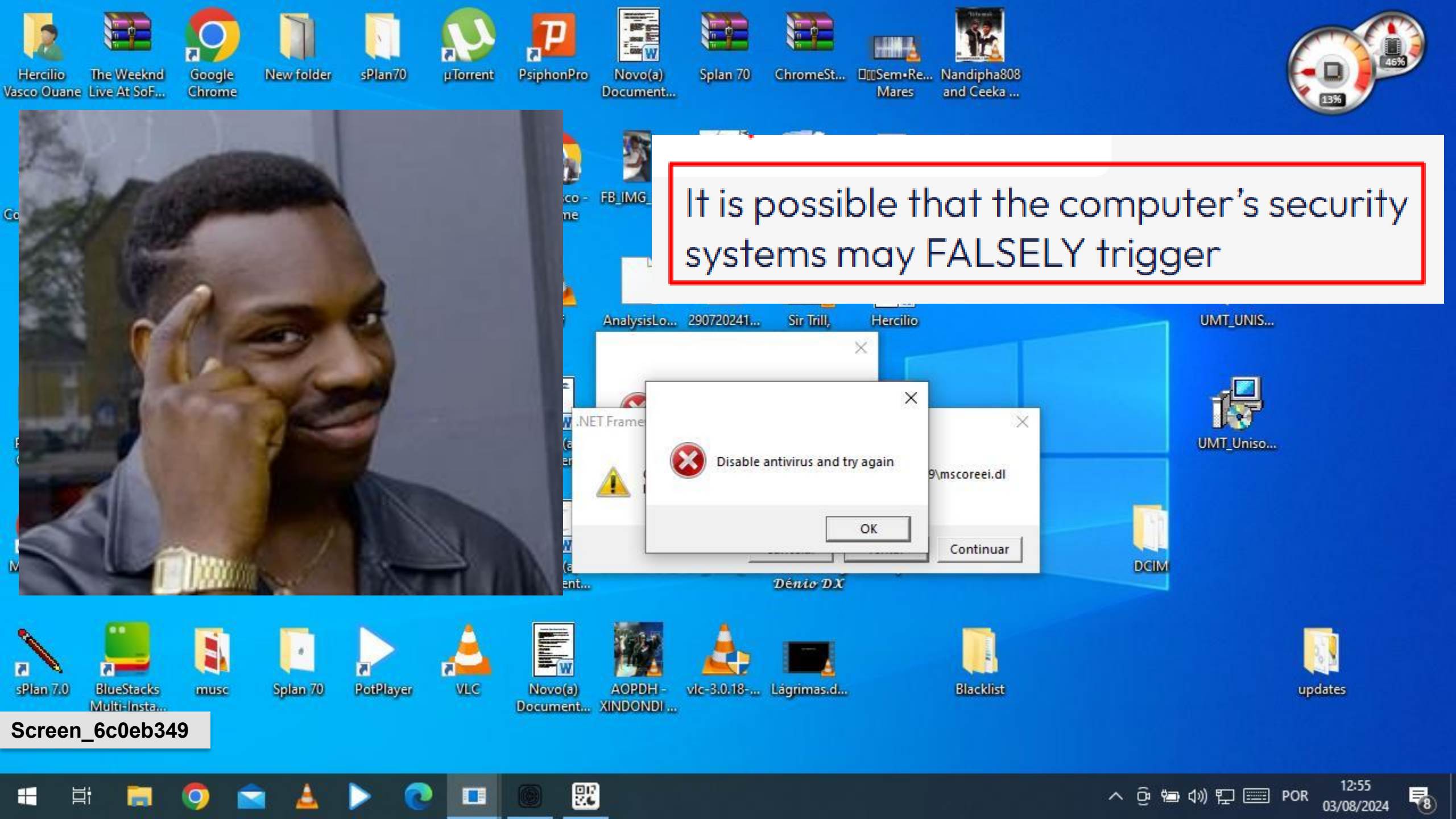
Why MidJourney?



Screen_6c0eb349



Screen_6c0eb349



It is possible that the computer's security systems may FALSELY trigger

Disable antivirus and try again

OK

Continuar

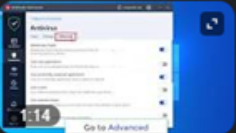
Screen_6c0eb349

12:55
03/08/2024

← → ↺ google.com/search?q=how+to+disable+bitdefender+antivirus+2023&ei=Syu_Y63QEI-crgSU7be4Cw&ved=0ahUKEwjtofWSvMD8AhUPJosKHZT2DbcQ4dUDCA8&uact=5&oq=how+to+disable+bitdefender+antivirus+2023&gs_lcp=... Schemavisare - Sko... CLIP STUDIO PAINT... YouTube Panzoid skola MP3 Downloader O... Blueprint Tracker ... rust NiceHash - Largest... https://arma3projec... MPGH - MultiPlayer... iSpace Mining Pool... Miscreated CVAR H... Övriga bokmärken

Google

how to disable bitdefender antivirus 2023



This video will show you how to temporary **disable or enable** ...
How to Temporarily **Disable or Enable Bitdefender TOTAL** ...
YouTube · Nam Anh Cap · 25 jan. 2022

Saknas: 2023 | Måste innehålla: 2023

https://www.youtube.com › watch

how to temporarily disable bitdefender 2023 - YouTube



ايداف برنامج **Bitdefender 2023** | كيف تعمل مؤقتا
... how to temporarily **disable bitdefender 2023** ...
بت ديفيندر ايداف
YouTube · MR BNA · För 1 månad sedan

https://clean-my-pc.com › how-to... › Översätt den här sidan

How to Disable Bitdefender Windows 10 - Clean my PC

Open the **Bitdefender Total Security** 2019 on your device. · Navigate to the Protection Features tab and click on the Settings icon under the Antivirus module.

https://www.prajwal.org › how-to... › Översätt den här sidan

How to Disable Bitdefender Notifications - Prajwal.org

17 dec. 2022 — Launch the **Bitdefender antivirus** or Total security tool. · Select Settings and click General tab. · **Turn off** the Special Offers and Recommended ...

https://www.safetydetectives.com › ... › Översätt den här sidan

How to Cancel Bitdefender Subscription (& Get a Refund) in ...

Find **Bitdefender** under your list of products and click **Stop** automatic subscription renewal. ... I recommend Norton — it's my favorite **antivirus** in 2023, ...
★★★★★ Ranking: 9/10 · 13 recensioner

Översättade sökningar

uninstall bitdefender bitdefender alert page

Screen_6cf1bf72

22:36
2023-01-11

⚙️ Virus & threat protection settings

View and update Virus & threat protection settings for Microsoft Defender Antivirus.

This setting is managed by your administrator.

Real-time protection

Locates and stops malware from installing or running on your device. You can turn off this setting for a short time before it turns back on automatically.



Cloud-delivered protection

Provides increased and faster protection with access to the latest protection data in the cloud. Works best with Automatic sample submission turned on.



Automatic sample submission

Send sample files to Microsoft to help protect you and others from potential threats. We'll prompt you if the file we need is likely to contain personal information.

⚠️ Automatic sample submission is off. Your device may be vulnerable. [Dismiss](#)



[Submit a sample manually](#)

Tamper Protection

Prevents others from tampering with important security features.

⚠️ Tamper protection is off. Your device may be vulnerable. [Dismiss](#)



Have a question?

[Get help](#)

Help improve Windows Security

[Give us feedback](#)

Change your privacy settings

View and change privacy settings for your Windows 10 device.

[Privacy settings](#)

[Privacy dashboard](#)

[Privacy Statement](#)

Screen_6ddb217b

⚙️ Virus & threat protection settings

View and update Virus & threat protection settings for Microsoft Defender Antivirus.

This setting is managed by your administrator.

Real-time protection

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Have a question?

[Get help](#)

Help improve Windows Security

[Give us feedback](#)

Change your privacy settings

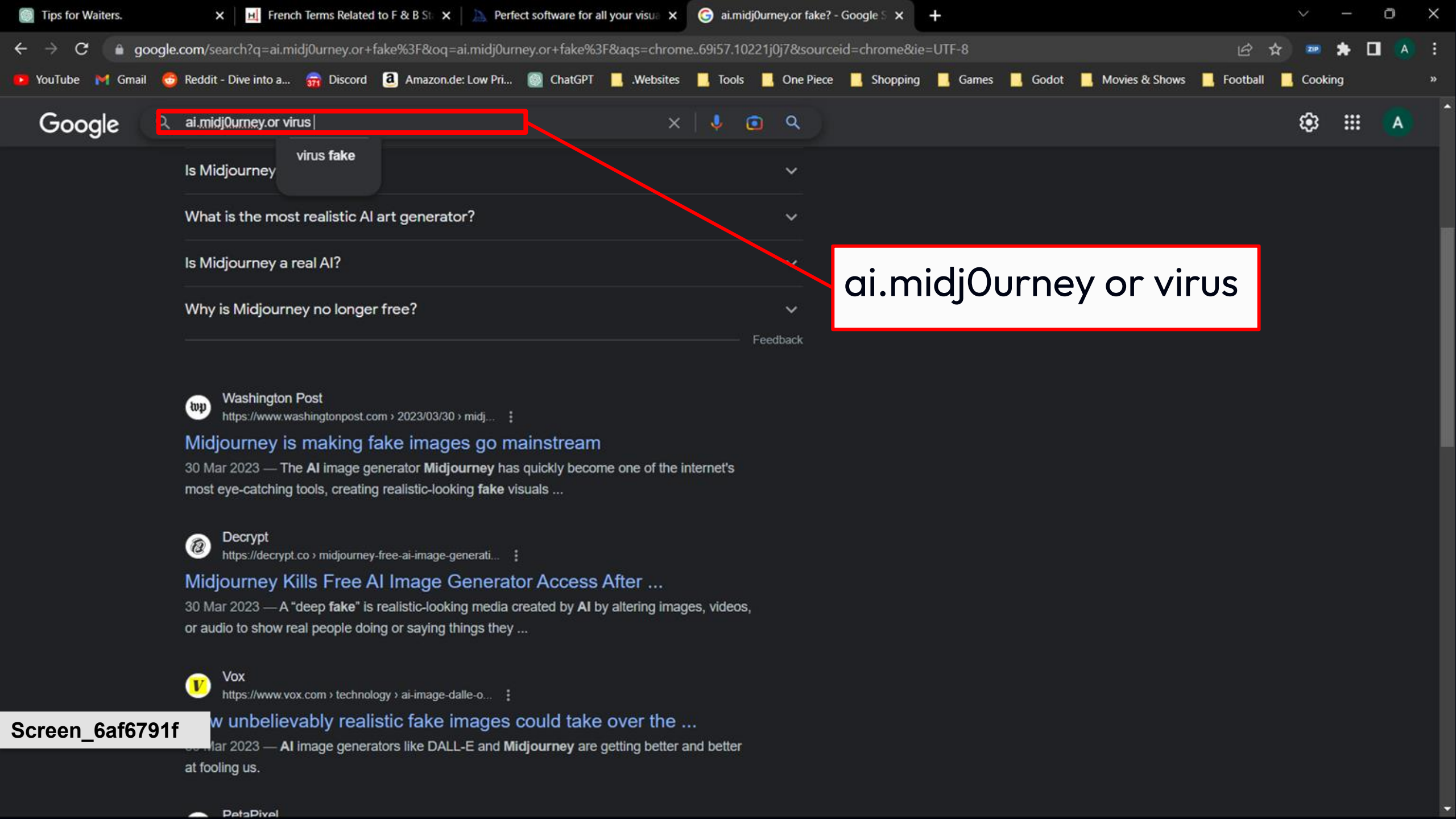
View and change privacy settings for your Windows 10 device.

[Privacy settings](#)

[Privacy dashboard](#)

[Privacy Statement](#)

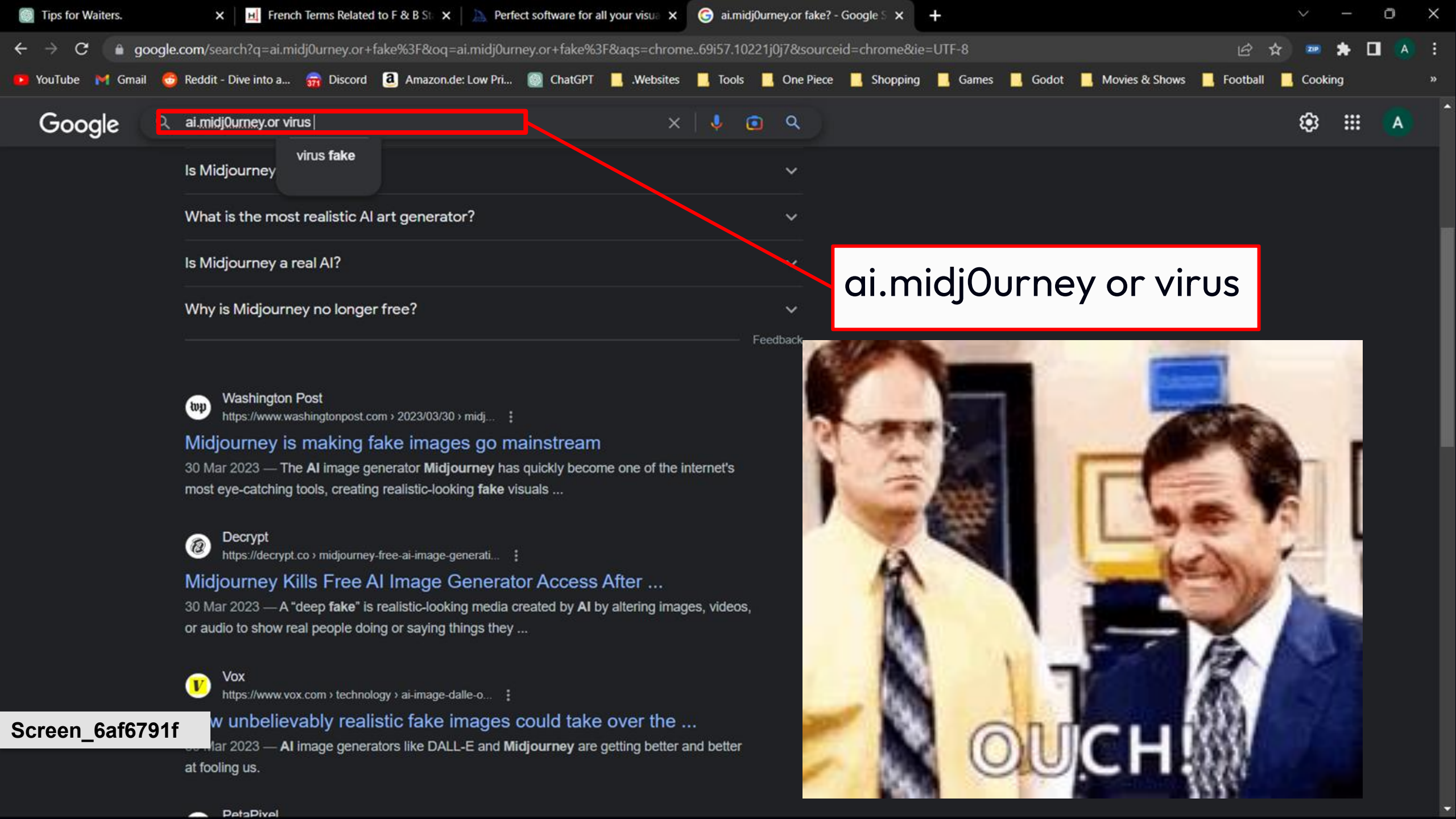
Screen_6ddb217b



ai.midj0urney.or virus

ai.midj0urney or virus

Screen_6af6791f



Google

ai.midj0urney or virus

Search icons: X, microphone, image, magnifying glass

Settings, App Drawer, Profile A

virus fake

Is Midjourney

What is the most realistic AI art generator?

Is Midjourney a real AI?

Why is Midjourney no longer free?

Feedback



Washington Post

<https://www.washingtonpost.com> › 2023/03/30 › midj...

Midjourney is making fake images go mainstream

30 Mar 2023 — The AI image generator Midjourney has quickly become one of the internet's most eye-catching tools, creating realistic-looking fake visuals ...



Decrypt

<https://decrypt.co> › midjourney-free-ai-image-generati...

Midjourney Kills Free AI Image Generator Access After ...

30 Mar 2023 — A "deep fake" is realistic-looking media created by AI by altering images, videos, or audio to show real people doing or saying things they ...



Vox

<https://www.vox.com> › technology › ai-image-dalle-o...

How unbelievably realistic fake images could take over the ...

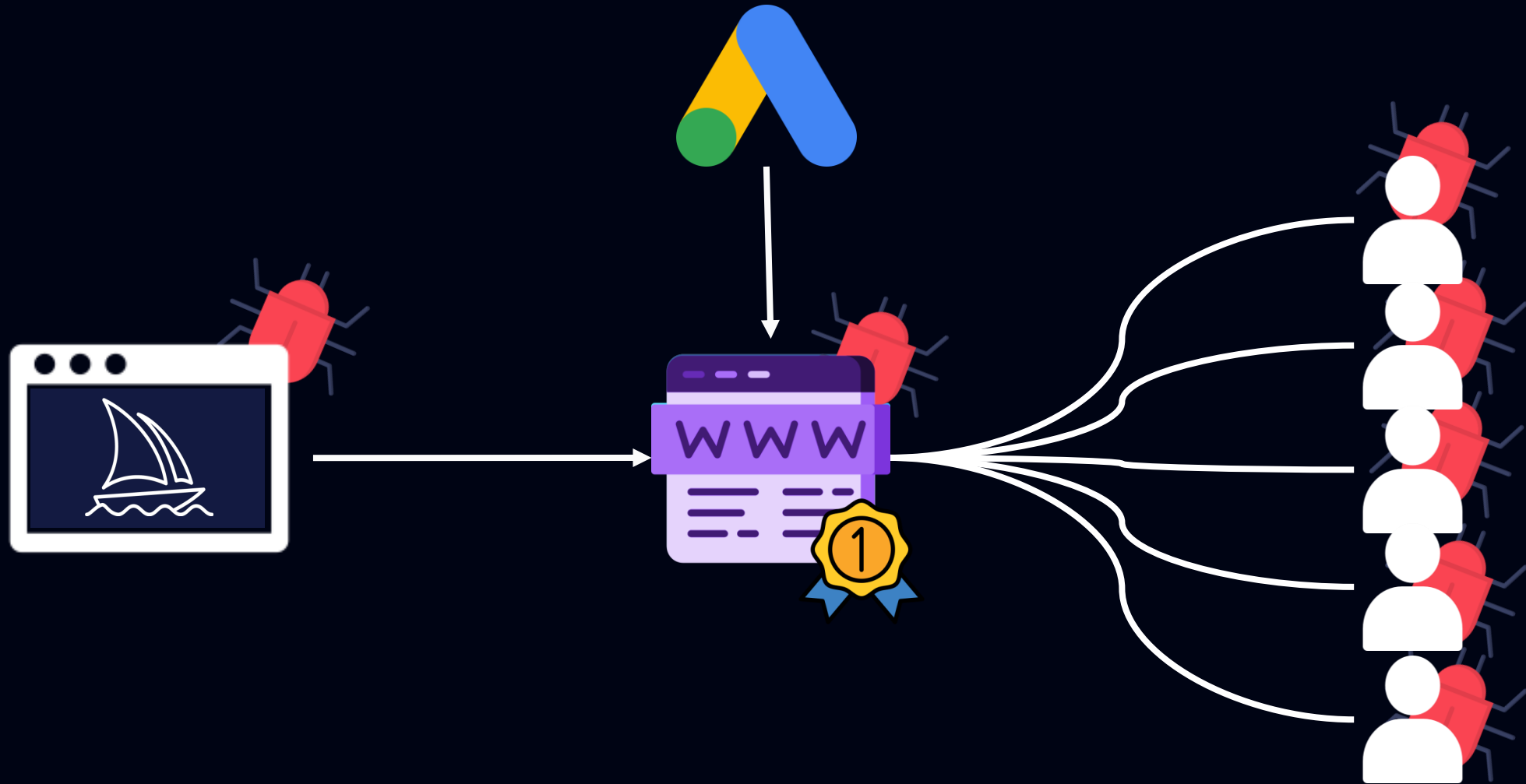
30 Mar 2023 — AI image generators like DALL-E and Midjourney are getting better and better at fooling us.

ai.midj0urney or virus



Screen_6af6791f

MidjØurney Campaign





MidJourney



Blitz Java



Battle.net



World of Warcraft



Epic Games Launcher



Fall Guys



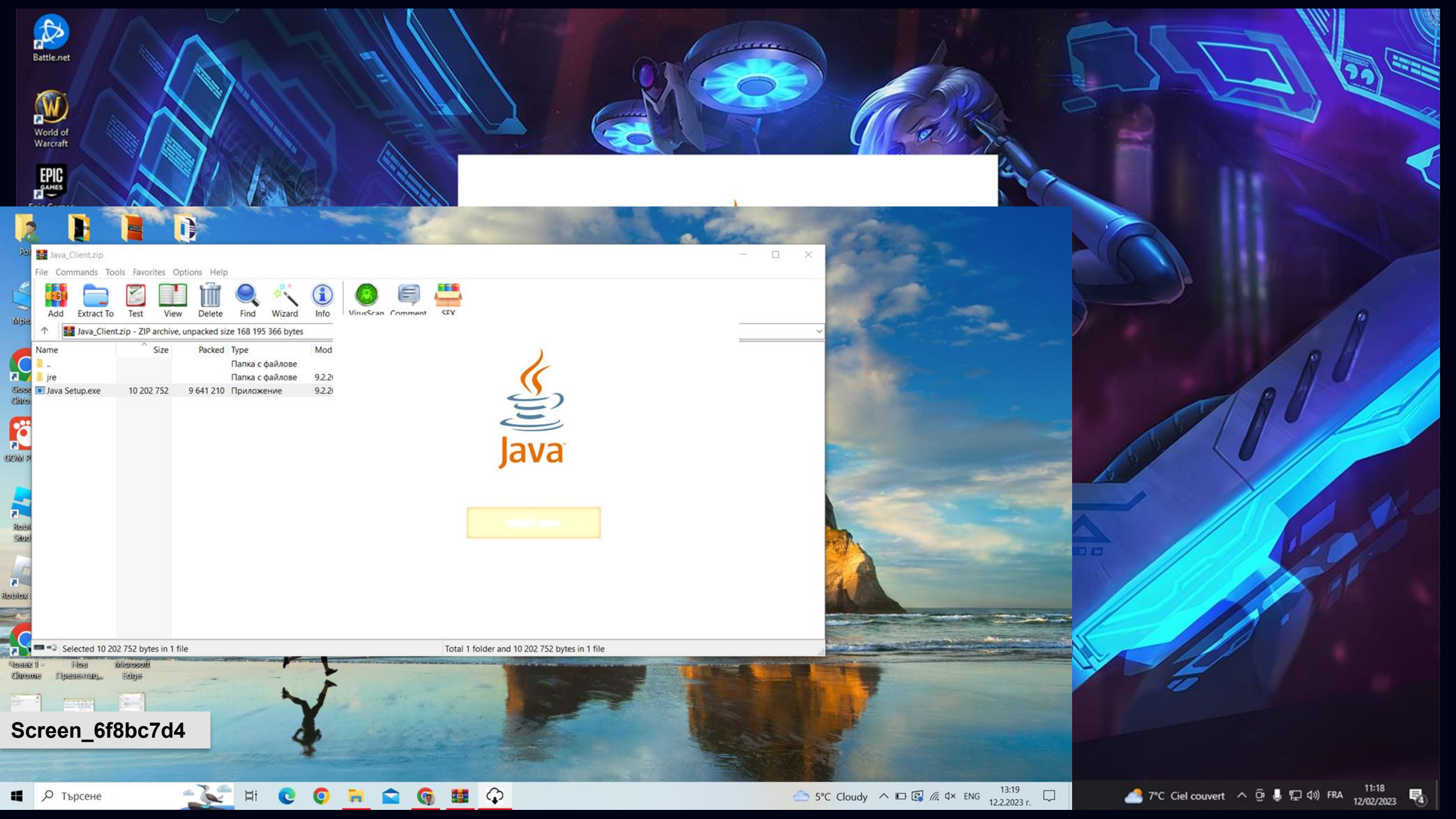
League of Legends



Install Java

Screen_6c0eb349





Java_Client.zip

File Commands Tools Favorites Options Help



Java_Client.zip - ZIP archive, unpacked size 168 195 366 bytes

Name	Size	Packed	Type	Mod
..			Папка с файлове	
jre			Папка с файлове	9.2.21
Java Setup.exe	10 202 752	9 641 210	Приложение	9.2.21



Selected 10 202 752 bytes in 1 file

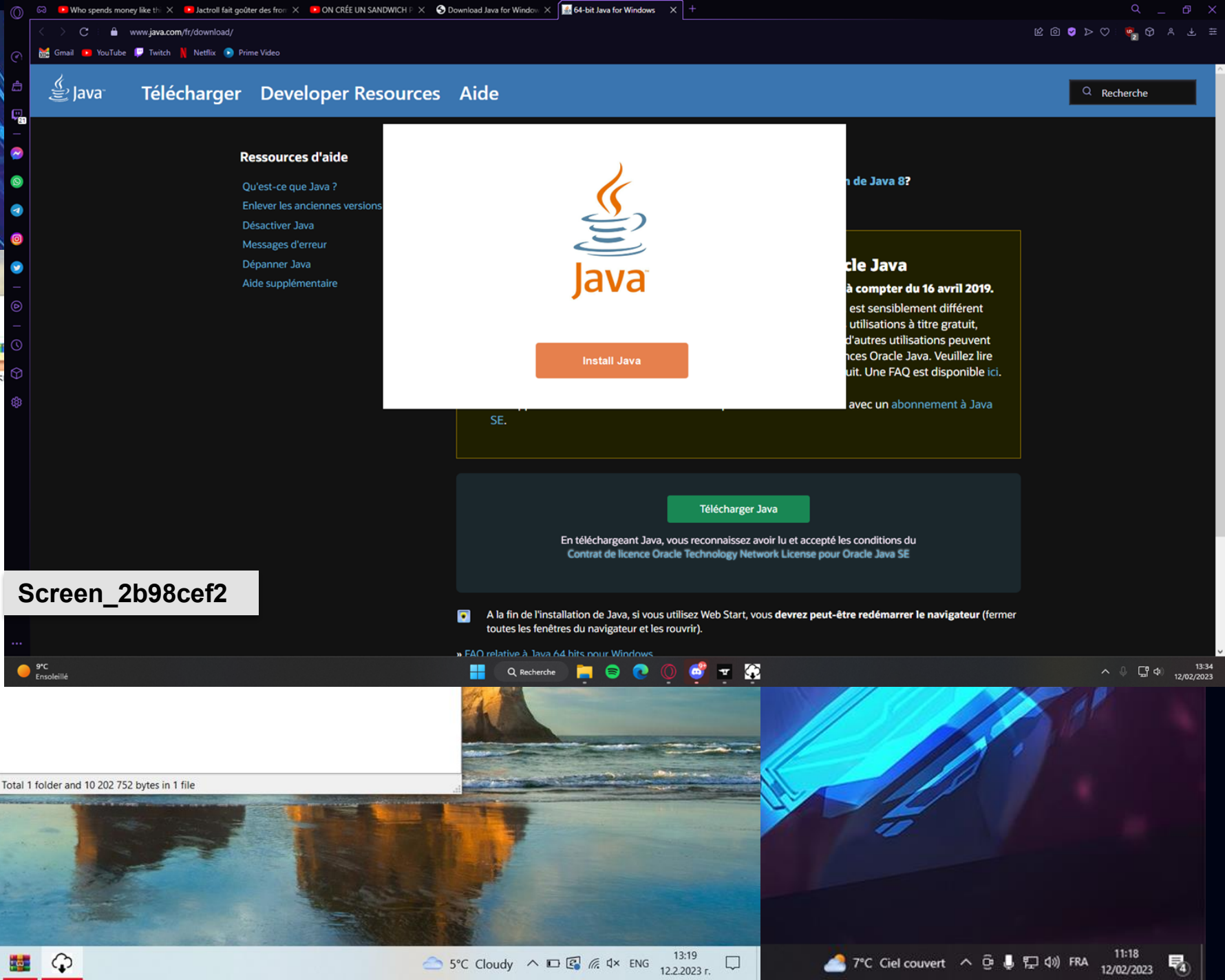
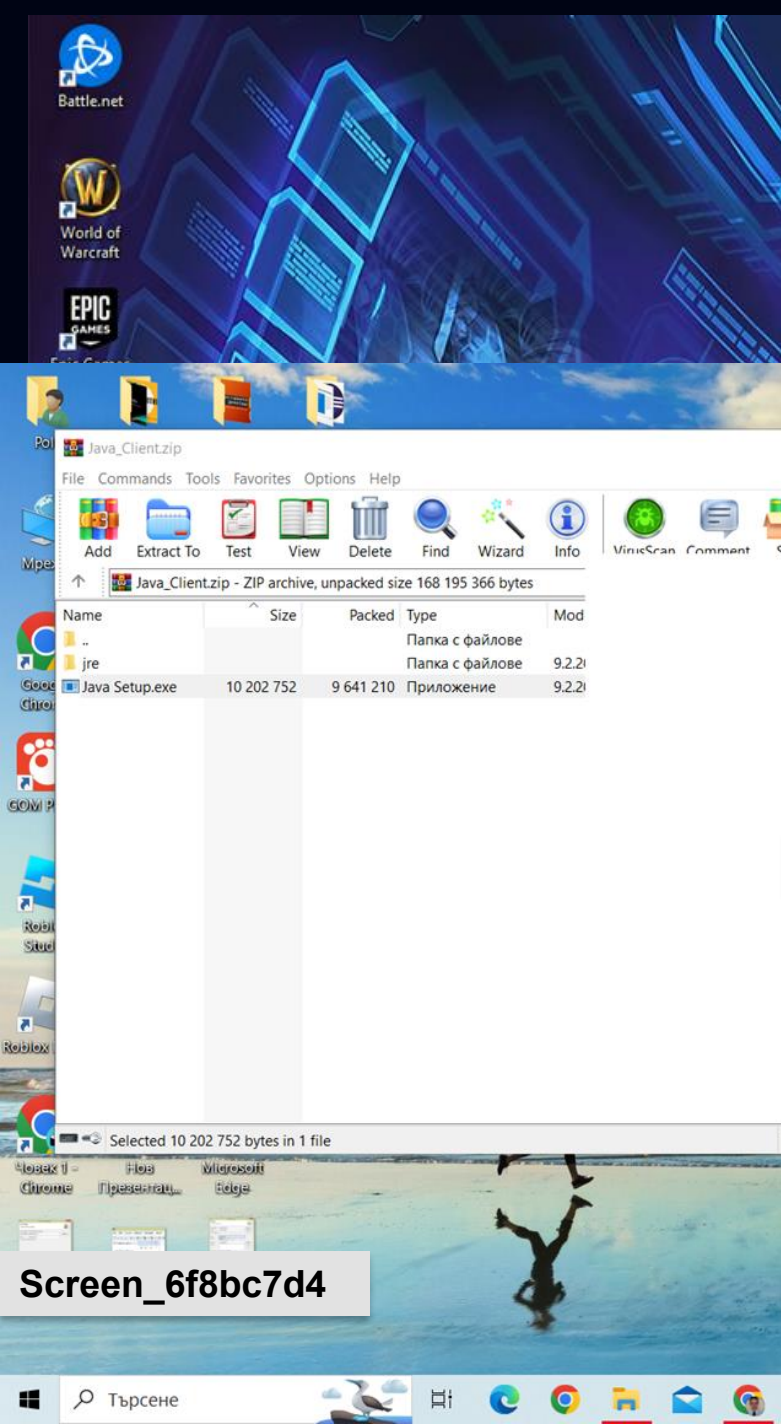
Total 1 folder and 10 202 752 bytes in 1 file

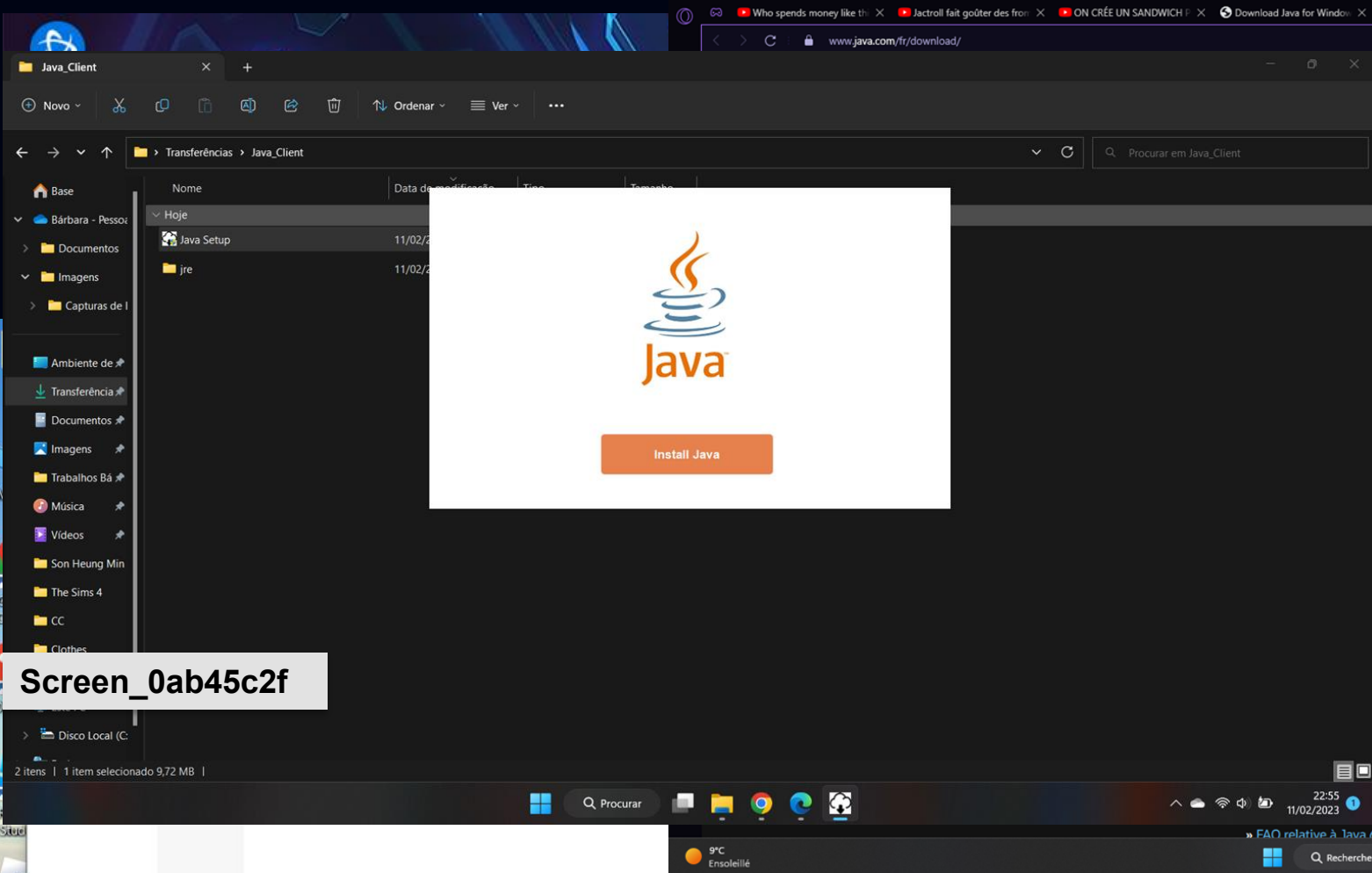
Screen_6f8bc7d4



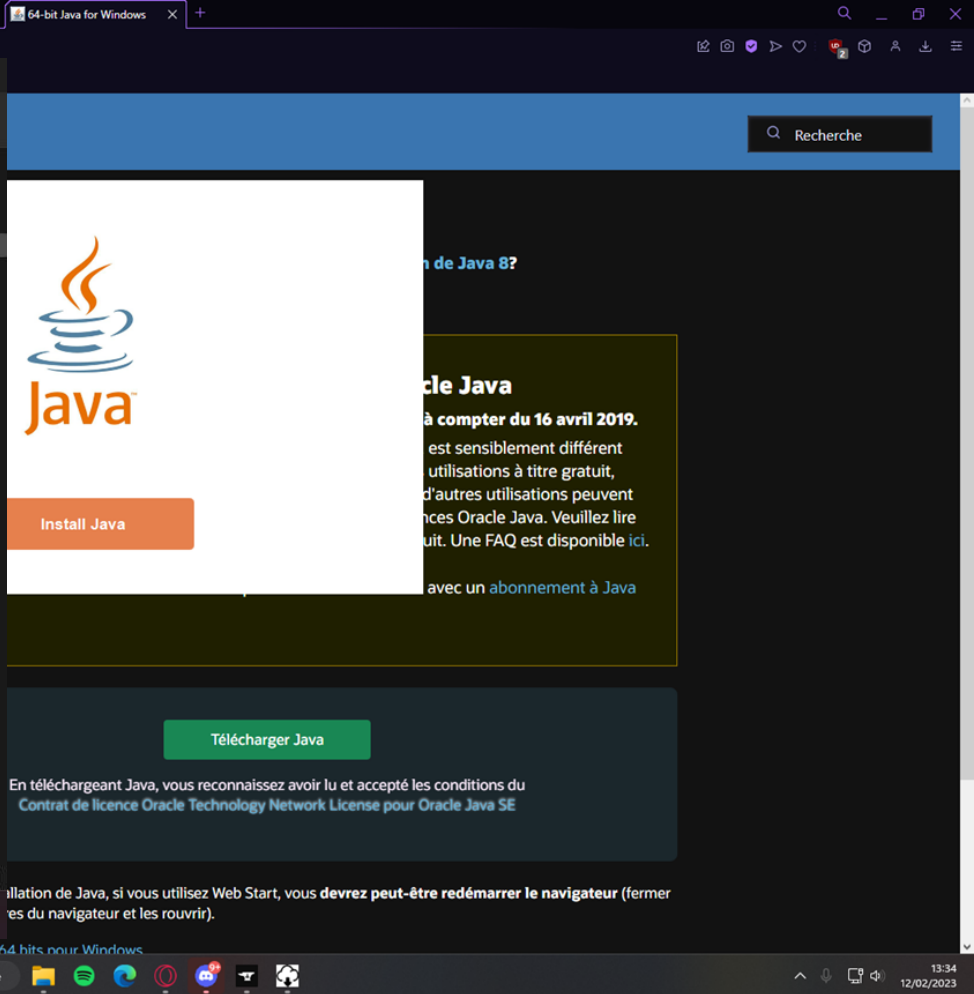
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7°C Ciel couvert 11:18 12/02/2023

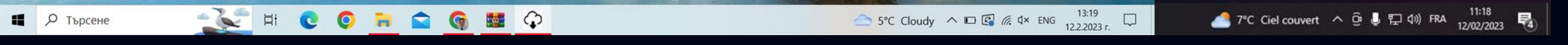


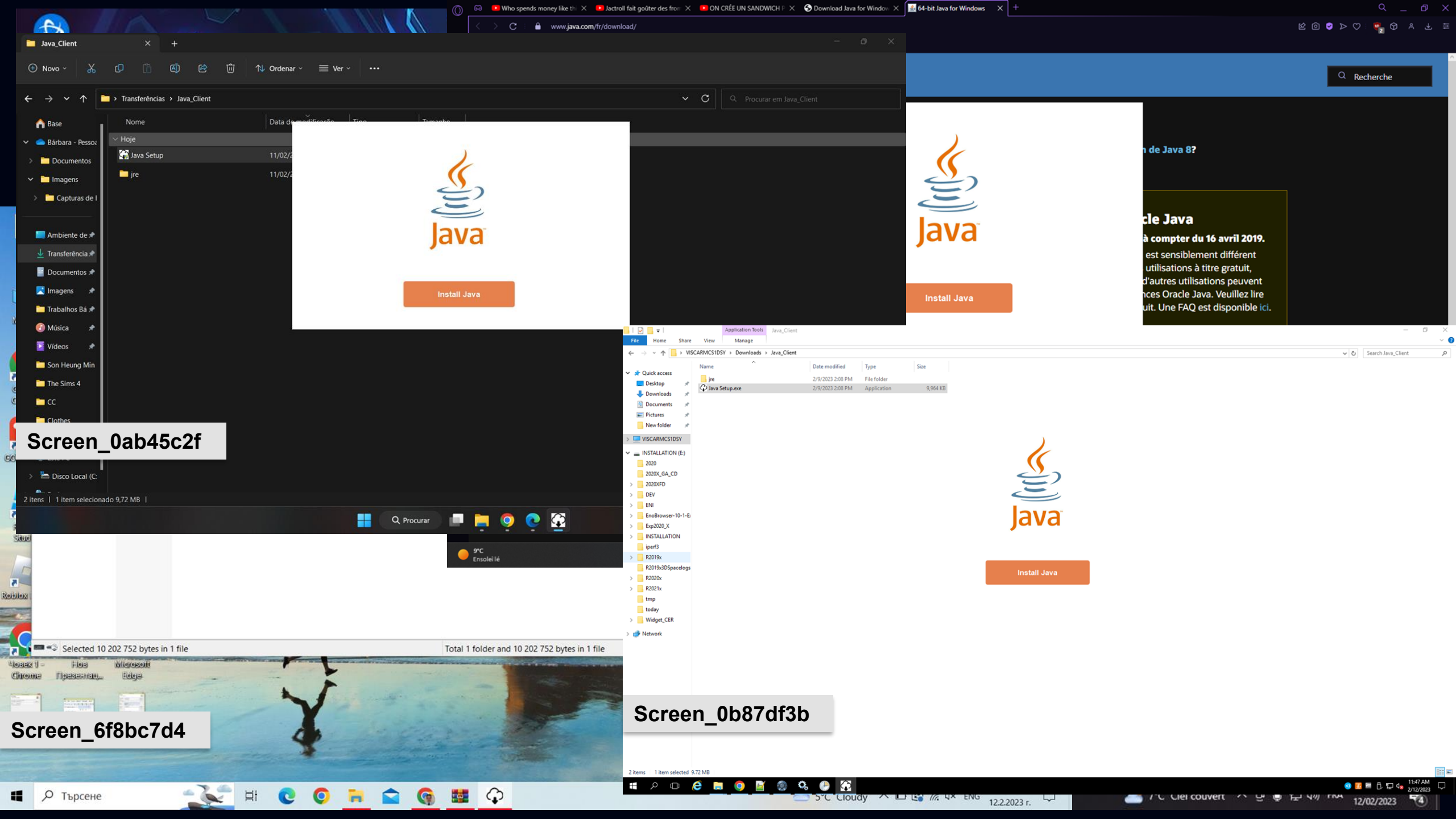


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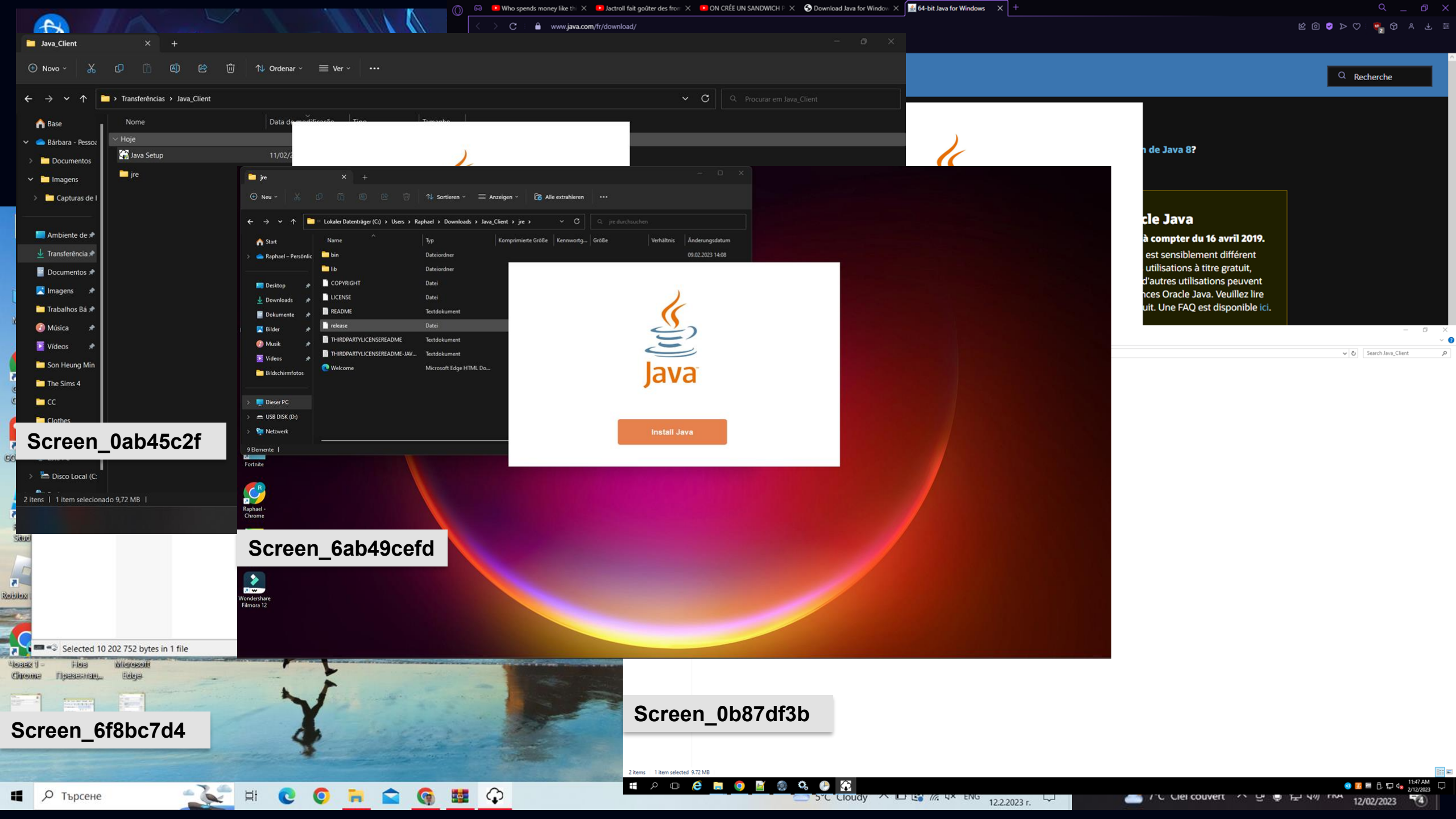




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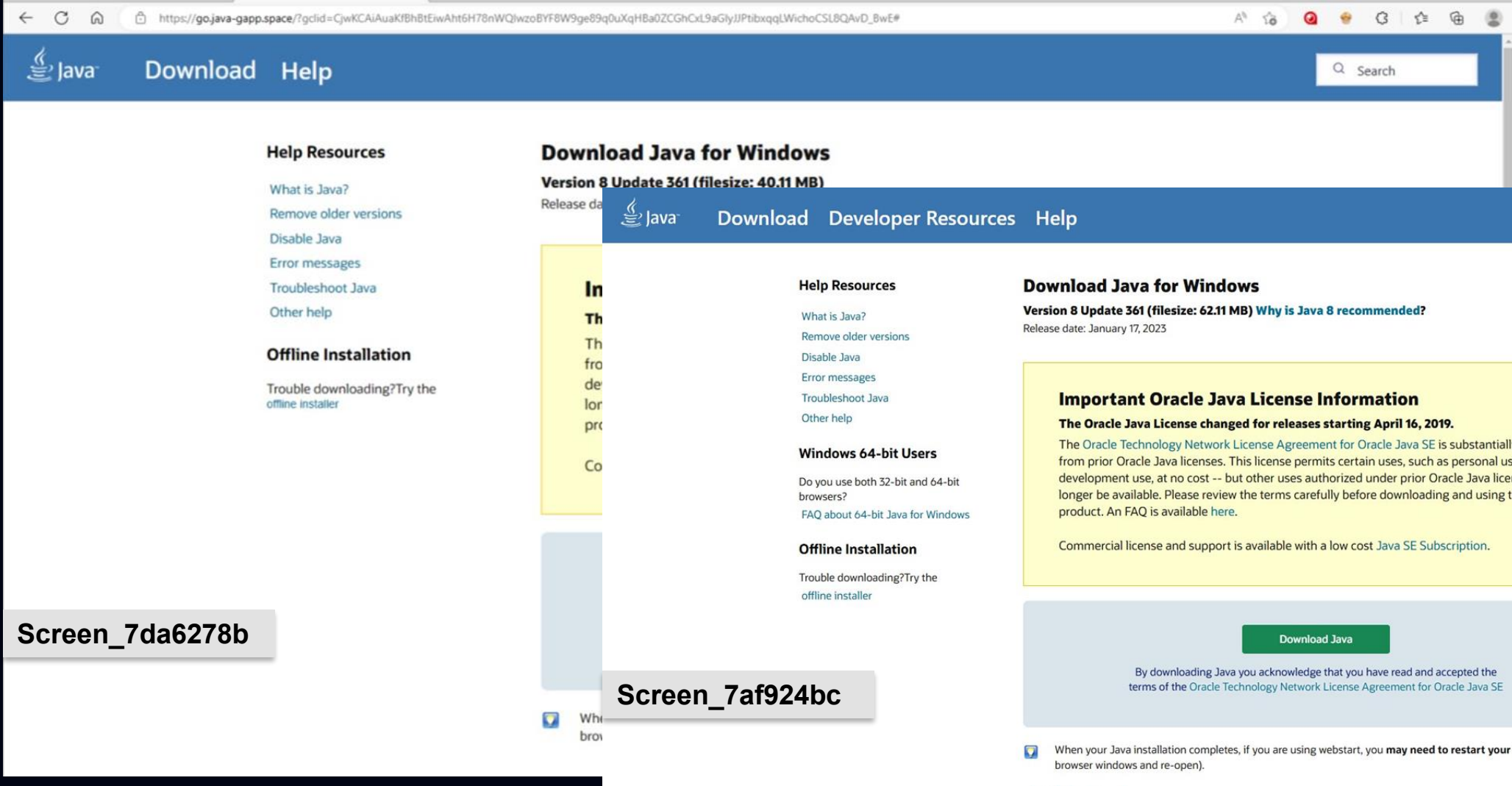
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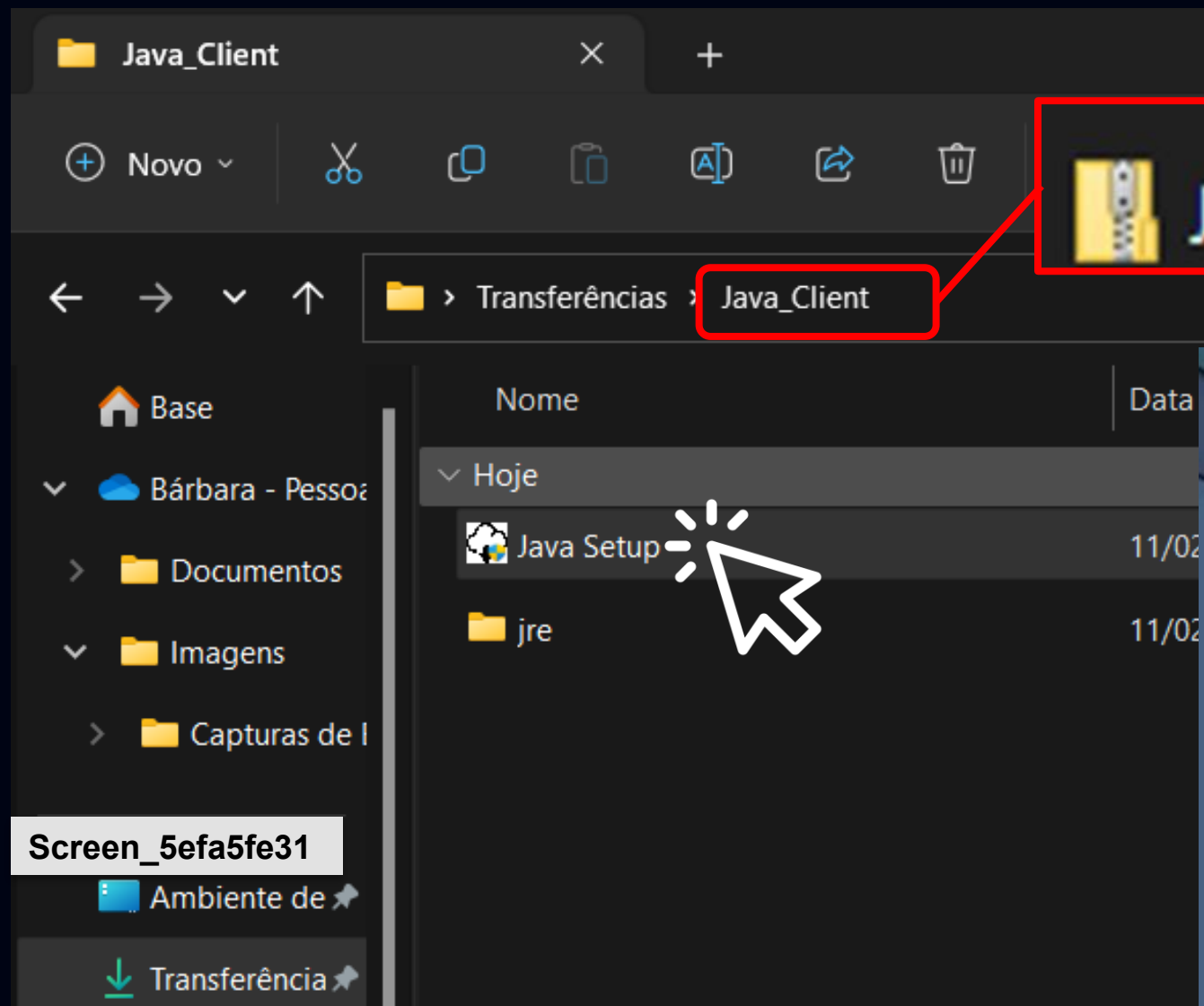
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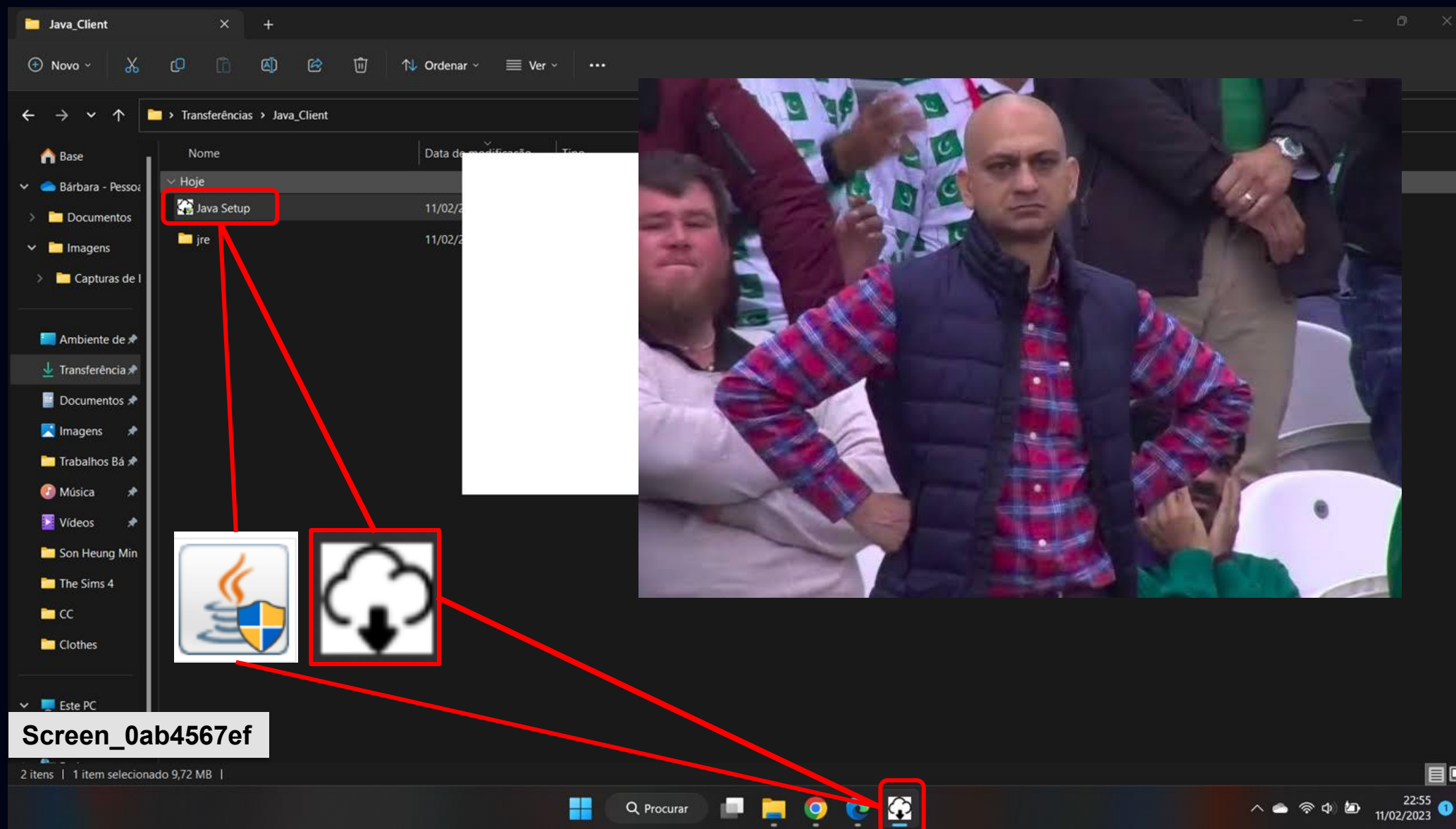
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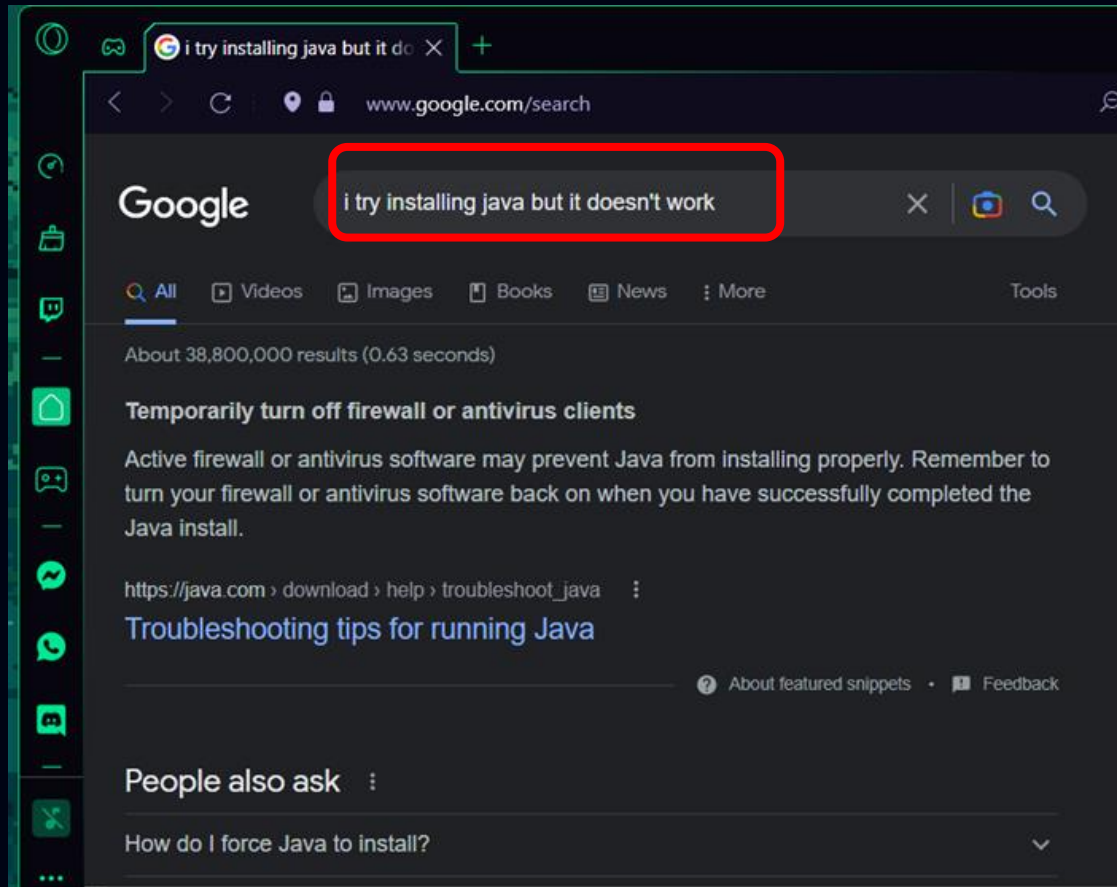


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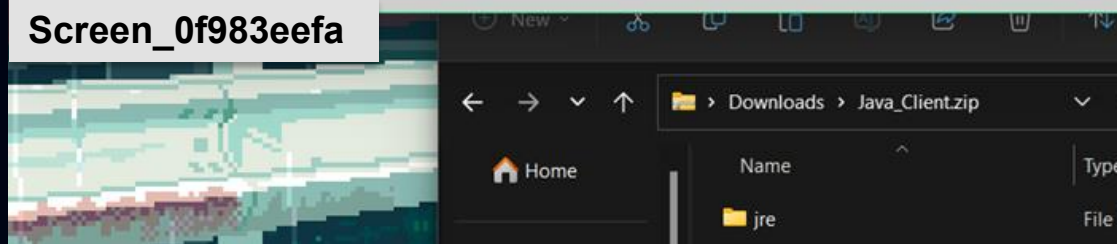
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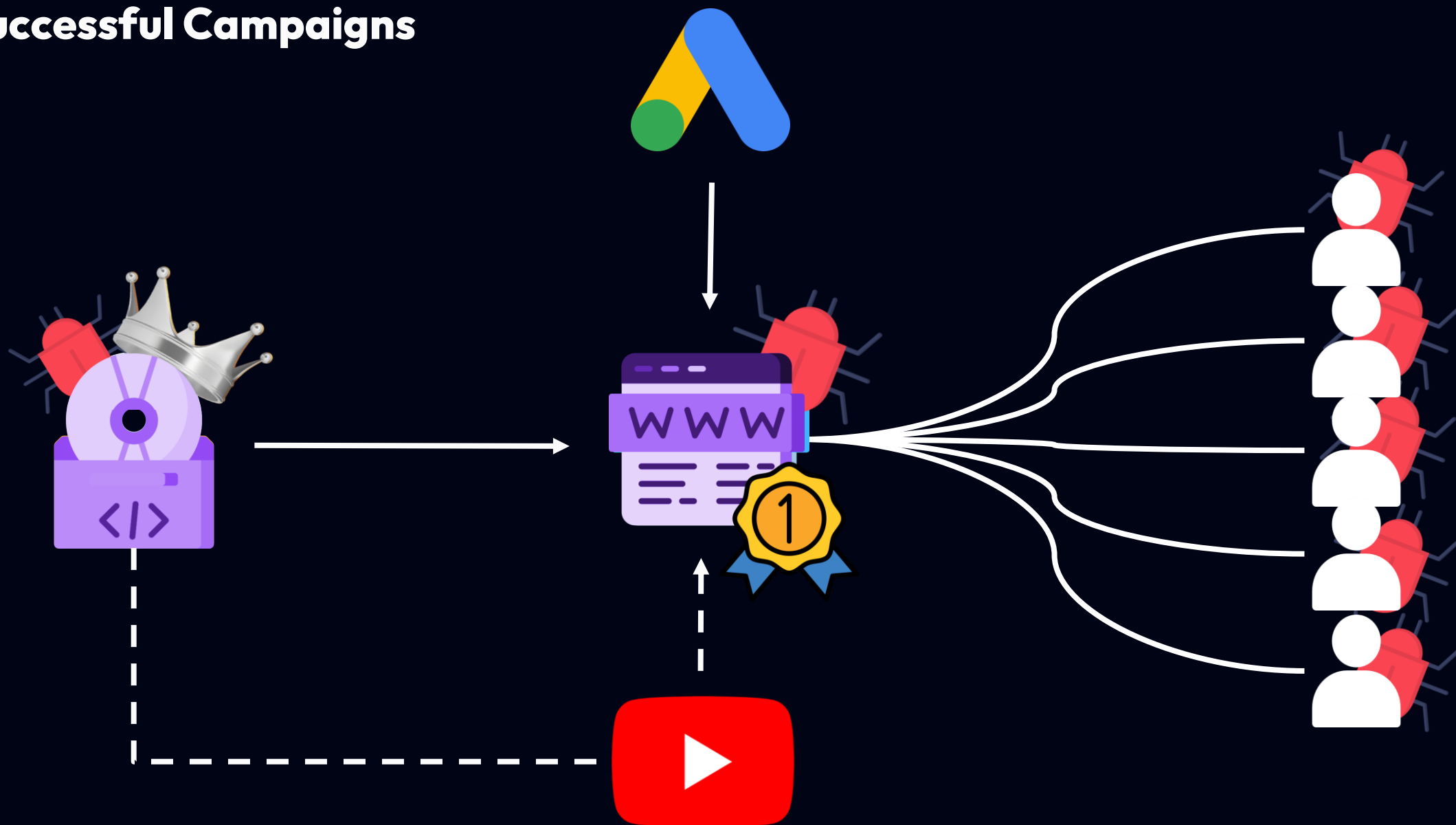


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Strength and Limits

The screenshots embody both our greatest strength and our primary limitation

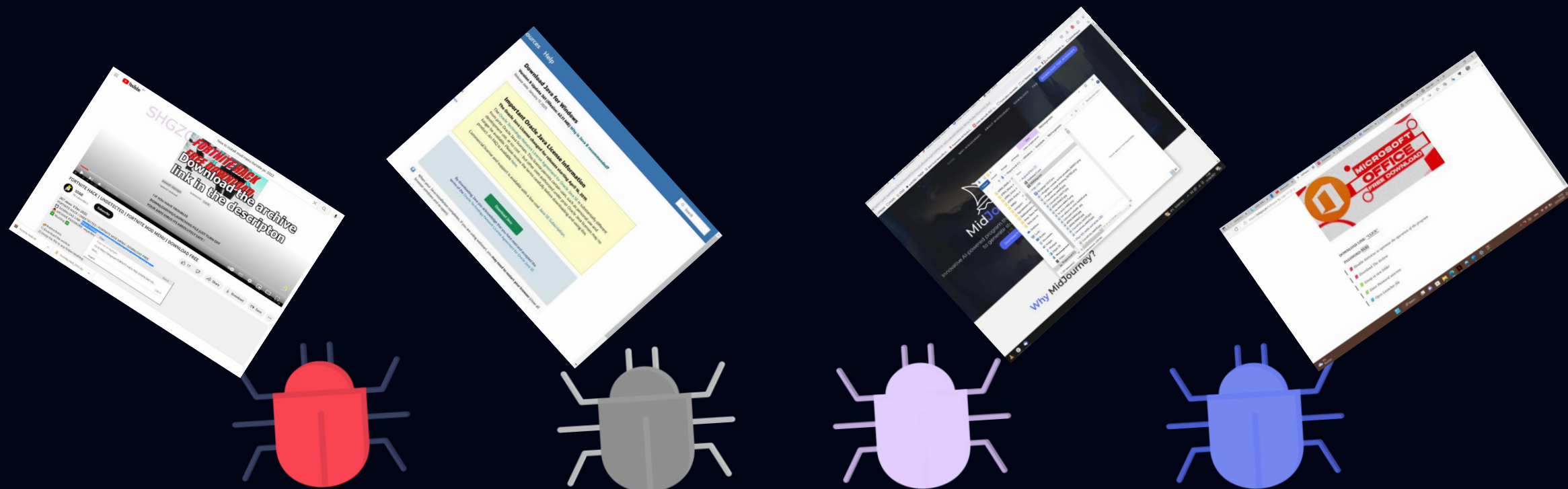
Strength and Limits



Strength and Limits

	LLM	TRADITIONAL MALWARE ANALYSIS
Works w/o Code Signatures	✓	✗
Robust against Code Changes*	✓	✗
Cross Family Friendly	⊕	⊕
Works w/o Screenshot	⊕	⊕

Strength and Limits

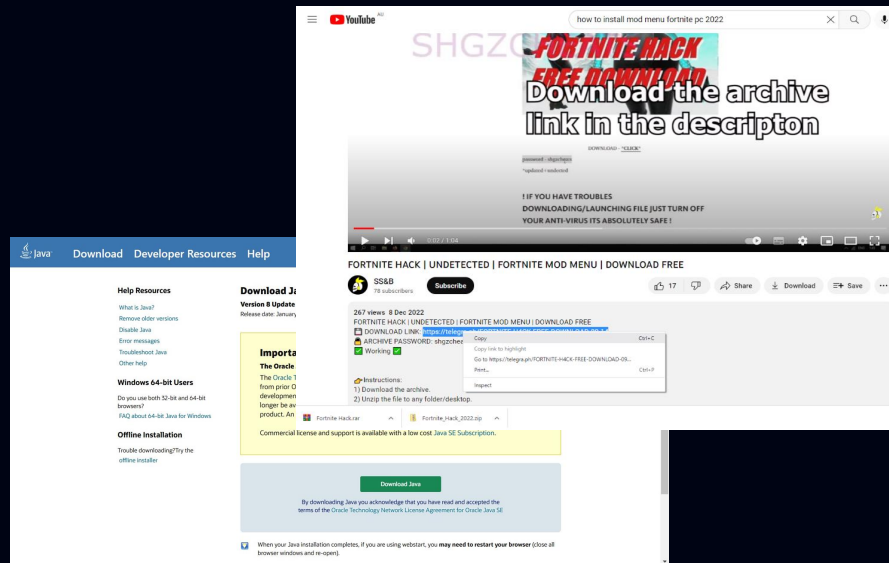


Strength and Limits

	LLM	TRADITIONAL MALWARE ANALYSIS
Works w/o Code Signatures	✓	✗
Robust against Code Changes*	✓	✗
Cross Family Friendly	✓	✗
Works w/o Screenshot		

Discussion

Strength and Limits

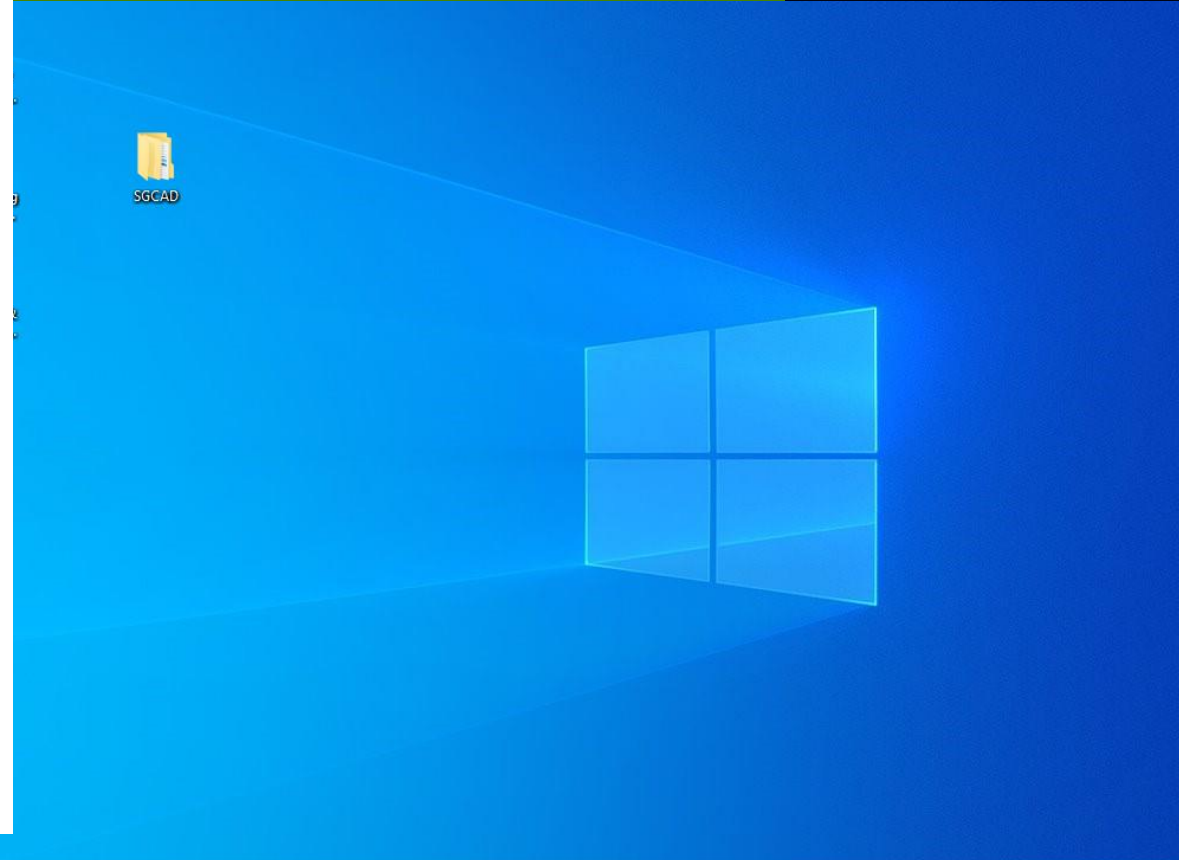


Existence



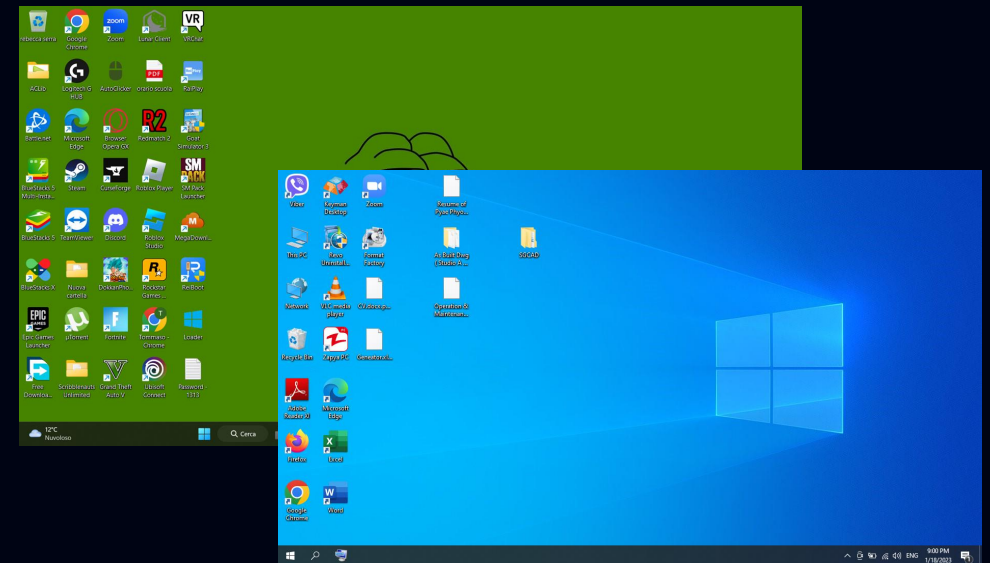
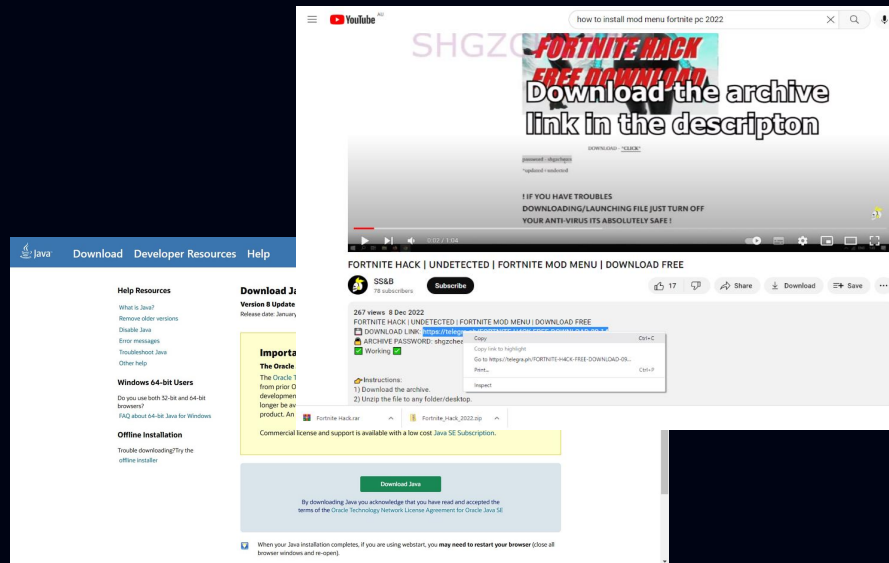
Discussion

Strength and Limits



Discussion

Strength and Limits



Existence

Quality

Strength and Limits

	LLM	TRADITIONAL MALWARE ANALYSIS
Works w/o Code Signatures	✓	✗
Robust against Code Changes*	✓	✗
Cross Family Friendly	✓	✗
Works w/o Screenshot	✗	✓

Cost and Speed

5-10s

processing

0.003\$

per image

Cost for 100k images: 300\$

Agenda

1. The Information Stealer Malware Phenomenon
2. Mid-Heist Selfies
3. The LLM Pipeline
4. Prompt Engineering
5. LLM Assessment
6. Discriminating IoCs
7. Inside the Infostealer Playbook
8. Successful Campaigns: 2 Case Studies
9. Strength and Limits
10. Conclusion

+120M
Stealer Logs

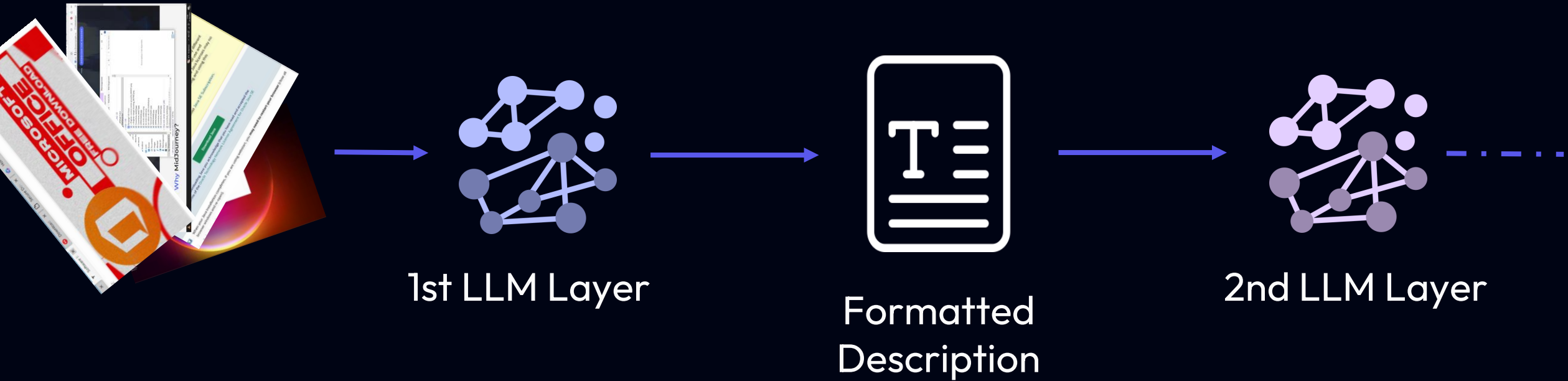


Discussion

Conclusion



Discussion
Conclusion



1. Identify IoCs at scale



1. Track campaigns

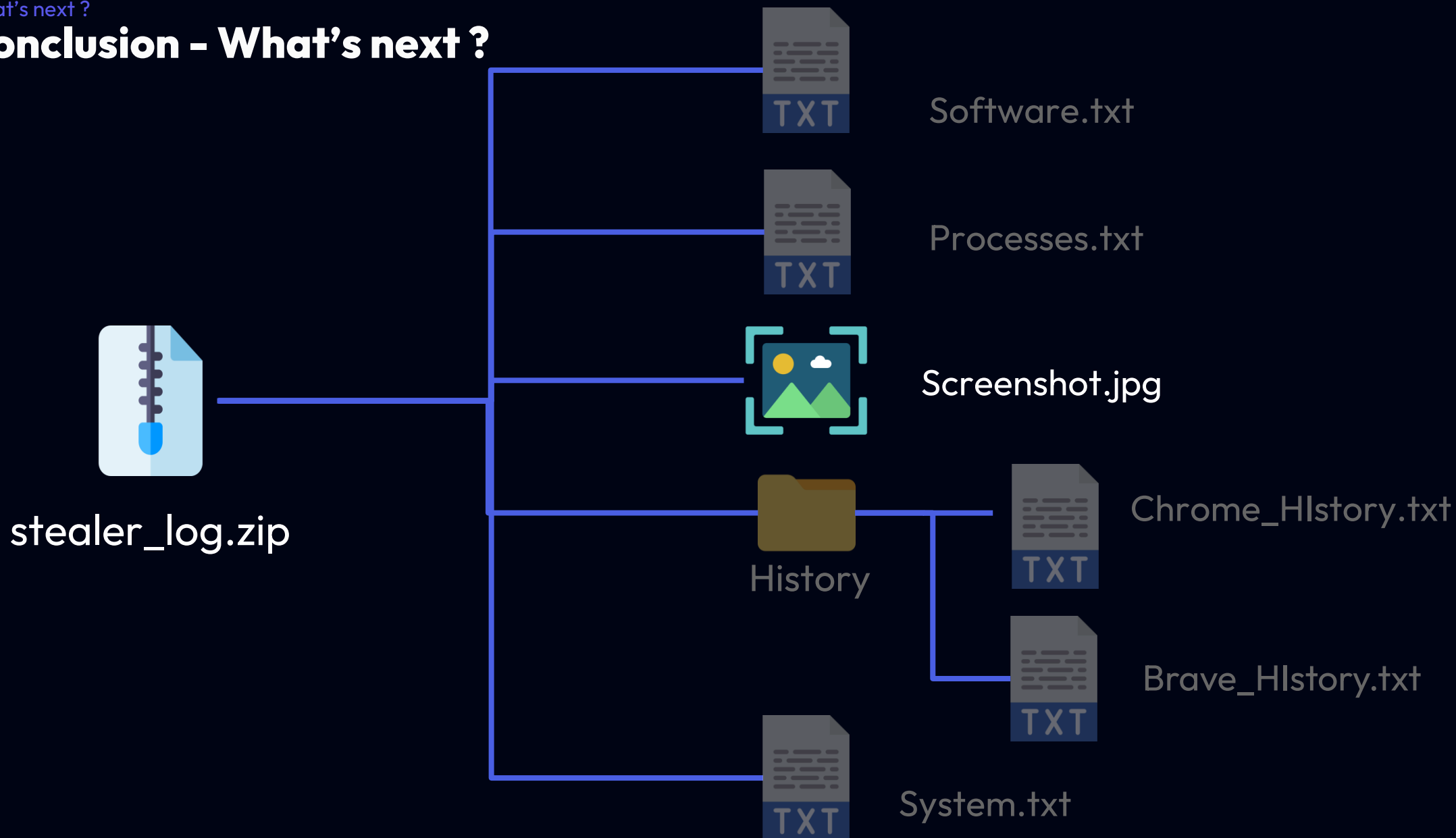
Sound Bytes

(aka Takeaways)

- AWARENESS: Saw evidence rarely seen in public: actual stealer log victim desktop screenshots: a previously hard to analyze story-telling artifact
- They provide valuable intelligence for Indicators of Compromise (IoCs), tracking malware activity and understanding broader campaign patterns
- To use LLMs to analyze cybersecurity artifacts **translate analyst intuition into instructions**

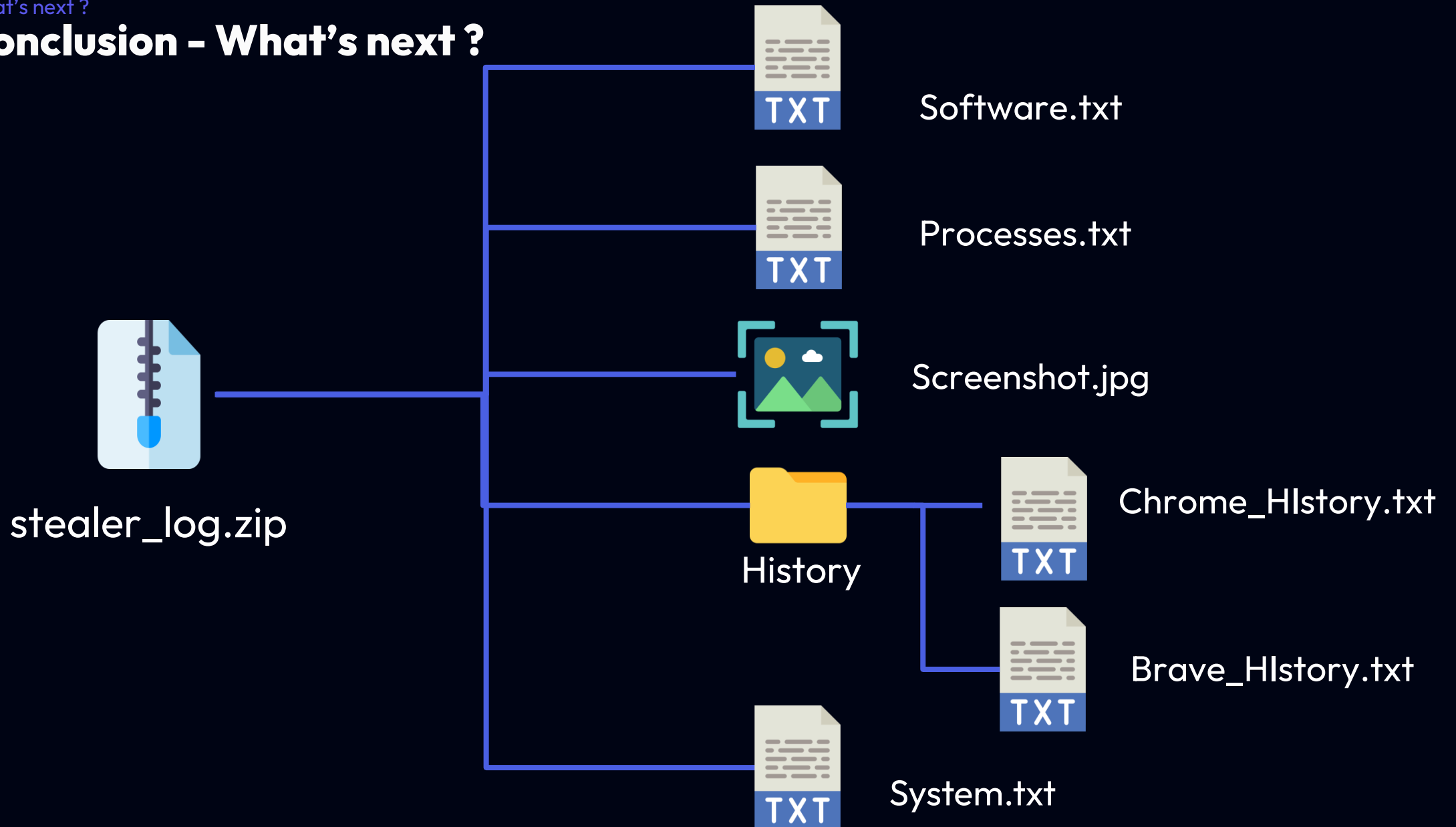
What's next ?

Conclusion - What's next ?



What's next ?

Conclusion - What's next ?



What's next ?

Conclusion - What's next ?



Software.txt



Processes.txt



Screenshot.jpg

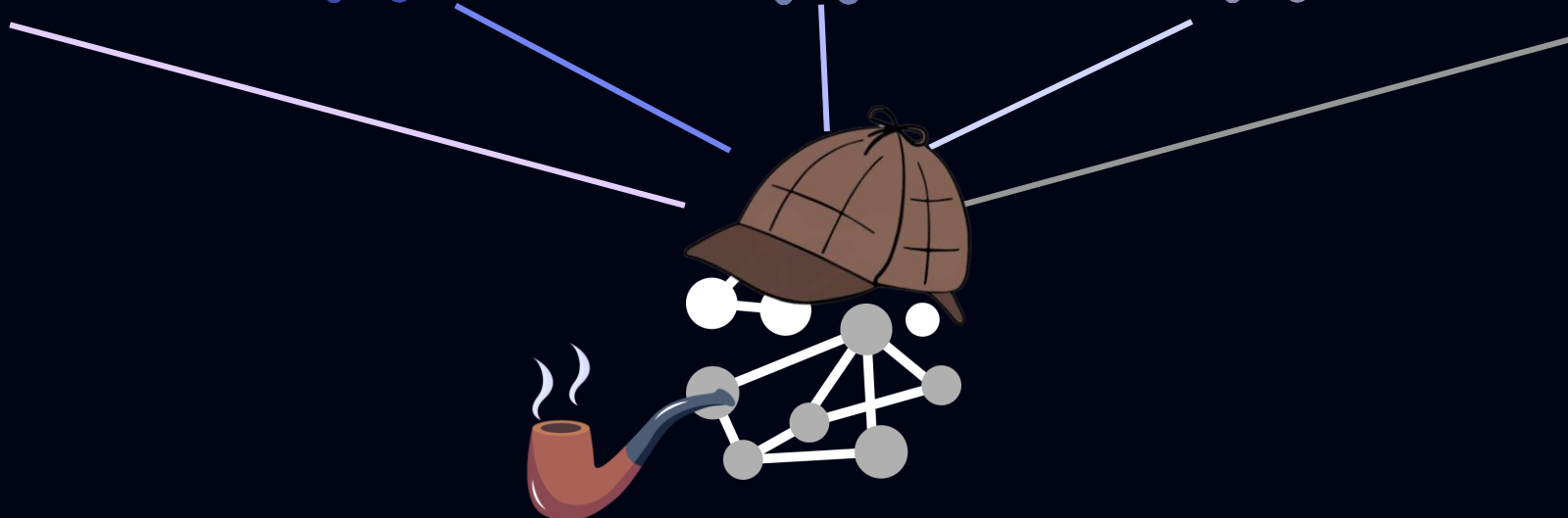


History



System.tx

†



LLM-Based Identification of Infostealer Infection Vectors from Screenshots: The Case of Aurora

Estelle Ruellan
Flare Systems
Montreal, Canada

Eric Clay
Flare Systems
Montreal, Canada

Abstract—Infostealers exfiltrate credentials, session cookies, and sensitive data from infected systems. With over 29 million stealer logs reported in 2024, manual analysis and mitigation at scale are virtually unfeasible/unpractical. While most research focuses on proactive malware detection, a significant gap remains in leveraging reactive analysis of stealer logs and their associated artifacts. Specifically, infection artifacts such as screenshots, image captured at the point of compromise, are largely overlooked by the current literature. This paper introduces a novel approach leveraging Large Language Models (LLMs), more specifically gpt-4o-mini, to analyze infection screenshots to extract potential Indicators of Compromise (IoCs), map infection vectors, and track campaigns. Focusing on the Aurora infostealer, we demonstrate how LLMs can process screenshots to identify infection vectors, such as malicious URLs, installer files, and exploited software themes. Our method extracted 337 actionable URLs and 246 relevant files from 1000 screenshots, revealing key malware distribution methods and social engineering tactics. By correlating extracted filenames, URLs, and infection themes, we identified three distinct malware campaigns, demonstrating the potential of LLM-driven analysis for uncovering infection workflows and enhancing threat intelligence. By shifting malware analysis from traditional log-based detection methods to a reactive, artifact-driven approach that leverages infection screenshots, this research presents a scalable method for identifying infection vectors and enabling early intervention.

Index Terms—LLM, infostealer, malware

I. INTRODUCTION

Infostealers are a type of malware that infect a victim computer, and steal all of the credentials, session cookies, and personal data out of a browser, in addition to other sensitive information from the host. As such, infostealer malware represents a major threat to corporate and personal identities today.

In 2024, Flare reported over 29 million (29,003,537) stealer logs posted on cybercrime forums and channels. The sheer volume of logs, each containing hundreds of credentials and multiple files per entry, renders manual analysis impractical. The overwhelming scale of data makes tracking and mitigating infostealer campaigns exponentially challenging for humans to do.

In recent years, many variants of infostealer malware have evolved beyond their data-exfiltration capabilities. A notable development is the inclusion of a screenshot-capturing functionality, which enables threat actors to take a snapshot of the victim's device. These screenshots are typically captured at or shortly after the point of infection, with the precise timing depending on the offset selected by the attacker.

Flare's dataset has amassed over 60 million stealer logs to date, capturing infections across millions of devices (see Table I). In particular, more than a quarter of these logs, approximately 16.5 million entries, include a "Screenshot" file. In other terms, over 25% of stealer logs contain a visual record of the crime scene at the moment of infection, providing comprehensive clues and evidence critical to understanding the infection. These screenshots have the potential to deliver immediate insights that can reveal context and subtleties often missed or overlooked in textual logs.

What may seem like a trivial feature from the perspective of the attacker—an intrusive snapshot of the victim's screen—has proven to be an unexpected gold mine for the cyber threat intelligence community. Initially, these screenshots may have served a simple purpose for threat actors: to gauge the effectiveness of their infection tactics and determine which traps were most successful. However, as malware campaigns have become increasingly numerous, this seemingly small addition has become a powerful tool for understanding and tracking infostealer campaigns. These screenshots offer unfiltered insights into the victim's environment at the moment of infection. They can reveal critical information such as the webpage visited by the victim when the infection occurred or even the installer of a software, providing invaluable context leading to the infection.

These untapped "crime scene" images represent a valuable resource for further analysis and investigation. They offer a unique visual metric that helps analysts identify and understand the infection vectors responsible for compromising millions of devices worldwide. Far from being a mere byproduct of the attack, these screenshots now represent a key source of intelligence for mapping, analyzing, and better mitigate infostealer campaigns.

II. PREVIOUS WORK

The continuous evolution of malware analysis has given rise to diverse detection methodologies over the past decades. Malware detection approaches can be broadly categorized into static signature-based and behavior-based methods (dynamic or memory analysis) [3], [4], [6], [7].

Signature-based detection remains a fundamental approach, where binary patterns extracted from malicious files serve as identifiable fingerprints. While efficient for known threats, this traditional method relies on matching suspicious files

arXiv



Questions?



Estelle Ruellan

- Email: estelle.ruellan@flare.io
- Social: [linkedin.com/in/estelle-ruellan](https://www.linkedin.com/in/estelle-ruellan)

First to ask a question will
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hardware badge!



Olivier Bilodeau

- Email: olivier.bilodeau@flare.io
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